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Seasonal changes in the biochemical profile of male fresh water fish *Channa punctatus*

Sharda & Arun Kumar*

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Received :20th April, 2019 ; Revised : 15th May, 2019

Abstract- Over the last few decades the contamination of fresh water with a wide range of pollutants has become a matter of concern. Metal contamination accounts for potential threat for aquatic organisms, which are exposed to significant amounts of heavy metals as a consequence of Industrial, Agricultural and Anthropogenic activities. Heavy metals at high concentration can be hazardous to metabolic, physiological and biochemical system of fish, which in turn affects human beings. The aim of the present study is to investigate the changes in biochemical parameters such as glucose and urea in *Channa punctatus*. *Channa punctatus* is commonly known as snake head fish of carnivore feeding habit. The cypermethrin pesticide is widely used in agricultural field for the control of insect pests and is coming under synthetic pyrethroids which have shown strong pesticidal activity in freshwater teleost fish, *Channa punctatus*.

Key words: Seasonal variation, glucose, urea, protein, *Channa punctatus*, biochemical parameters

INTRODUCTION

Seasonal variations in the various biochemical constituents were studied in the *Channa punctatus*. The chemical composition in fish is traditionally used as an indicator of the nutritional value as well as the physiological condition of fish and its habitat[1,2]. The present paper deals with similar studies carried out on the fresh water snake headed fish *Channa punctatus*. This fish is one of the commonest fish in India. It occurs throughout the country and forms the mainstay of inland fishery in many parts of India. It is popular item of diet amongst the poor class of people and is sold rather cheaply. Among these fishes its nutritive value is higher than that of *M.seenghala*. It has been fairly well established that in many fishes, both from freshwater and marine environments marked changes occur in the levels of the chemical constituents from season to season[3-5]. This fish is one of the common fishes of India. It is a popular item of diet and its nutritive value is high. The present paper describes the monthly variations in the biochemical composition of this fish. The perusal of the available literature shows that few workers have reported the seasonal variations in the blood glucose and urea levels. However, the seasonal variation in the blood glucose levels in *Channa punctatus* (Bloch) is reported[7]. The seasonal fluctuations and normal levels in the blood glucose in both fresh water and marine fishes. In the present study, attempts have been made to investigate to study the changes in biochemical parameters of *Channa punctatus*.

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The physiological and biochemical alterations observed in an animal under any physiological stress can be correlated with the structural and functional changes of cellular proteins. The normal blood glucose level of *Channa punctatus* undergoes a seasonal change. A post-spawning hyperglycemia is found during August, September, October and November. Fish kept out of water show increase in blood glucose level upto 24 hours.

When returned to water, the asphyxial hyperglycemia subsides after 24 hours. Starvation does not appear to affect the blood glucose in this fish. Basic ecological factors, such as feeding regime and stocking density, also have a direct influence on certain biochemistry parameters. The variation in protein, cholesterol and glucose level is directly to sex, size and age of the fishes[8-10].

MATERIALS AND METHODS

Sampling area

For the present study the carp, Snake head fish (*Channa punctatus*), male were collected from local market of Madhepura during different seasons of the year 2018-2019 and kept in laboratory aquaria. Healthy fishes were used for the estimation of glucose and urea level. The fishes weighing about 200-250g (Fig-1) were chosen as per availability in the market for the study. Before use fish were allowed to recover from asphyxia suffered.

Fish sample collection and maintenance

The blood was collected from the heart for the detection of urea and glucose for each month for a period of two years to detect the seasonal changes. For the estimation of urea, urease nesslerization method was adopted. The effect of asphyxiation was studied by keeping fish out of water in a glass trough having moist cotton at its bottom. The blood glucose was determined at varying intervals. Effect of starvation was observed in a batch of 30 fish kept in a large aquarium. Six fish were sacrificed at 7 day interval for one month and six fish after 60 days starvation. In all the cases fish were killed by a blow on the head and blood sample was taken by cutting the hepatic veins. The blood glucose was determined according to the method of Nelson as below: 0.1 ml of blood was mixed with 3.1 ml of distilled water and deproteinized with 0.4 ml of 0.3 N Barium hydroxide and an equal amount of 5 % zinc sulphate solution[11].

OBSERVATION

In male fish, the glucose content varies from 73.60mg/100ml to 87.75mg/100ml of blood. The maximum amount of blood glucose (87.75/100ml in male) is recorded during the winter, while minimum values (73.60mg/100ml in male) are obtained during summer. The highest values in blood urea (27.12 mg% in males) are recorded during monsoon while minimum values (25.22 mg% in males) are obtained during summer. Urea content during different seasons ranges from 25.22 mg% to 27.12mg% males.

Blood glucose

Seasons	Male		
	Range (mg%)	Mean (mg%)	S.E.M
Winter	76.0-100.0	87.75	±3.271
Summer	50.0-100.0	73.60	±3.902
Mosoon	64.0-100.0	83.40	±2.902

Blood urea

Seasons	Male		
	Range (mg%)	Mean (mg%)	S.E.M
Winter	20.0-37.0	26.66	±1.358
Summer	21.0-32.0	25.22	±1.230
Mosoon	20.0-34.0	27.12	±0.960

DISCUSSION

The result of the present study indicate that blood glucose and urea of *C. punctatus* changes seasonally and this is probably connected with the temperature, gonadal cycle, metabolic activities, feeding intensity etc. It shows that seasonal variations in blood glucose level are mainly due to thermal variation and gonadal cycle[12-13]. It is observed experimentally that lowering of temperature causes hyperglycemia in *Opsanus tau*.[14] They recorded that serum glucose level increased with the fall of temperature. They observed highest blood glucose level during winter and lowest during summer and correlated it with the lowering of temperature and gonadal development. Experimentally they observed that the fish showed a considerable rise in its blood glucose level when transferred from water at 35°C to water at 5°C. The present study also revealed that the fish, *C. punctatus* has highest blood glucose level during winter. Thus, the authors agree with the views of the above authors.

Thus from the above observations it may be concluded that temperature, gonadal development, metabolic activities, feeding intensity etc., play an important role in the regulation of blood glucose and urea levels.

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Effect of water pollutants on biochemicals of gut of some aquatic insects

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Abstract : The two important pollutants DDT and Detergent were taken; these pollutants affect and decline the Biochemicals like carbohydrate, Protein of gut of aquatic insects. The concentrations of carbohydrate were changed with the treatment of pollutant compared with controlled condition

Key words: Water pollutants, biochemicals, gut, DDT, detergent

INTRODUCTION

Some water pollutants liked DDT and Detergent, having the poisoning effects on the insects because DDT and Detergent is non-biodegradable substances, which can not be decomposed and persist in environment for long time of periods. These pollutants accumulate in the tissue of insects and alter the conformation of histology and change the concentration of biochemicals like carbohydrate, protein and enzyme.

These pollutants not only affect the aquatic insects but also other animals like fish, Human, etc in ecosystem they affect the whole food chain and food web. DDT the first of chlorinated organic insecticides was originally prepared in 1873, but it was not until 1939 that Paul Muller of Geigy Pharmaceuticals in Switzerland discovered the effectiveness of DDT as an insecticide; he was awarded the Nobel Prize in medicine and Physiology in 1948 for this discovery.

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The use of DDT increased enormously on a worldwide basis after World War II, primarily because of its effectiveness against the mosquito that spreads malaria and lice that carry typhus. The World Health Organization estimates that during the period of its use approximately 25 million lives were saved. DDT seemed to be the ideal insecticide is cheap and of relatively low toxicity to mammals (oral LD50 is 300mg/kg). However, problems related to extensive use of DDT began to appear in the late 1940s. Many species of insect developed resistance to DDT, and DDT was also discovered to have high toxicity toward fish, Update by Darl Harrison (Molecule of the Month for August 1997)

In insects, like other animals, glucose occupies a central role in carbohydrate metabolism participating in different biochemical process however, in a majority of insect's species; amount of the free glucose is quite small.

The most characteristic sugar in insect's haemolymph and other body tissue is Trehalose, a disaccharide, of glucose. It is non reducing sugar by virtue of anomeric carbon atoms of both glucose molecules are bonded in

glycosidic linkage. In haemolymph usually there is relatively low level of glucose along with higher trehalose content (Friedman, 1970; Chippendale, 1978).

It has been suggested by Mordeou *et.al* (1984) that body carbohydrate concentration in insects fluctuates, with food intake and starving condition, other available information in this regard is reported for *Anthrenus vorak* (George and Nayer 1964) *Trogoderma granarum* (Koarnavas and Nair 1969; Rao and Aggarwal 1970). *Philosoma ricini* (Plant and Moris 1972, Pant and Kumar, 1979) *Callyphora erythrocephala* (Durve 1977) *Bombyx mori* (Nandi 1979); *Tribolium estaneum* and *Tribolium confusum* (Islam and Roy 1982), *Hydrophilus olivaceous*, Srivastava, 1991) *Spherodema rusticum* (Sharma and Ehteshamuddin, 1992).

A sub lethal dose of insecticides although is a very low concentration apparently do not cause mortality of aquatic insects, crustaceans fish etc, but they cause alteration in their metabolic process including carbohydrate metabolism O'Brion(1960,1967) gave review of insecticides action of lamarrei upto 72 hours of exposure after which hypoglycaemia response were recorded, Khan(1994) reported that in *Cybister confusus*, DDT and EDTA has glycemic effects on haemolymph, fat body and ovary of both immature and mature insects preventing carbohydrate yolk deposition in ovary thus affecting ovarian maturation.

MATERIALS AND METHODS

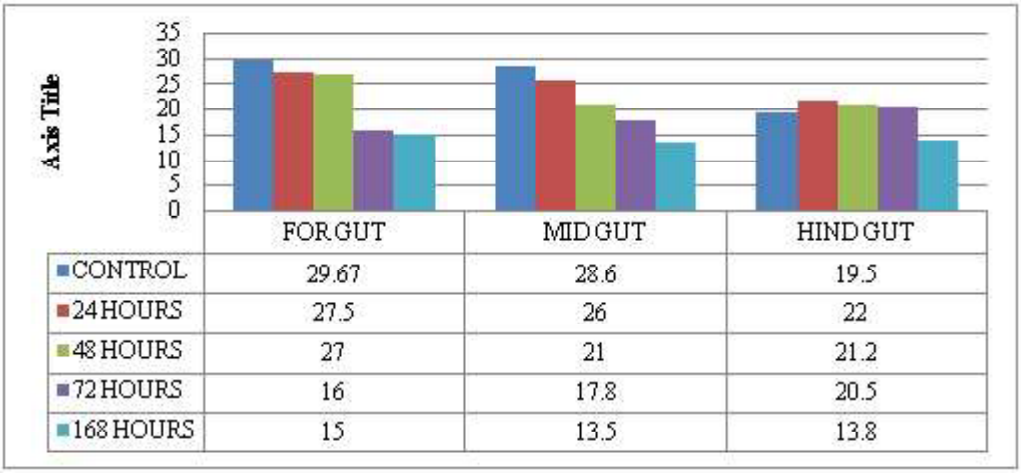
Quantitative estimation of carbohydrate of the alimentary canal of aquatic insects of both control and treated insects was done by Anthron reagent method (Carrol *et al.* 1956) which is a modified method of Roe (1955)

1. Insects were dissected in normal saline alimentary canal was taken our Fore gut, mid gut and hind gut were taken separately. Each part was homogenized in 1ml dist.water with homogenizer.
2. This homogenate was collected in centrifuge tube and was centrifuge at 3000 r.p.m in the centrifugal machine.
3. 1ml of the supernatant 4ml of 02% Anthron reagent was added and kept on hot water bath for 15minutes. Green colour developed. It was called down to room temperature.

RESULT AND DISCUSSION

In the present investigation total carbohydrate concentration in the different gut of control aquatic beetle was 29.67+4.8 S.D mg/g in the foregut, 28.6+4.5 S.D mg/g in midgut19.5+4.0 S.D mg/g in the hindgut. Both DDT and detergent treatments resulted into the gradual but significant decline in carbohydrate concentration of the mid gut, in the fore and hind gut, however the total carbohydrate concentration after DDT and Detergent treatment showed an initial rise and subsequent decline. Both detergent and DDT affect carbohydrate concentration of the gut adversely. (See graph plate)

Graph 1. Level of concentration of carbohydrate in control and DDT treated aquatic insects (water beetle)



	FOR GUT	MID GUT	HIND GUT
CONTROL	29.67	28.6	19.5
24 HOURS	27.5	26	22
48 HOURS	27	21	21.2
72 HOURS	16	17.8	20.5
168 HOURS	15	13.5	13.8

Graph 2. Level of concentration of carbohydrate in control and Detergnet treated insects (water beetle)



	FORE GUT	MID GUT	HIND GUT
CONTROL	29.67	28.6	19.5
24 HOURS	21.5	21	21.5
48 HOURS	16.4	16	20.5
72 HOURS	26.2	21.5	21.8
168 HOURS	25	19	20

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Effects of water quality on the growth of freshwater fishes in Medininagar, Palamu, Jharkhand, India

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Abstract : This paper represents the effects of water quality on the growth of freshwater fishes have immense significance for Medininagar, Jharkhand where large population depend on fishing not only as food source but also for their livelihood. Fish is the main aquatic vertebrate as far as nutritional point of view and concerned due to its high protein content. It is required to correlate the growth of fish with respect to local environment that surround the water body and influence water quality. The members of Cyrinidae family dominate in fish population. There are few ponds which contain water throughout the year and forms only source for supply of fishes in Palamu. The ponds and other water bodies are not fully exploited for fish culture. It has been suggested that water pollution is the leading world wide cause of death and diseases. However, fishes are very sensitive to wide variety of toxicants especially pesticides which cause deleterious effects through accumulation[1].

Key words: Freshwater fishes, water quality, Cyprinidae, Palamu, Growth, Population

INTRODUCTION

It is observed that there is great effects of water quality on the growth of freshwater fishes in Medininagar, Palamu, which causes serious health hazards to living system because of their rapid fat solubility and bioaccumulation in non target organisms. Even at low concentration, pesticides may exert several adverse effects which could be monitored at biochemical, molecular and behavioural levels. The factors affecting water quality with pesticides and their residues include drainage, rainfall, microbial activity, soil temperature, treatment surface, application rate as well as the solubility, mobility and pesticides. It is seen that water pollution is a major cause of global concern as it leads to onset of numerous fatal diseases. Pesticides have been associated with reduced hatching blue sac disease and swim-up syndrome mortalities in lake trout eggs yet,[2] these studies indicated that the concentration of pesticides necessary to effect eggs weremuch higher than the level found in fish eggs. Accumulation of pesticides in the water bodies primarily effects in the non-target organism especially fish and yet deposited. These fishes through food chains effects humans and causes deleterious effects. Thus, it has been suggested that water pollution i.e. water quality is the leading world wide cause of death and disease. However, effect of water quality on the growth of freshwater fishes have immense significance for Jharkhand where large population defend

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on fishing not only as food source.but also for their livelihood.

The members of Cyprinidae family dominate the fish population. The survey is based on present observations and earlier reports. There are few ponds which contains water throughout the year and forms only source for supply of fishes is Palamu. The ponds and other water bodies are not fully exploited for fish culture.

MATERIALS AND METHODS



Fig. 1- *Cirrhinus mrigala*

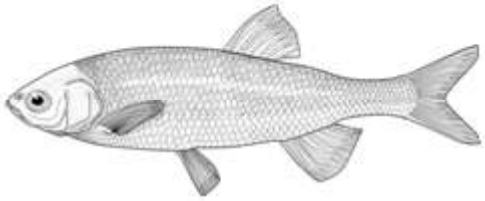


Fig. 2- *Catla catla*

Periodical estimates of fish growth and yield in relation to water samples were collected and analysed for computation of fish population dynamics and impact of physico-chemical complexes on fish yield of the water bodies. The whole exercise also involved network analysis and rendering of food chains/food web of the existing ecosystems. Analyticdal sutdies in the selected ponds were carred out over 3 years (2015-2018 in different seaons. Mean values are presented here physico-chemical parameters and biomass of most organisms were studied and analyzed as given by Saxena.[3] Identification of fish was assisted by the publication of Chakraff.[4] Fish species *Labio rohita*, *Catla catla, Cirrhinus mrigala* were selected for study. Purification of ponds is carried out by using calcium carbonate of phosphate after every 15 days. For oxygen generation, potassium permagnate is used. Temperature is normally around 20°C except in cold weather. Temperature of the water affects the activity, behaviour, feeding, growth and reproduction of all fishes. Metabolic rates in fish double for each 18°C rise in temperature. Temperature also determines the amount of dissolved gases in the water. The cooler of water the more soluble the gas. Temperature plays a major role inthe physical process called thermal stratification water has a high-heat capacity and unique density qualities. Water has its maximum density at 39.2°F. In spring, water temperature are nearly equal at all pond depths. As a result nutrients, dissolved gases and fish wastes are evenly mixed throughout the pond. As the days become warmer, the surface water becomes warmer and lighter while the cooler-denser water form a layer underneath. Circulation of the coldeer bottom water is prevented because of the different densities between the two layers of water. Dissolved oxygen levels decrease in the bottom layer since photosynthesis and contact with the air is reduced. The already low oxygen levels are further reduced through decomposition of water products, which settle to the pond bottom.

RESULTS AND DISCUSSIONS

Knowing biological information about growth reflects the general well-being and fish adaptation to the environment.[5] The behavioural response of fish towards toxicant was grossly dependent on concentration and length of exposure. Freshwater fishes relies more on aquatic gas exchange obtaining about 95% oxygen from water as only about 5.0% of obtained is obtained from air under normal bimodal conditions. Fishes were often observed swimming with jerky movements on the surface of water and tried to jump out of aquaria. For optimal growth of most fish species, desirable pH range is 6.5 – 9.0 and levels below 6.5 leads to a decrease of reproduction.[6] Also, a literature review concerning pH values for lakes indicates that usually are found in the range 7.3 – 9.2.[7] A value of pH lower than than 8.4 could indicate that the respiratory processes in a water reservoir are greater than the photosynthetic processes, while alkaline pH higher than 8.8 could produce death of the fish.[8] Yang et al; (2011) claimed that the increase in ammonia concentration in pond water may cause toxicity to aquatic organisms. However, toxicity of ammonia to fish varies with fish species, age or other quality parameters.[6,9] Both human and fish health could be affected by microorganisms and infections can represent the cause of losses in freshwater fish.[10]

The microbial pollution in the aquatic ecosystem gets accumulated in the tissues of fishes and thus they become unfit for human consumption.[11] In adddition, it has been reported that fungal contamination of fish may constitute a public health hzards, due to my cotoxin production.[12] This fact could be a general interest if one takes into account that fishermen encountered in the studied area use to keep, cook and eat their catch.

CONCLUSION

Assuming the values of all the abiotic and biotic parameters, the studied area was found to be not within the suitable range for well-being of both analyzed fish species. Although recreational fishing is often practiced in the study site, we strongly recommended caution regarding the consumption of water and fish caught from ponds. However, fish represents an important part of a healthy diet, can be eaten by people who catch it.

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Fish diversity and abundance in relation to water quality of Gaya reservoir, Bihar (India)

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Abstract : Freshwater fish diversity, abundance and richness status of Gaya reservoir (Bihar) was studied monthly from September 2017 to October 2018 in three sampling sites. The present study has shown that Gaya reservoir supported 3 fish species belonging the same class, order and family. Fish diversity was assessed by calculating the various diversity indices such as Shannon-Weiner Index (H), Simpson's Dominance Index (D), Simpson's Index of diversity (I-D), Pielons Evenness and Margalef Index of species richness and seasonal water quality parameters such as water, temperature, pH, electrical conductivity, free CO_2, DO, BOD hardness, chloride, nitrate and phosphate has been recorded and were found suitable for fish population. It can be concluded that reservoir supported rich fish population. It needs proper management and utilisation of these fish's wealth and sustainable steps to monitor and conserve these fish health. However, the present paper deals with fish diversity in the freshwater fishes of Gaya, as the water bodies harbour a large number of fish fauna. The survey is based on present observations and description of fishes as recorded in Systems Nature by Carls Linnaeus.

Key words: Freshwater fishes, water quality, biodiversity indices, Cyrinidae, Growth, Gaya, Bihar

INTRODUCTION

Fish perform all their body functions in water because it is totally dependent upon water to breathe, feed and grow, excrete wastes and reproduce understanding the physical and chemical qualities of water. Water has very unique density qualities. Most liquids become denser as they become cooler. However, water gets dense as it cools until it reaches a temperature of approximately 39°F. As it cools below this point, it becomes higher until it freezes (32°F). As ice develops, water increases in volume by 11 per cent. The increase in volume allows ice to float rather than sink, a characteristic that prevents ponds from freezing solid. It is the fact that water can dissolve more substances than any other liquid. This is why, it is called universal solvent. After oxygen, water temperature may be the single most important factor affecting the welfare of fish. Fishes are cold-blooded organisms and assume approximately the same temperature as their surroundings. The temperature of the water affects the activity, behaviour, feeding, growth and reproduction of all fishes. Metabolic rates in fish double for each 18°F rise in temperature. Temperature also determines the amount of dissolved gases (oxygen, carbondioxide, nitrogen etc.) in the water. The cooler the water the more soluble the gas. Water has its maximum density at 39.2°F. In spring season, water temperatures are nearly equal at all ponds depths. As a

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result, nutrients, dissolved gases, and fish wastes are evenly mixed throughout the pond. As the day become warmer, the surface water become warmer and lighter while the cooler-denser water forms a layer underneath. Circulation of the cooler bottom water is prevented because of the different densities between the two layers of water. Dissolved oxygen levels decrease in the bottom layer since photosynthesis and contact with the air reduced. The already low oxygen levels are further reduced through decomposition of waste products, which settle to the pond bottom. Localized dissolved oxygen depletion poses a very real problem to the fish culturists.

MATERIALS AND METHODS

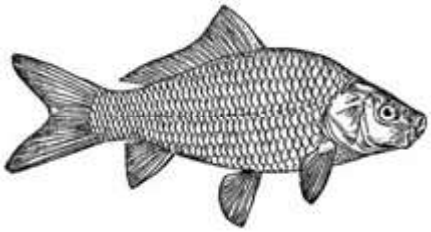


Fig. 1- *Labio rohita*

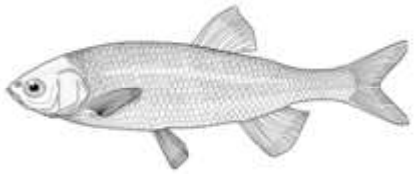


Fig. 2- *Catla catla*



Fig. 3- *Cirrhinus mrigala*

1. Study Area:

The study area is situated at Gaya (Bihar). Gaya is surrounded by hills on three sides and river on the fourth side. The climate of Gaya is seasonable. Its climate is characterised by relatively high temperature and evenly distributed rainfall throughout the year. The water bodies like ponds, lakes are very few in Gaya. There are few water bodies which contain water throughout the year and there are only source for supply of fishes in Gaya. The fishes feed on these biomass and mass efficiently convert them into animal protein in order to help farmers for fish culture in this area because of its high protein content. So a detail study is urgently required to correlate the growth of fish with respect to local environment that surround the water body and influence water quality. These works have immense significance for Bihar where large population depend on fishing not only as food source but also for their livelihood.

2. Fish Samples Collection

Fish samples were collected from different selected localities during the study period from September 2017 to October 2018 with the help of local fishermen using different types of nets namely gill nets, cast nets and dragnets. Immediately photographs were taken prior to preservation since formation decolorises the fish colour on the long preservation. 10% formalin solution was prepared for preservation of fish samples. Fish brought to the laboratory were fixed in this solution in separate jars according to the size of species. Smaller fishes were directly placed in the formalin solution while large fishes were given an incision on the abdomen before they were fixed.

3. Fish Identification

The fishes were identified by using Datta Munsi & Srivastava (1988) Talwar and Jhingran (1991) and Jayaram (1999).

4. Diversity Indices

Species diversity can be managed separately either as species richness or evenness or diversity as a whole. Species richness was measured by Index of Richness (denoted by R) given by Margalef (1958). Species evenness was measured with evenness index (denote by E) given by Hill (1973). Diversity of the species calculated directly with a variety of indices, of which two commonly used are Shannon-Weiner Index or simply the Index of diversity as given by Shannon and Weiner, 1963) and Index of dominance given by Simpson (1949). Shannon's index has a direct relationship with the species diversity, whereas Index of dominance has an inverse relationship.

5. Water sample collection for physico-chemical analysis

Surface water samples were collected in polythene cans from different sampling sites of the Gaya reservoir. Air and water temperature, pH, DO and free CO_2 were determined on the spot itself and for BOD determination water samples were collected in BOD bottles. For further analysis, samples were transported to laboratory and analyzed other parameters using standard procedures (APHA, 1985, 2005).

Periodical estimates of fish growth and yield in relation to water samples were collected and analysed for computation of fish population dynamics and impact of physico-chemical complexes on fish yield of the water bodies. The whole exercise also involved network analysis and rendering of food chains/food web of the existing ecosystems.

Analytical studies in the selected water bodies were carried out two years 2016-2018 in different seasons such as prebreeding, breeding and post breeding. Mean values are presented here. However, methodology of APHA, 1985, 2005 used for experimental work. Three different seasons have been considered to determine that water condition affect the body of fishes, physiology taking growth index as measurement.

RESULTS AND DISCUSSIONS

The present investigation of fish fauna in Gaya reservoir showed that most of the fishes recorded were widely distributed in the lentic and lotic water bodies of Gaya. The members of Cyprinidae family dominate the fish population. The economically important and cultivated fishes were also recorded and include *Labio rohita, Cirrhinus mrigala, Catla catla*. The economically important and cultivable above fishes and reservoir can exploit for commercial production of fishes for better improvement of socio-economic condition of local people.

Table – 1 : Fish abundance and biodiversity status in Gaya reservoir, Gaya

Sl.No.	Name of Bony fish	Zoological Name	Abundance	Biodiversity	Habitat
1	Rohu	*Labio rohita* (Ham)	A-2	LR-nt	Lentic and Lotic
2	Mrigala (Naini)	*Cirrhinus mrigala*	A-2	LR-nt	Do
3	Katla	*Catla catla*	A-2	VU	Do

A – 2 : Common
VU : Vulnerable
LR-nt : Lower risk near threatened

As far as biodiversity status (IUCN-1994) is concerned different diversity indices were calculated as per standard methods. The Shannon-Weiner fish diversity index of Gaya reservoir ranged from 2.4 to 3.0. The Simpson's dominance index values ranges between 0.08 and 0.2. Simpson's index of Diversity (1-D) also ranges between 0 &1 but now the greater the value the greater the sample of diversity. This makes more sense. In the present study, the Simpson's index of Diversity (1-D) values fluctuated between 0.89 and 0.95. However, Pielou's evenness values were ranges from 0.6 to 0.9. The Margalef index of species richness values revealed 1.48 - 2.4 (Table 2).

Table – 2 : Fish diversity indices of Gaya, Bihar

Diversity indices	Range
Shannon - Weiner Index (H)	2.4- 3.0
Simpson's Dominance Index (D)	0.08-0.2
Simpson's Index of Diversity (I-D)	0.89- 0.95
Pielou's evenness	0.6- 0.9
Margalef index	1.48 -2.4

Table 3: Seasonal variation of physico-chemical parameters in the water samples of Gaya reservoir during 2017-2018

Parameters	Site 1			Site 2			Site 3		
	Winter	Summer	Rainy	Winter	Summer	Rainy	Winter	Summer	Rainy
Air Temp.	28 ± 2.16	33 ± 0.81	28.5 ± 2.51	28 ± 2.16	33.5 ± 0.57	28.25 ± 3.30	29 ± 2.44	32.75 ± 1.25	28.75 ± 3.77
Water Temp.	25.25 ± 0.5	29 ± 2.16	26.75 ± 1.70	25.25 ± 0.5	29.75 ± 1.89	26.5 ± 1.91	26 ± 0.81	30.25 ± 1.71	27.25 ± 2.87
Turbidity	30.82 ± 22.35	21.75 ± 10.37	52.62 ± 29.23	15.35 ± 3.92	28.7 ± 10.83	49.82 ± 30.57	15.5 ± 4.20	26.25 ± 6.34	56.5 ± 28.38
pH	7.43 ± 0.15	7.62 ± 0.09	7.47 ± 0.15	7.52 ± 0.06	7.63 ± 0.08	7.52 ± 0.14	7.43 ± 0.23	7.65 ± 0.07	7.40 ± 0.22
Alkalinity	51.11 ± 6.61	59.25 ± 2.27	42. 78 ± 4.45	S6.13 ± 8.33	61.45 ± 1.47	42.58 ± 13.41	55.03 ± 5.92	61.50 ± 2.08	40.08 ± 13.21
EC	133.75 ± 8.38	165.5 ± 13.72	121.35 ± 37.42	136.75 ± 6.89	167 ± 7.39	120.67 ± 35.93	155.75 ± 34.27	I 9.5 ± 9.74	140.5 ± 21.11
CO_2	2.47 ± 0.74	1.4 ± 0.52	2.55 ± 0.92	2.54 ± 0. 91	1.45 ± 0.58	2.63 ± 0.83	2.31 ± 0.47	1.49 ± 0.53	2.68 ± 0.72
DO	7.31 ± 1.53	6.18 ± 1.l5	5.79 ± 1.10	7.52 ± 1.02	5.75 ± 0.62	5.72 ± 0.71	8.28 ± 1.50	6.38 ± 0.83	5.72 ± 1.26
COD	8.8 ± 6.34	16.1 ± 9.46	4.50 ± 1.73	12.5 ± 2.88	12.25 ± 6.94	11.5 ± 5.51	10.22 ± 9.25	15.5 ± 9.98	14.75 ± 9.28
BOD	1.01 ± 0.68	2.47 ± 1.40	0.83 ± 0.04	1.17 ± 0.47	1.31 ± 0.67	1.21 ± 0.44	1.18 ± 0.84	2.20 ±1.39	1.77 ± 1.23
Hardness	50.5 ± 5.0	62.0 ± 11.54	44.00 ± I 0.70	47 ± 5.29	62.5 ± 9	48 ± 11.77	56.25 ± 9.53	67.5 ± 12.33	52.5 ± 9.84
Chloride	21.98 ± 0.81	26.59 ± 4.68	20.27 ± 3.00	21.45 ± 2.19	26.57 ± 5.12	21.62 ± 4.81	24.89 ± 1.71	27.65 ± 5.37	23.04 ± 1.36
NO_3	0.108 ± 0.059	0.165 ± 0.03	0.328 ± 0.221	0.13 ± 0.09	0.162 ± 0.04	0.34 ± 0.25	0.120 ± 0.06	0.142 ± 0.026	0.340 ± 0.285
PO_4	0.011 ± 0.008	0.012 ± 0.007	0.009 ± 0.005	0.004 ± 0.002	0.012 ± 0.006	0.007 ± 0.004	0.006 ± 0.003	0.013 ± 0.006	0.009 ± 0.004

All The parameters are expressed in mg/L except temperature (^oC) turbidity (NTU), pH, electrical conductivity (μmhos/cm)

2. Water Quality Parameters

The physico-chemical characteristics of water have an important role in supporting fish diversity freshwater ecosystems. The seasonal average values of physico-chemical parameters of the reservoir are depicted in the Table 3. In the present study, air temperature was recorded in the range of 28.00±2.16 during winter season to 33.50±0.58^{o}C at Site 1. Water temperature was varied between 25.25±0.50^{o}C at Site 1 and 2 during winter season and 30.251.71^{o}C at the site 3 during summer season. pH was in the range of 7.40±0.22 at site 3 during rainy season and 7.66±0.07 at site 3 during summer season. A minimum value of electrical conductivity was recorded at site 120.68±35.93 mmhos/cm during rainy season and a maximum of 169.50±9.75 mmhos/cm at site 3 during summer season. Free CO_2 was observed in the range of 1.41±0.52 at site 1 during summer season to 2.68±0.72 mg/L at site 3 during rainy season. DO level was varied between 6.19±1.16 mg/L at site 1 during summer season to 8.28±1.50 mg/L at site 3 during winter season. Biological oxygen demand for 3 days at 27°C was fluctuated between 0.83±0.04 mg/L at site 1 during rainy season and

2.48±1.41 mg/L at site 1 during summer season. Total hardness was registered isn the range of 44.00±10.71 mg/L at site 1 during rainy season to 67.50±121.37 mg/L at site 3 during summer season. Nitrate concentration was observed in the range of 0.108±0.06 mg/L at site 2 during rainy season. Phosphate content was fluctuated between 0.012±0.01 mg/L at site 2 & 3 during summer season.

The diversity index indicates good correlation with overall species richness across the sites and could be utilized by the biodiversity conservation managers for prioritization of sites of conservation and habitat restoration. Bergerot *et al.* (2008) developed indices of fish biodiversity conservation concern, rarity index and fish magnitude values for prioritization of sites' for large scale European freshwater basin in France. Lasne *et al.* (2007) used fish zonation and identified indicator species for the evaluation of the ecological status of the Loire basin (France). They also carried out a discriminant analysis on environmental variables revealed that they could be mainly determined by the slope, temperature, and depth.

During study it was found that abundance and diversity of fishes were found to be very high in respect to extent of water body. The maximum numbers of species were recorded from low land areas. According to Shaikh *et al*. (2011) in low and middle land areas fresh water fish diversity was found to be very high. It is due to deep water bodies allow niche segregation in order to enable the fishes to live without facing more intra and inter specific competitions. During summer when maximum level of water decreased due to hot air and high temperature most of fishes migrated toward low land for survival. But during winter season diversity of fish fauna abundant due to clear water, preference of maximum amount of phyto and zooplankton as complain to rainy season. The presence of exotic species in the reservoir may be due to carriage from nearby water bodies by flood water.

The mean air and water temperature values were similar and slightly high during summer season. pH is considered as an indicator of overall productivity that causes habitat diversity (Raj and Jayasekhar, 2007). The pH of most the natural water is varied between 6.0 and 8.5. In the present study, the seasonal mean pH was close to neutral. According to Goldman and Horne (1983), low pH < 5.0 can severely reduce aquatic species diversity. However, the range of water pH was reasonably good for fish population. The conductivity values were higher in the summer season due to decreases in the total volume of water, whereas; low in rainy season because of dilution factor (Alam *et al*., 2007). CO_2 has a great effect on photosynthesis which affect again on fish growth. There is no much variation in the concentration of free CO_2 throughout the study period. Fafioye et al (2005) have observed similar trend in Omi waterbody Ago Iwoye (Nigeria). Further, free CO_2 more than 20 mg/L is toxic to fish (Battul et al., 2007). The observed values were below 20 mg/L of free CO_2 hence, water is more suitable for fish diversity. Dissolved Oxygen is a primary and comprehensive indicator of water quality in surface water. The decline of dissolved oxygen level has a serious implication for the health of aquatic system. The optimum value for good water quality is 4 to 6 mg/l of DO, which ensures healthy aquatic life in a water body (Santosh and Shrihari, 2008). In the present study, DO level was more than 4 mg/L at all sites. Further, DO level was high in winter followed by summer and rainy season. The reason could be low temperature, turbulence of surface water by high wind action etc. And its level drops in summer due to high metabolic rate of organisms (Salve and Hiware, 2006) and limited turbulence in the reservoir (Mwaura, 2006). BOD value can be used as a measure the degree of water pollution and is useful in evaluating self-purification capacity of a water body.

The seasonal BOD values were slightly high in summer low during winter and rainy season. Higher values of BOD in summer season due to higher microbial activity and elevated temperature (Patel, 1999). As per the hardness values, water belonging to soft (0-60 mg/L) and moderate (60-120 mg/L) category. Seasonal variation of total hardness shows that its concentration was high in summer and more or less similar in other seasons. The high hardness during summer season could be due to reduced water level and increased salts content in water (Kemdirim, 2005). Nitrate (NO_3) is an essential nutrient for aquatic plants in natural waters. An excess concentration of nitrate tends to stimulate algal growth and leads to eutrophic conditions. Seasonal nitrate concentration was high during rainy season at all stations and remained similar in winter and summer season. The higher concentration of nitrate during rainy season could be due to leaching of nitrate from agricultural fields. Phosphorus concentrations above 0.02 mg/L in

reservoirs tend to produce algal blooms. However, in the present study, phosphate was below 0.02 mg/L. Its concentration was high in rainy season than summer and winter season. The overall water quality parameters indicated that water is suitable for fish life in the reservoir.

This ichthyofaunal study indicates that this waterbody is rich in diversified fish fauna consists of native species, economical, cultivable. Changes in fish community, directly or indirectly affect other components of the reservoir ecosystem including physical, chemical and biological characteristics. Habitat loss and environmental degradation has seriously affected the fish fauna. Conservation of fish diversity assumes top most priority under changing circumstances of gradual habitat degradation. Therefore, a sustainable strategies needs to explore more fish species, utilization and save fish community of this reservoir. The study will provide future strategies for development and fish conservation.

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Role of earthworms in heavy metal accumulation

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Abstract:- Due to rapid increase in urbanization in the whole world, there arises the problem of management for organic wastes. The process of vermicomposting is the method by which such waste materials can be remediated easily and the compost which is released can be used in growing food crops for human consumption. By using earthworms, the management of hazardous wastes with high metal contents can be done easily, because earthworms soak up or bioaccumulate these toxic heavy metals into their body tissues and make the soil and surrounding pollution free. The importance of earthworms for bioaccumulation and monitoring of metal pollution was widely recognized in terrestrial ecosystem. They have a great power of accumulation of these hazardous metals as a natural ecological indicator of soil contamination. The contamination which was done by heavy metals can cause severe impact on the functioning of soil ecosystem by affecting the activities of soil fauna as well. Therefore, vermibiotechnology is the best technique for management of biological waste and accumulation of heavy metals from soil by in-situ application of earthworms. Vermicompost is a good alternative of chemical fertilizers for production of crops and it also enhances the soil fertility. This is eco-friendly management process and protects our soil and ecosystem free from heavy metal contaminations by the vermiremediation process of earthworms.

Key words: Earthworms, Heavy metals, Bioaccumulation, Soil, Organic wastes, Vermiremediation.

INTRODUCTION

Environmental quality of urban area is vital to be investigated as large number of human population's lives in the cities. However, due to rapid increase in urbanization and industrialization in urban areas cause a severe adverse effect on environment as well as on humans. Due to this factor there is an enhancement of pollution levels in our environment, especially due to the heavy metals which are serious source of contamination throughout the world. The advances of humans need towards industrialization caused the production and entrance of hazardous pollutants such as carcinogenic compounds, toxins and heavy metals into the environment. The main pollutants present in the environment are heavy metals which cause big problem due to their toxicity and accumulation in the environment. Due to the rapid increase in population, there is requirement also enhanced for food, agricultural land per unit area needs to achieve maximum efficiency. Agricultural productivity and its quality depend on nutrition of the plant. There is only one way to achieve this target is abundant use of fertilizers to supply these nutrients. Excess fertilizers may cause impact of heavy metal accumulation. The presence of heavy metal affects the plant growth.[1] There are several heavy metals like Nickel, Cadmium, Boron and Iron, they have tendency to accumulate in human bodies. Due to these properties they are called important pollutants which

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cause chronic effects.[2-6] When these heavy metals inters into soil cause adverse environmental issues and due to their contamination level there is serious issue arise regarding to human health and other organisms.[7-8] Since, all the agricultural products are linked directly to the soil and because of the widespread human use of these products and the high potential, of these soils for contamination with heavy metals. In the decontamination of soils contaminated with heavy metals, stability and high resistance to degradation of the metals is considered to be one of the most challenging issues.

Heavy metal contamination in soil is a most common problem in terrestrial ecosystems which are affected by atmospheric deposition and industrial activities.[9-10] Heavy metal which is present in environment up-taken by crop plants, due to the process of food chain.[11] Left-over materials which are released from urban systems are termed as urban waste, which usually consist of household garbage such as kitchen waste, green waste, street sweepings, sanitation residues,etc.[12] There is a rapid increase in these urban wastes nowadays and their disposal is of great concern as it causes serious management problems particularly in developing countries like India, where the management of waste are mostly unsystematic and unscientific.[13-14.] These urban wastes usually consist of hazardous persistent pollutants such as heavy metals, which cause severe threat to ecosystem as well as on human beings.[15-16] Heavy metals are persistent in the environment, contaminate the food chains, and various health issues are also occurred due to their toxicity. Metal concentrations above threshold levels affect the microbiological balance of soil and can reduce their fertility.[17]

Heavy metals are persistent in the environment; they can easily accumulate in living organisms and are transferred from one trophic level to another trophic level into the food chain. Heavy metals are transferred from soil-to-plant and it is very important step in the trophic transfer of such metals in food chains. Plants uptake these metals from polluted soil and easily transferred to higher level of environment.[18] Human beings are considerably exposed to toxic heavy metals in the environment through different pathway including ingestion, inhalation and dermal absorption. Most of the people are exposed to heavy metals in developing countries.[19] Earthworms play a vital role in metal pollution monitoring and widely recognized in terrestrial ecosystems. Earthworms have a great tendency for bioaccumulation of heavy metals in their tissues and can be used as an ecological indicator of soil contamination. According to their virtue habitation earthworms participate in the formation of soil structure and regulate dynamics of soil organic matter, usually earthworms directly or indirectly modulate the transfer of organic and inorganic chemicals (toxicants) in soils. Due to this reason, earthworms are considered as useful biological indicator of several metals in soils.[20] Earthworms are considered as important bioindicators for risk assessment and they are used in ecotoxicological studies to assess the power of bioaccumulation of heavy metals in food chain.[20-23] Earthworm bodies are externally and internally always came in contact with soil, due to this factor they readily take up and bioaccumulate many heavy metals including Cd, Zn, Cu and Pb.[24-25] Different species of earthworm show different tolerance level to contaminated soils and presence of heavy metals consistency.[26-31]

Heavy Metal

When heavy metal are found at above threshold concentration they are considered as one of the most major pollution which act as threat to man and other organisms by the route of food chain. Pollutants such as heavy metals, pesticide, polychlorobiophenys (PCBs), etc. are occur most dangerous, due to this factor they show severe impact on the human race and environment.[32] The term 'heavy metals' mostly refers to those metals having specific weights more than 5 g/cm^3. The densities of these metallic elements are relatively high however; being a heavy metal has more to do with their chemical properties. These metals are usually found dispersed in rock formations and about 40 different metals have been come under this category. In the environment due to rapid rise in industrialization and urbanization, heavy metals levels are also increased. Heavy metals cause serious health issues to humans and perturbing the environment, because these metals are spread broadly over the globe. Urbanization, land use changes, and industrialization, are major root cause for this heavy metals pollution, especially in developing countries with extremely high populations, such as India and China.[33] Non-essential to metabolic and other biological functions are classification for several hazardous heavy

metals and metalloids (e.g., As, Pb, Cd, and Hg). Those metals are deleterious in various respects.[34] and they metals have therefore been included in the top 20 list ofdangerous substances by the United States Environmental Protection Agency and the Agency for Toxic Substances and Disease Registry.[35-39] Due to the result of human caused activities these heavy metals become concentrated, but naturally they are found in the earth. Some common sources are as mining and industrial wastes; vehicle emissions; lead-acid batteries; fertilizers, paints and treated woods.

Entry routes

Due to the process of air inhalation, diet and manual handling heavy metals enter plant, animal and human tissues. Motor vehicle emissions are considered asa major source of airborne contaminants including arsenic, cadmium, cobalt, nickel, lead, antimony, vanadium, zinc, platinum, palladium and rhodium.Another potential source of heavy metal contaminationis also caused due to absorption through skin contact with soil. Heavy metals are hard to metabolize, that's why they are accumulated in organisms.

Detrimental effects

Heavy metals can easilybind to the vital cellular components, such as structural proteins, enzymes, and nucleic acids, and therefore do interfere with their functioning." Long-term exposure in the presence of heavy metals can cause carcinogenic, central and peripheral nervous system and circulatory effects. From thousands of years these heavy metals have been used by humans. When heavy metals are up taken by any means of food consumption will show confirmative negative impact towards the human health.[40]

Earthworms in vermicomposting

Vermicompostingis a type of process which utilizes earthworms; it is an eco-bio-technological process that transforms energy rich and complex organic substances into stabilized humus like product [41]. Vermicomposting is a type of kindred process to composting, by the addition of different species of earthworms used to enhance the process of waste conversion and produce a better end product. All the urban waste which consists of organic component can be bio-converted and recycled into a useful end-product called compost and Vermicompost when processed by earthworms. Not only are the organic fractions of urban wastes converted into available nutrients with the help of earthworms.[14] But also consequently earthworms remediate the persistent heavy metals from the wastes by bioaccumulation into their bodies via Vermicomposting process.

The transformation of organic waste into Vermicompost is a bio-oxidative mesophilic process in which detritivorous earthworm species interact with micro-organisms strongly affecting decomposition processes, and also accelerating the stabilization of organic matter and greatly modifying its physical, chemical and biological properties. Earthworms are ecological engineers that through remediation process involved in the indirect stimulation of microbial populations through fragmentation and ingestion of contaminated soil and organic matter. The potential of metal accumulation is maintained by earthworms and make the ecosystem in a balanced state. Moreover, earthworms ingest soil and various forms of biomass to produce vermicasts. Thus adopting the vermicomposting process not only provides availability of plant mineral nutrients but also act as a best method for agricultural benefits and bioaccumulation of various heavy metals present in the soil.

Earthworms

Earthworms are macroscopic clitellate oligochaete annelid that usually lives in soil. They are segmented worms, bilaterally symmetrical, with an external gland for production of egg case (cocoon), with present of sensory lobe in front of the mouth and an anus at the end side of the body, along with small number of setae on each segment. The distribution of earthworms in soil are influenced by several factors such as soil texture, aeration, temperature, moisture, pH, organic matter, dung and reproductive potentials.[42-43] For the earthworm population external biotic parameter and nutritive resources of the soil are primary controlling factors.[44-45] During the process of feeding and digestion earthworms are act as an aerator, chopper, crusher, chemical degrader and a biological stimulator.[46] Aristotle mentioned earthworms 'territory intestines' in 330 B.C, and he confront that soil is an organic entity and very important role played by an earthworms for maintaining the life of soil. Earthworms are burrowing animals which are grouped into 13 families. Although about

5,000 species have been described, most of the species are still unknown, mainly in the tropics.[47] These worms are found in soil ecosystem of many parts of the world, but these earthworms are abundantly found in temperate and tropical regions, due to their large fraction of living biomass, providing agro-ecosystem sustainability.[48]

The earthworms are members of large, ubiquitous group with evolutionary history. About 6000 species of earthworms are found on all continents except Antarctica. Most of the species of the earthworm dwell in soil, whereas some live in leaf litter, decaying log and river banks, as well as lives in trees and even along the seashore. Earthworms are major terrestrial ecosystem engineers and hence their economic impact is immense. Although earthworms are the members of most familiar and economically important groups of large invertebrates, their evolutionary history is not well understood. Earthworms leave an impressive record for trace of fossils, but it is not easy to determine which species made a particular set of fossilized burrows, since body fossils of these earthworms are extremely rare.Some analyses reveal that the ancestor of all the living earthworms probably lived over 209 million years ago, this quality makes earthworms about as old as mammals and dinosaurs.

The reputation of earthworms was rehabilitated when Darwin published his book in-titled "The Formation of Vegetable mould through the action of worms with observations on their habits" in 1881. Due to their antiquity and relative stability in soil, those that persist today, especially in primitive families, are more akin to "living fossils". The terms "alien", "exotic" and "invasive" are often described earthworms that are dispersed over a wide range and in geographically remote localities, they are widely distributed in many parts of the world or merely present in more than one country, but with the ecological implication of passive or unintentional transportation.[49]

Although Darwin mainly commented on their intimate ecological relationship with soils and plant growth, his contributions to earthworm studies extended into the realms of anthropology and archaeology. Earthworms usually have remarkably higher rate of biodiversity. The natural spread of earthworms can either be a slow or negligible, the dispersion of exotics appears to closely follow cultural and technological trade and exchanges gradually in pre-agricultural societies. Thus, earthworm transportation is considered to be mainly human mediated with exotic species, origins often corresponding with provenances of traditional agricultural and horticultural crops. There are about 3,320 species of earthworms are distributed all over the world.[50] Whereas, in India approximately there are about 590 species of earthworms present.[51] But their functional role of the majority of the species and their influence on habitat are lacking. India is a very diverse country and thus showed a very rich diversity of earthworms, mostly present in Western Ghats and Eastern Himalayas. The scientific exploration of earthworm's diversity in India dates back to the 19th century.

The naming of first earthworm species in the Indian subcontinent credit goes to Templeton (1844) when he discovered *Megascolex caeruleus* from Sri Lanka.[52] Whereas, Perrier (1872) was the first to describe earthworm species from the Indian mainland.[53] Earthworms not only play important role for agricultural maintenance but also they act as a nutritive content for the growth of the plants as well. They are the most attention seeker organisms on this earth due to their activities.

The digestive systems of the worms are apparently capable of detaching heavy metal ions from the complex aggregates and humic substances in the waste as it rots. Various enzyme drivenprocesses seem to lead for assimilation of the metal ions by all earthworms so that these metals lock up in the tissues of the organism rather than being released into the compost as worm casts.Soils are contaminated with metals and their accumulations are reported in earthworms many times.[54] Aristotle mentioned earthworm as "intestine of the earth". Heavy metals are usually remaining in the soil immobilized by the addition of various chemicals (solidification/stabilization).

When earthworm treatment was introduced into the soil, the distribution of heavy metals gradually dropped due to action of earthworm activity. The earthworms are considered as ecosystemivorous feeding on entire soil microbial ecosystems.[55] Metals are up taken by earthworms into their gut are bounded by a protein called 'metallothioneins'. The chloragogen cells which are present in earthworms appear to accumulate heavy metals absorbed by the gut and cause metals immobilize in small spheroidal chloragosomes.[56] Excreted material of earthworms such as mucus and urine into the soil environment which caused increase in microbial activity into the soil ecosystem.And

as well as also release other stimulating substances in addition to the mucus and urine.[57] This statement is very clear and understood that earthworms are very important soil fauna which act as an indicator organism for recycling of organic wastes which are present in soil environment and as well as accumulates heavy metals into their body and protect our various food products from the contamination of these hazardous heavy metals and make our environment clean and safe from these toxic metals because heavy metals are poisonous for consumption of human beings when they enter our food chain.

Heavy metals can enter the soil ecosystem from different sources. Fertilizers, pesticides, organic and inorganic amendants, wastes and sludge residues can contain variable amount of these metals. Accumulation of these metals by earthworm species occurs through two ways.[25] Which include absorption directly into their body tissues or either by ingestion of organic matter.[58-59]

The direct measurement of heavy metal concentrations in earthworm tissue could provide a means of assessing environmental pollution levels, given the demonstrated correlation between soil contamination and earthworm metal bioaccumulation.[60] Cadmium, Copper, Lead and Zinc are some heavy metals which are generally accumulated by the earthworms.

Various earthworms' species are crucial drivers of the remediation process as they are involved in the indirect stimulation of microbial population through ingestion and fragmentation of contaminated soil and organic matter. And most importantly, the potential rates of earthworms for bioaccumulation of heavy metals have maintained our ecosystem in a well-mannered and balanced state.

By accumulation of these heavy metals into their tissues, they maintained nutrient enriching abilities of the soil ecosystem. The vermicasts of most earthworm species are known to contain hormones and enzymes which stimulate plant growth and discourage pathogens. Earthworms have the potential for safety management of all municipal and industrial organic wastes and convert them from ending up in the landfills. Thus vermicomposting technology acts as a best method for bioaccumulation of heavy metals from the soil ecosystem by the activities of earthworms.

Earthworms are known to cause contribution to soil formation through consumption process of dead plant and animal matter. Earthworms are also known to cause recycling of carbon and nitrogen into the ecosystem. The cast of the worms have higher available nitrogen, potassium, calcium and phosphorous content than the surrounding soils as well as higher cation exchange capacity. Soil passed through gut of earthworm has a neutral pH.

Earthworms excrete material that has concentration of beneficial microbes that help decompose crop residue. Earthworms play a major role in monitoring soil structure and fertility because it increase mineralization and humification of organic matter by food consumption, respiration and may indirectly stimulate microbial mass and activity as well as mobilization of nutrients.

The accumulation was occurred in the yellow cells of body tissues of earthworms. Depending upon the concentration of heavy metals, the body of earthworm gets affected. Hg, Co, Pb and Zn are some toxic metals which cause severe impact on the growth of earthworms. Whereas, Hg is more toxic for earthworms species because it effects cocoon production, coelomocytes, body weightand their length as well.

The earthworm digestive system is apparently capable of detaching heavy metal ions from the complex aggregates between these ions and humic substances in the waste as it rots. The separation of dead worms from compost is a relatively straight process allowing the heavy metal to be removed from the organic waste.

Heavy metals may cause damage to an organ, inhibition of an enzymatic activities and significant alteration in their various metabolic activities. On the basis of morphological characteristics earthworms have been classified into three categories.[61] These categories are- Epigeic, Endogeic, and Anecic.

Epigeic

Epigeic earthworm species are non-burrowing litter and dung-feeding earthworm's habitat is in organic horizons and near the surface litter. By commuting the litter, they modify its physico-chemical characteristics, usually reducing its C/N ratio, making it suitable for further microbial breakdown. These species are known as "litter transformers". Their body size is about small-to-medium, uniformly pigmented species, they have dorsoventrally flattened body, tolerance to disturbance, high metabolic

rate, high fecundity, short life cycle and they also have high power of regeneration. They are r-selected species earthworms. Epigeic earthworms form the forest floor community of temperate and tropical countries. When the environmental conditions are not favorable or their food is limited, Epigeic species are difficult to find. Examples include *Eisenia fetida, Perionyx excavatus, Eudrilus eugeniae, Dendrobdena rubida* (Savigny) etc. Epigeic earthworms are of often bright red or reddy-brown in colour, but they are not stripy. These earthworms usually do not form permanent burrows.

Anecic

Anecic earthworm's species live in permanent burrows as deep as 3m below the soil surface. These earthworms species are called the 'ecosystem engineers' because they play a great role in soil turn over, and improve water holding capacity, aerobicity and as well as exerting regulatory force in soil functioning. These earthworm species lives in vertical burrow system that usually open up to the soil surface. Their casts are formed on the surface of soil.They gradually feed on decaying plant litter with little amount of soil, so they are called as geophytophagous. Anecic earthworms are large in size, moderate to heavily pigmented and intolerant to disturbances. These worm species are found in temperate part of the world. Their reproductive rates are relatively low. Examples of anecic earthworms are *Lampito mauritii, Metaphire houlleti, Lumbricus terrestris, Drawida grandis* etc. Anecic species have dark coloured at the head end (red or brown) and have paler tails. They have long life cycle with the limited power of regeneration.

Endogeic

Endogeic species of earthworms are geophagous because they live deeper in the soil profile and feed primarily on both soil and related organic matter. They form both sub-surface and surface casts. These species of worms are non-pigmented or lightly pigmented and form horizontal burrows in top and sub-soils. These species have symbiotic relationship with the gut micro-flora. Endogeic earthworms are found in tropical parts of the world. They have long life cycle with the limited power of regeneration and also have diverse effects on soil properties. They increase the proportion of large aggregates along with bulk density of soil. These species are apparently of no major importance in litter incorporation and decomposition since they feed on subsurface material, and also play a vital role in other soil formation process. These species are concerned with formation of humus in the surface soil. These endogeic species also increase overall soil porosity and enhance water infiltration. Species such as *Polypheretima elongata, Metaphire posthuma, Allobophora caliginosa, Octolasio ncyaneum* (Savigny) are included in this group. Endogeic earthworms are often pale in colors, grey, pale pink, green or blue.

These three different categories of earthworms thus contribute differently to soil fertility process and structure maintenance. They may be manipulated in a given area in a specific agricultural situation.

***Metaphire posthuma* (Vaillant, 1869)**

Metaphire posthuma is an endogeic species mostly common to tarai region, which lives deeper in the soil profile and feed primarily on both soil and associated organic matter i.e. geophagous and they form both sub-surface and surface casts. They are generally found near anthropogenic habitats like paddy fields, ridges of paddy fields, and in wastewater saturated soil from households. In India, *Metaphire posthuma*are distributed in Jharkhand, Rajasthan, Uttar Pradesh, Pakistan.[62-64] They are dark brown in colour. Length 110-158mm, clitellum width 3-8mm, segment number 91-124. Prostomium epilobous or tanylobous, setae 111-130 in 8, 71-83 in 20, 18-19 between male pores.Clitellum 14-16, annular, setae absent, dorsal pore absent.Their cast is thick pellets like. These worms are very active and twist their body away when touched. Among the physico-chemical parameters, only moisture was observed to affect the density of *Metaphire posthuma*. It is widely distributed in Southeast Asian countries. Coelomocytes, the chief immune effector cells of earthworms perform diverse physiological functions under the challenge of toxic heavy metals. These worms are highly resistance to heavy metals so that it is called as heavy metals bioindicators, capable of being a waste processor or Vermicomposting.[65-68]

***Eisenia fetida* (Savigny, 1826)**

Eisenia fetida corresponds to the striped or banded morph; they are reddish-purple in colour. The original range where this species is supposed to be from the Caucasus to the forest-steppe zone of Russia where these species

are occurring under bark of ballen trunks and in decaying organic material.[69] In India, they are distributed in Himanchal Pradesh, Punjab, Pondicherry, Uttarakhand, Jharkhand and in Guana, Iran, Spain as well. *Eisenia fetida* are generally found in paddy field, grassland and sugarcane field. Their casts are very small pellets like. Length is about 60-90mm; diameter 3-4mm; with red purple or brown segmented bands over dorsum separated by paler intervals. Prostomium epibolous ½.Dorsal pore from 4/5. Setae slender, ornamental, closely paired.

Clitellum from xxiv, xxv or xxvi to xxxii (=7-9). Male pore with fairly large raised areas which do not transgress the limits of xv. Spermathecal pores two pairs, in 9/10 and 10/11, near the middorsal line. The growth and reproduction of this earthworm species is affected by several factors such as food quality, moisture, temperature and population density. *Eisenia fetida*are very affecting earthworm for management of waste via vermicomposting process and bioaccumulation of heavy metals are also often done from contaminated field and soils.[68]

***Lampito mauritii* (Kinberg, 1867)**

Lampito mauritii was found in all habitats such as groundnut field, millet field, garden, grassland, cultivated field, paddy field, sugarcane field and vegetable field. In India, this species are most widely distributed in different agro-systems of Punjab, Rajasthan, Tamil Nadu, Pondicherry, Uttar Pradesh.[70-73,62] This is one of the most common earthworm species in the loamy soils of Gorakhpur, Deoria districts as well. This species is abundantly distributed throughout the country *Lampito mauritii* are dark yellow in colour. Length 92-168mm, diameter 5.0-5.3mm, total segments 126-170. Colour dorsally is greyish, brownish or yellowish with purplish tinge at anterior end. Prostomium is epilobic with tongue 1/2, closed. First dorsal pore start on 10/11 or 11/12. Setal arrangement is perichaetine, 40-50 setae per segment in the preclitellar region. No genital markings are there. Their casts are granular like.

***Perionyx excavatus* (Perrier, 1872)**

Perionyx excavates are cosmopolitan species. In India they are found in Himanchal Pradesh, Uttarakhand, Mizoram, Tripura, Pondicherry, Tamil Nadu, Punjab, and Uttar Pradesh. As well as also in Sri Lanka, Southeast Asia, Taiwan, Hawaii, West Indies, Madagascar, Samoa, Fiji, U.K, U.S.A, Australia and in New Zealand. They are widely distributed in Paddy field, sugarcane field, garden millet field, groundnut field, cultivated field, grassland and vegetable field. Their casts are loose pellets like. This species are commonly found in compost heaps. They are reddish brown in colour. Length 65-150 mm, diameter 3-4.5mm, total segments 90-155. Colour deep purple to reddish brown dorsally but pale ventrally.

Prostornium epilobous. Setae small, perichaetine 42-52 per segment. Clitellum thin often indistinct. Body dorso-ventrally flattened and the posterior portion more tapering than the anterior. No genital marking. Gizzard is not prominent. This is an Epigeic species that lives solely in organic wastes, and high-moisture contents and adequate amounts of suitable organic material are required for populations to become fully established and to process organic wastes efficiently. This tropical earthworm is extremely prolific, and it is easy to handle and very easy to harvest.

Its main drawback is its inability to withstand low-temperature conditions, but for tropical conditions it seems an ideal species. They have shorter maturation and incubation time period. *Perionyx excavatus*does not grow much at low temperatures although it can survive those4℃ (39.2 °F), but it is less susceptible to high temperatures over 30 (86°F). Even in tropical areas, *Perionyx excavates* does not survive the high summer temperatures. The life cycle of *Perionyx excavates* takes 40-50 days. Sexually maturity is attained within 20-28 days.[68]

***Eutyphoeus waltoni* (Michaelsen, 1907)**

*Eutyphoeus waltoni*are distributed throughout in India such as Punjab, Sikkim, Jharkhand, Himanchal Pradesh and Uttar Pradesh. Their casts are tower-like. This species are usually found in cultivated field, grassland, groundnut field, millet field, paddy field, sugarcane field and vegetable field. They are brownish in colour. Length is about 100-210mm, diameter 4.5-6.5mm, and body segments 190-210, body colour brownish to violet-grey dorsally. Dorsal pores from 12/13 or 11/12. Setae rather smaller, paired but not closely. Clitellum ring shaped, but thinner ventrally, 1/2 xiii-xvii. Septum-7 is the first and being thickened. Septa, calciferous glands, and vascular system as usual in mentandric species.From the eighth to the twelfth segment also the internal and external segmentation do not

correspond; and septum-9 is actually situated in segment x as delimited externally by the furrows, if not on a level with groove; Gizzard large. Intestine are begins from xiv.[68]

Earthworms are called as hermaphrodite, they have both male and female sex organs but they need to the same species to produce off springs due to different maturation period of sperms and ova. Earthworms can breed throughout the year under ideal conditions, but cocoon output is known to decrease rapidly after a period of prolific production.

Therapeutic properties of earthworms

Since, from centuries medicinal values of earthworms are known. The earthworms are usually acted as a source of proteins, peptides, enzymes and physiologically active substances. The pastes which are prepared from earthworm tissue are used in the treatment of numerous diseases. Earthworms are also been used as a drug for the improvement of blood circulation for centuries. Worms have wide variety of biologically active properties such as Nitrogen, Antibacterial, Antioxidative, and Antipyretic as well as wound healing abilities. Earthworms are also used in pharmaceutical industries as well. Ashes of earthworm have been used as an stimulant for hair growth in head and as tooth powder.[74] And earthworm pastes are also used for the treatment of piles, fever, small pox, jaundice and removal of stones in bladder.[75] The 'leucocytes' of earthworm can recognized human cancerous cells as 'foreign' and can kill them.[76] Paste or extract which are obtained from earthworms are used to cure thrombotic diseases, arthritis, diabetes mellitus, pulmonary heart disease, mumps, exzema, anemia, and paralysis of limbs.[77] On the basis of above these accounts, the main aimof the present investigation is to study about the earthworms and their role in heavy metal accumulations.

Influence of Environmental factors on survival and growth of earthworms

Different conditions of earthworms such as cocoon production, rates of development and as well as growth are critically affected by environmental conditions. Epigeic earthworms are said to be relatively tolerant to the environmental conditions of the organic wastes. However, these earthworms are tolerance to other environmental parameters, such as moisture and temperature. If these factors are greatly exceeded, earthworms may move to more suitable zones in the waste, leave it or die. The most important abiotic factors which include combination of different organic wastes, aeration, pH, temperature, moisture, C: N ratio etc.

A-Temperature:

Earthworms have complex responses to changes in temperature. The suitable temperature range which is best for earthworms during vermicomposting process is about 12-28°C). Their activities are significantly influence by variation in temperature. During, winter time period their temperature are maintained above 10°C and in summer the temperature are maintained below 35°C.[78] When earthworms are affected by temperature below 10 (50°F), then generally reduced or little feeding activity occurs and below 4 (39.2^{o}F) cocoon production and development of young earthworm ceases completely.When temperature was extreme, earthworms tend to undergo hibernation and migrate to deeper layers for protection. *Eisenia fetida* can survive properly in very harsh environmental condition especially temperatures (5-43°C).[79]

B-Moisture:

The growth rate and maturation of earthworms was said to be best at 20°C and 85% moisture content. During the vermicomposting process, the optimum range of moisture for most of the species has been concluded to be between 50% and 90%. Whereas, the optimum moisture range up to 50% and 90% are suitable for *Eisenia fetida* and *Eisenia andrei* for their survival but they grow more rapidly between 80% and 90% in organic wastes.[80-81] During vermicomposting process, poor aeration may arise due to high moisture greasy and oily wastes in the vermibed.[82]

C-pH:

Most of the species of the earthworms are relatively tolerant to variation in pH range and can tolerate the pH levels of 5-9, but they prefer mostly pH range of 5.0, they move toward the more acid material. The pH is usually a parameter which greatly influences the influences the process of vermicomposting and life cycle of earthworms. Most majorities of species prefer slightly alkaline soils with pH ranging between 7.4 and 7.8. *Eisenia fetida* and *Perionyx excavatus* prefer pH range of 8.1 in dung heap for better growth rate.

D-Aeration:

The earthworms are usually aerobic organisms, and they need oxygen for their vital activity which is very important for their growth and reproduction process. Because earthworms lack specialized respiratory organs, and oxygen and carbon dioxide diffuse through their body wall. Earthworms are said to be very sensitive to anaerobic conditions. For better aeration conditions, vermibed should be properly aerated either mechanically or manually and cause better growth of earthworms.[78,82]

Effect of heavy metals on earthworms

Heavy metals according to their non-biodegradable nature is said to be remain attached to soil for a very long time, and these metals can easily bioaccumulate into soil biota, they can also leached into the underground water and caused a severe threat to the environment biodiversity and public health.[83] Whereas, earthworms are very important components of soil zone. In various, food chain they are abundantly distributed and act as a key organism for determining soil function and health.[84]

Earthworms have highly tendency of bioaccumulation of heavy metals such as cadmium, mercury, lead, copper, manganese, calcium and zinc into their tissues without affecting their physiology and this particularly when the metals are mostly non-bioavailable.[85] The presence of excess heavy metals in the soil is cause lead to increased mortality of the worms.[86-88]

If the organic material are present in soil is low, so that earthworms was unable to digest the soil and, as a result, the consistency of toxicity of cadmium increases in their body, and the mortality rate as well as disorder in reproduction system rised.[89] Toxic metal-induced oxidative stress including lysosomal membrane stability and also altered their genetic expressions.[90]

Heavy metal ions-induced DNA damage, DNA repair system impairment and as well as alteration in coelomocytes frequency in earthworms *Eisenia Andrei.*[91] *Metaphire posthuma*, an endogeic earthworm, cause encounters a substantial range of physiological stress due to exposure of metals like cadmium, chromium, lead and mercury which are abundantly present in the soil of industrial and agricultural regions of India. The bioaccumulation of heavy metals was occurred in the yellow cells of body tissues of earthworms. Mercury is more toxic for earthworm species because it effects cocoon production, coelomocytes, body weight and their length as well. Heavy metals may cause damage to an organ, inhibition of an enzymatic activities and significant alteration in their various metabolic activities.

The effects of sub-lethal concentrations of lead nitrate are shown on reproduction and growth rate of *Perionyx excavates*.[92] Earthworm populations was having tendency of mechanism by which they are able to tolerate or resist the negative effect of metal induced stress. Such tolerance levels of earthworms are acquired by through their variation in their genetic structure or changes in their physiology patterns. The bioaccumulation of methyl-mercury in the *Eisenia fetida*and its effect on regeneration after excision of the caudal end.[93]

Effect of heavy metals on other organisms

The presence of very high poisoning level of heavy metals causes negative impact on the environment as well as aspects of life. [94-95.] There are about 13 heavy metals which are listed as a hazardous environmental pollution such as As, Be, Cd, Cr, Sb, Cu, Hg, Ni, Pb, Se, Ag and Zn according to the U.S. Environmental Agency (USEPA). These heavy metals are very dangerous for human health and can show continuity throughout life in the neighborhood. The individual those who are exposed to contaminants in their environment and show responses towards that in a measurable and predictable way, and these responses are observed at different level of biological organization.[96]

Among these short lists of heavy metals, some are carcinogenic in nature such as Cd, Ni and Cr. Accumulation of heavy metals had a wide range of toxicity, neurotoxicity and mutagenic or teratogenic.[97] Bioaccumulations of heavy metals are also entered in food crops and cause severe impact on human health and this problem is a great concern across worldwide.

The main reason behind is due to rapid rise in population growth and industrialization which cause excessive changes in land-use pattern and addition of sustained efforts are enhanced the agricultural productivity of limited areas for production of adequate quantities of food. Along with fertilizers and other sources these heavy metals enter into irrigation zone because they act as low-cost sources for production of food crops but caused

negative impact on health of organisms those who consume these food crops.

In India, due to use of long term wastewater irrigation shown contamination of food crops with heavy metals and causes severe health hazards.[98-102] Heavy metals are present in soil and transfer to crops via accumulation and these metal can further transferred to other media through the food chain. Contamination of Lead can cause adverse effect on mental growth, neurological and cardiovascular disease also occurs in humans as well as in children.[103-105] Excessive higher concentration of Arsenic in soil, food crops can cause cancer, dermal problem, respiratory complications, and many other diseases.[104,106] Contamination with Cd heavy metals in food crops and their impact on human health are also reported extensively.[107] When Zn concentration will occur higher in human body, it can disturb the immune system.[104] When humans consume vegetables which are contaminated with heavy metals, are more deleterious for health.[108] Heavy metals are usually entered the cells of food crops via metal transporters such as phyto-siderophores.[109,39] Toxicity level of metal and behavior in soil are influenced by soil pH, carbonate contents, and their level of contaminations.[110]

CONCLUSION

From the findings of the present work and that of others, it can be clearly concluded that It is very well vermicomposting is the best way for remediation of soil from heavy metals like nikle, cadmium, boron and iron. It can also protect environmental damage through the implementation of earthworms. This vermiremediation process is an eco-friendly way. By accumulation of heavy metals into the body tissues of earthworms, they make soil free from metal contamination and improve soil quality. If earthworms are applied in in-situ application, they minimized the impact of heavy metals hazards and make the soil pollution free. But while application of earthworm, the problem will be reduced. An earthworm produces vermicompost from organic materials and improves soil fertility level. All different species of earthworm play very important role for bioaccumulation of heavy metals, but while accumulation process the earthworms are also affected in term of their survival growth and rate of reproduction. The present study will be helpful and told about the role of earthworms for accumulation of heavy metals by biological means. Thus, earthworms can safely manage all municipal and industrial organic wastes and divert them from ending up in the landfills and makes the soil pollution free.

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Daidzein affects leptin hormone secretion in the male Wistar rat

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Abstract : Daidzein (7, 4 dihydroxy isoflavone) is a naturally occurring isoflavone present in a number of plants, especially in soybeans and soy-derived products. The influence of this isoflavone, daidzein on leptin hormone was studied in sexually mature Wistar rats. Rats were divided into three groups: Control (received no daidzein), Sham control [received vehicle as ethanol: saline water (1:9)] and Experimental group [received daidzein 15mg/kg body weight (bw)]. Daidzein was administered intragastrically, i.e. by using a cannula inserted via oesophagus into the stomach once a day for 15, 30 and 60 continuous days. In animals treated with daidzein blood leptin-hormone level was significantly reduced. After 15 days, the effect was less significant ($p<0.05$) as compared to that observed after long duration administration for 30 and 60 days ($p<0.001$).This latter effect was probably due to direct inhibitory influence of daidzein on leptin-hormone secretion from adipocytes. Results obtained in the study indicate that daidzein may affect leptin-hormone responsible for metabolism, energy expenditure and food intake in Wistar rat.

Key words: daidzein, testosterone, Wistar rat, isoflavone

INTRODUCTION

Daidzein is an isoflavone found in high concentrations in soy. It plays an important role in prevention of many diseases and affects a number of metabolic pathways. The isoflavones belong to a large group of compounds classified as phytoestrogens[1], which have a chemical structure similar to estrogen and can bind to estrogen receptors.[2,3] Daidzein has an aromatic ring with two hydroxyl groups (other isoflavones such as genistein, formononetin and biochanin A have one or three hydroxyl groups). This structure enables it to bind to the estrogen receptor (ER). Daidzein has an affinity to both ER subtypes -ER α and ER β, but its affinity for ER β is significantly higher.[2,4]

The soy- derived isoflavone demonstrates pleiotropic action in the organism. It has both estrogenic and anti-estrogenic effects.[5] Adipose tissue has been shown to be a major endocrine system that plays an important role in energy homeostasis, lipid metabolism, immune response and reproduction.[6,7] Phytoestrogens promote, maintain and control the typical distribution of body fat and adipose tissue metabolism, through a still unknown mechanism. Daidzein and genistein were also found to inhibit lipogenesis and stimulate lipolysis in rat adipocytes. This activity was manifested by the direct effect of these compounds on isolated fat cells.[8]

The literature data reveal that the effects of daidzein on metabolism have not been fully elucidated. Many experiments performed to test the direct influence of daidzein and genistein on metabolism of fat cells suggests the possibility of its effect on leptin hormone secretion. This problem seems to be important since the proper leptin secretion constitutes an important factor regulating the energetic status of the whole organism.

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The aim of the present study was to determine the influence of daidzein administration for 15, 30 and 60 days on leptin hormone parameter in mature male rat.

EXPERIMENTAL DESIGN

Animals and measurement

The experiment was performed according to the guidelines accepted by the Ethics Committee, Ranchi University, Ranchi for investigations on animals.

Male and female Wistar rats weighing about 150g were used in the experiment. The animals were kept under standard conditions, at a constant temperature (21±4°C) with a 12-h dark-light cycle. Rats were fed a soy- free diet *ad libitum*. Males were divided into three groups of 10 animals each. Two female rats were introduced in each group. Animals in the control group received no daidzein, whereas, those in the second group received the vehicle, i.e. saline water: ethanol mixture (1:9 v/v). The third group (experimental) received daidzein dissolved in the vehicle in the amount of 15 mg/kg BW. The vehicle and daidzein solutions (Sigma) were given intragastrically (0.5ml/150g BW) once a day for 60 consecutive days. The males were anesthetized (di-ethyl ether) and their blood serum were collected after 15, 30 and 60 days of the start of the experiment. It was stored (-80°C) until analysis.

ANALYSIS

Leptin hormone was assayed by Radio immunoassay kit.

RESULT

Results obtained for leptin hormone content of blood serum after daidzein administration at a dose of 15 mg/kg BW for 15, 30 and 60 consecutive days are presented in Table- 1.

A significant reduction was noted in the leptin levels of blood serum after 15 days ($p<0.05$), 30 days ($p<0.001$) and 60 days ($p<0.001$) of daidzein treatment, as compared to the controls. Leptin levels in sham-control animals were not significantly different from those of controls, in all cases.

Table 1 -Effect of intragastric administration of daidzein on serum leptin level (ng/ml) of male Wistar rat

Treatment	15 days	30 days	60 days
Control	0.31±0.01	1.34±0.02	2.24±0.01
Sham Control	0.29±0.02	1.31±0.02	2.26±0.01
Treated	0.26±0.02*	0.29±0.03**	1.89±0.02**

Daidzein was dissolved in saline water: ethanol mixture (9:1 v/v) and was administered intragastrically (0.5 ml; 15 mg/kg BW) for 15 days, 30 days and 60 days.
Values are means ±SEM, Significance of difference from control: *, $p<0.05$; ** $p<0.001$.

DISCUSSION

The present results indicate that intragastric administration of daidzein at a dose of 15 mg/kg BW causes a significant reduction in the serum leptin concentration of the rat. This effect of daidzein was observed after 15 days of treatment although the effect was less significant ($p<0.05$) as compared to that observed after long duration administration for 30 and 60 days ($p<0.001$). An effect of genistein, administered to male rats for a period of three and seven days at a dose of 5 mg/kg BW, in substantially diminishing blood leptin levels, has been reported.[9, 10] However, such an effect was not observed, by these authors, in animals receiving daidzein at the similar dose. Neither genistein nor daidzein consumed by rats affected leptin gene expression in adipocytes isolated from these animals.[9] The present results are different from these and showed significant decrease in blood leptin levels after daidzein administration to rats, in all cases. It can be opined that perhaps a duration of three and seven days was not sufficient for the expression of the effect of daidzein on leptin level of blood in rat, and that a longer period of treatment was required to produce any effect.

Leptin is a hormone that plays an important role in regulation of food intake, body weight and energy status

of the organism. [9, 11] Daidzein has been found to inhibit lipogenesis and stimulate lipolysis in rat adipocytes. This activity is manifested by the direct effect of daidzein; on isolated fat cells.[8] This direct influence of daidzein on metabolism of fat cells suggests the possibility of its effect on leptin secretion.

It has been found that inhibition of some enzymes in adipocytes substantially abates leptin secretion in spite of unchanged expression of its gene.[12]

The results clearly show an inhibitory effect of daidzein on leptin level of blood under the present experimental conditions. It may be suggested that this effect of daidzein was possibly due to its ability to restrict glucose transport due to conformational changes in GLUT 4, as reported for genistein.[13] The effect may also be a result of restricted glucose transport through GLUT 1. Hence, inhibition of glucose transport in adipocytes could be inhibiting secretion of leptin. Such an effect has been reported for genistein in case of rat adipocytes.[14] Further, it has also been demonstrated that GLUT 1 levels are decreased after treatment with daidzein.[15] GLUT expression is dependent on the concentration of compounds used and their inhibition constant. Compounds that block glucose transport also decrease glycosylation levels, and as a result, GLUT levels are lower.[16] This observation demonstrates that effects of daidzein on adipocytes are responsible for decreased leptin release and this significantly diminishes the blood leptin level.

The inhibition of glucose transport and/or glycolysis in adipocytes strongly prevent the secretion of leptin.[17] These observations allow to suppose that daidzein induced drop of leptin secretion in the present study could be due to inhibitory influence of this phytoestrogen on glucose transport.

The decreased blood leptin level observed in the present experiment after 15, 30, and 60 continuous days administration of daidzein to mature male rats may also have resulted from the direct influence of daidzein on fat tissue. Daidzein has been reported to inhibit lipogenesis and stimulate lipolysis in rat adipocytes.[18,19] This direct influence of daidzein on metabolism of fat cells suggests the possibility of its effect on leptin secretion.

Phytoestrogens are endogenous estrogens. They can act as weak estrogens and bind to the estrogen receptor in various tissues.[20] It has been demonstrated that the appetite repressing action of estrogen causes reduction in food intake.[21] Dietary phytoestrogens decrease feed intake and hence decrease body weight and abdominal fat mass.[22] Phytoestrogens are capable of inducing apoptosis of adipocytes suggesting that, at least, part of the weight loss is due to destruction of fat cells which could result in better maintenance of weight loss. [23] The present observation that dietary daidzein significantly ($p<0.001$) depressed serum leptin concentration after continuous administration for 15 to 60 days allows to believe that the effect of this phytoestrogen daidzein may be due to its direct influence on adipocytes which are the main source of leptin.[9] Phytoestrogens such as daidzein and genistein inhibit enzyme activity in adipocytes which decreases leptin secretion.[12]

It has been documented that factors augmenting cAMP content in adipocytes decrease blood leptin concentration[24,25] and reduce its release from fat cells.[26,27,28] Lipolysis is enhanced by the activation of protein kinase A (PKA) in the presence of increased c AMP in fat cell.[29] An inhibitory action of genistein on cAMP phosphodiesterase has been found in adipocytes[30], neural cell line[31] and hepatocytes[32]. Further research in this direction is needed to ascertain if daidzein exerts a similar effect on adipocytes. It was demonstrated that different lipolytic agents, including cAMP phosphodiesterase inhibitors, substantially reduced secretion of leptin from isolated adipocytes.[26] Thus, it may be possible that the inhibition of cAMP phosphodiesterase by daidzein contributes to restriction of leptin secretion.

Results obtained in the present study clearly indicate an inhibitory effect of dietary daidzein on leptin secretion in male Wistar rat. It is suggested that this inhibitory effect of daidzein may be due to the restriction of glucose metabolism. The decrease in serum leptin could be the effect of daidzein action on adipocytes such as inhibition of glucose transport, restriction of insulin action or inhibition of cAMP phosphodiesterase activity. The mechanisms involved in the expression of these effects, by daidzein, remain yet to be investigated.

CONCLUSION

The results obtained in this study show that intragastric administration of daidzein for 15 to 60 days

significantly affects leptin hormone levels in male rats. This compound essentially decreased leptin concentration in blood serum. Mechanisms responsible for the detected activity of daidzein require further investigations.

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Fish diversity and abundance in relation to water quality of Medninagar, Palamu, Jharkhand (India)

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Abstract : This paper represents the Freshwater fish diversity, abundance and richness status of Medninagar, Palamu, Jharkhand was studied in three sampling sites. The present study has shown that Nawatoli pond, Kachrwa dam and Bara talab supported 3 fish species belonging the same class, order and family. Fish diversity was assessed Shannon-Weiner Index (H), Simpson's Dominance Index (D), Simpson's Index of diversity (I-D), Pielons Evenness and Margalef Index of species richness and seasonal water quality parameters such as water, temperature, pH, electrical conductivity, free CO_2, DO, BOD hardness, chloride, nitrate and phosphate has been recorded and were found suitable for fish population. It can be concluded that reservoir supported rich fish population; it needs proper management and utilization of these fish's wealth and sustainable steps to monitor and conserve these fish health. However, the present paper deals with fish diversity in the freshwater fishes of Medninagar, Palamu, as these water bodies harbour a large number of fish fauna. The survey is based on observations and description of fishes as recorded in Systems Nature by Carls Linnaeus.

Key words: Freshwater fishes, water quality, biodiversity indices, Cyprinidae, Growth, Palamu, Jharkhand

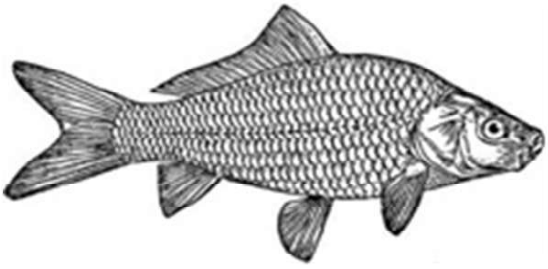
INTRODUCTION

Fish perform all their body functions in water because it is totally dependent upon water to breathe, feed and grow, excrete wastes and reproduce understanding the physical and chemical qualities of water. Water has very unique density qualities. Most liquids become denser as they become cooler. However, water gets dense as it cools until it reaches a temperature of approximately 39°F. As it cools below this point, it becomes higher until it freezes (32°F). As ice develops, water increases its volume by 11 per cent. The increase in volume allows ice to float rather than sink, a characteristic that prevents ponds from freezing solid. It is the fact that water can dissolve more substances than any other liquid. This is why, it is called universal solvent. Beside oxygen, water temperature is one of the important factors that affect the growth of fish. Fishes are cold-blooded organisms and acquire approximately the same temperature as their surroundings. The temperature of the water affects the activity, behaviour, feeding, growth and reproduction of all fishes. Metabolic rates in fish double for each 18°F rise in temperature. Temperature also determines the amount of dissolved gases (oxygen, carbon dioxide, nitrogen etc.) in the water. The cooler the water the more solubility of the gases. Water has its maximum density at 39.2°F. In spring season, water temperature is nearly equal at all ponds depths. As a result, nutrients, dissolved gases, and fish wastes are evenly mixed throughout the pond. As the day become warmer, the surface water becomes warmer and lighter while the

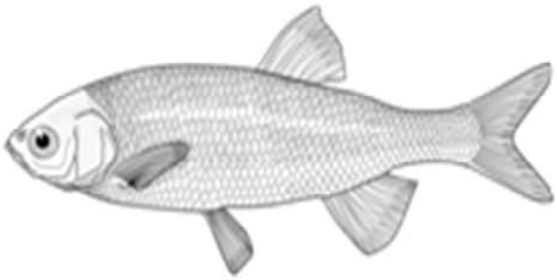
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cooler-denser water forms a layer underneath. Circulation of the cooler bottom water is prevented because of the different densities between the two layers of water. Dissolved oxygen levels decrease in the bottom layer since photosynthesis and contact with the air reduced. The already low oxygen levels are further reduced through decomposition of waste products, which settle to the pond bottom. Localized dissolved oxygen depletion poses a very real problem to the fish culturists.

MATERIALS AND METHODS



Labio rohita



Catla catla



Cirrhinus mrigala

1. Study Area

The study area was situated at Medininagar, Palamu, Jharkhand. Palamu district is located in the South-West of Jharkhand State. It is a part of Chotanagpur highlands, covered with green bush vegetation. Due to maximum height at Pat region, it gives pleasant situation. Here high rainfall occurs in comparison to other parts of Jharkhand. It has rugged topography. Region is covered with lava, where the ocean lava was erupted; the region is formed by lava deposit of igneous and metamorphic rocks. Due to the mechanical action of the agents of demidatran volcaner rocks have been eroded and transformed into lalurope soil.

Due to leeward action of the water on the surface, rugged topography of granite and other metamorphic rocks are seen. North Koel River is the most popular river of this region. Budhaghagh and Gautam dhara are two famous waterfall situated in this region. There are a number of ponds in this region like Nawatoli, Kachrwa dam, Bara talab. All these tanks are utilized for various purposes by the local people. Water is fresh and aquatic creatures are utilized by human beings. The climate of this region is monsoon type, but due to the highland area, this region is cooler than the other parts of Jharkhand. It has low temperature in summer season and receives high rainfall in rainy season as comparison to the other parts of Chhotanagpur highlands. This region seems very beautiful in rainy season; Green vegetation and cloudy weather attract tourism in this region.

The fishes such these biomass and efficiently convert them into animal protein in order to help farmers for fish culture in this area because of its high protein content. So a detail study is urgently required to correlate the growth of fish with respect to local environment that surround the water body and influence water quality. These works have immense significance for Jharkhand where large population depends on fishing not only as food source but also for their livelihood.

2. Fish Samples Collection

Fish samples were collected from different selected localities during the study period from September 2017 to October 2018 with the help of local fishermen using different types of nets namely gill nets, cast nets and

dragnets. Immediately photographs were taken prior to preservation since formalization decolorizes the fish colour. 10% formalin solution was prepared for preservation of fish samples. Fish brought to the laboratory were fixed in this solution in separate jars according to the size of species. Smaller fishes were directly placed in the formalin solution while large fishes were given an incision on the abdomen before they were fixed.

3. Fish Identification

The fishes were identified by using Dutta Munsi & Srivastava (1988), Talwar and Jhingran (1991) and Jayaram (1999) methodology

4. Diversity Indices

Species diversity can be managed separately either as species richness or evenness or diversity as a whole. Species richness was measured by Index of Richness (denoted by R) given by Margalef (1958). Species evenness was measured with evenness index (denote by E) given by Hill (1973).

Diversity of the species calculated directly with a variety of indices, of which two commonly used are Shannon-Weiner Index or simply the Index of diversity as given by Shannon and Weiner, 1963) and Index of dominance given by Simpson (1949). Shannon's index has a direct relationship with the species diversity, whereas Index of dominance has an inverse relationship.

5. Water Sample collection for physico-chemical analysis

Surface water samples were collected in polythene cans from different sampling sites of the Medininagar reservoir. Air and water temperature, pH, DO and free CO_2 were determined on the spot itself and for BOD determination water samples were collected in BOD bottles. For further analysis, samples were transported to laboratory and analyzed other parameters using standard procedures (APHA, 1985, 2005).

Periodical estimates of fish growth and yield in relation to water samples were collected and analysed for computation of fish population dynamics and impact of physico-chemical complexes on fish yield of the water bodies. The whole exercise also involved network analysis and rendering of food chains/food web of the existing ecosystems.

Analytical studies in the selected water bodies were carried out for two years 2016 to 2018 in different seasons, prebreeding, breeding and post breeding. Mean values are presented here. However, methodology of APHA – 1985, 2005 used for experimental work. Three different seasons have been considered to determine that water condition affect the body of fishes, physiology taking growth index as measurement.

RESULTS AND DISCUSSIONS

The present investigation of fish fauna in Medininagar reservoir showed that most of the fishes recorded were widely distributed in the lentic and lotic waterbodies of Medininagar reservoir. The members of Cypriinidae family dominate the fish populaation.

The economically important and cultivated fishes were also recorded and includes *Labio rohita, Cirrihinus mrigala, Catla catla*. The economically important and cultivable above fishes and reservoir can be exploited for commercial production of fishes for better improvement of socio-economic condition of local people.

Table - 1 : Fish abundance and biodiversity status in Medininagar, Palamu, Jharkhand

Sl.No.	Name of Bony fish	Zoological Name	Abundance	Biodiversity	Habitat
1	Rohu	*Labeo rohita (Ham)*	A-2	LR-nt	Lentic and Lotic
2	Mrigala Naini	*Cirrhinus mrigala*	A-2	LR-nt	Do
3	Katla	*Catla catla*	A-2	VU	Do

A – 2 : Common
VU : Vulnerable
LR-nt : Lower risk near threatened

As far as biodiversity status (IUCN-1994) is concerned Different diversity indices were calculated as per standard methods. The Shannon-Weiner fish diversity index of Medninagar reservoir ranged from 2.4 to 3.0. The Simpson's dominance index values ranges between 0.08 and 0.2. Simpson's index of Diversity (1-D) also ranges between 0 and 1 but now, the greater the value, the greater the sample of diversity. This makes more sense. In the present study, the Simpson's index of Diversity (1-D) values fluctuated between 0.89 and 0.95. However, Pielou's evenness values were ranges from 0.6 to 0.9. The Margalef index of species richness values revealed 1.48 to 2.4. (Table No.2).

Table-2 : Fish diversity indices of Medninagar, Palamu, Jharkhand

Diversity indices	Range
Shannon - Weiner Index (H)	2.4- 3.0
Simpson's Dominance Index (D)	0.08-0.2
Simpson's Index of Diversity (I-D)	0.89- 0.95
Pielou's evenness	0.6- 0.9
Margalef index	1.48 -2.4

Table 3 : Seasonal variation of physico-chemical parameters in the water samples of Medninagar, Palamu, Jharkhand during 2017-2018

Parameters	Site 1			Site 2			Site 3		
	Winter	Summer	Rainy	Winter	Summer	Rainy	Winter	Summer	Rainy
Air Temp.	28±2.16	33±0.81	28.5±2.51	28±2.16	33.5±0.57	28.25±3.30	29±2.44	32.75±1.25	28.75±3.77
Water Temp.	25.25±0.5	29±2.16	26.75±1.70	25.25±0.5	29.75±1.89	26.5±1.91	26±0.81	30.25± 1.71	27.25±2.87
Turbidity	30.82±22.35	21.75±10.37	52.62±29.23	15.35±3.92	28.7±10.83	49.82±30.57	15.5±4.20	26.25±6.34	56.5±28.38
pH	7.43±0.15	7.62±0.09	7.47±0.15	7.52±0.06	7.63±0.08	7.52±0.14	7.43±0.23	7.65±0.07	7.40±0.22
Alkalinity	51.11±6.61	59.25±2.27	42. 78± 4.45	S6.13±8.33	61.45±1.47	42.58±13.41	55.03±5.92	61.50±2.08	40.08±13.21
EC	133.75±8.38	165.5±13.72	121.35±37.42	136.75±6.89	167±7.39	120.67±35.93	155.75±34.27	I 9.5±9.74	140.5±21.11
CO_2	2.47±0.74	1.4±0.52	2.55±0.92	2.54±0 91	1.45±0.58	2.63±0.83	2.31±0.47	1.49±0.53	2.68±0.72
DO	7.31±1.53	6.18±1.l5	5.79±1.10	7.52±1.02	5.75±0.62	5.72±0.71	8.28±1.50	6.38±0.83	5.72±1.26
COD	8.8±6.34	16.1±9.46	4.50± 1.73	12.5±2.88	12.25±6.94	11.5±5.51	10.22±9.25	15.5±9.98	14.75±9.28
BOD	1.01±0.68	2.47±1.40	0.83±0.04	1.17±0.47	1.31±0.67	1.21±0.44	1.18±0.84	2.20±1.39	1.77±1.23
Hardness	50.5±5.0	62.0±11.54	44.00± I 0.70	47±5.29	62.5±9	48±11.77	56.25±9.53	67.5± 12.33	52.5±9.84
Chloride	21.98±0.81	26.59±4.68	20.27±3.00	21.45±2.19	26.57±5.12	21.62±4.81	24.89±1.71	27.65±5.37	23.04±1.36
NO_3	0.108±0.059	0.165±0.03	0.328±0.221	0.13±0.09	0.162±0.04	0.34±0.25	0.120±0.06	0.142±0.026	0.340±0.285
PO_4	0.011±0.008	0.012±0.007	0.009±0.005	0.004±0.002	0.012±0.006	0.007±0.004	0.006±0.003	0.013±0.006	0.009±0.004

All The parameters are expressed in mg/L except temperature (^oC) turbidity (NTU), pH, electrical conductivity (μmhos/cm)

Water Quality Parameters

The physico-chemical characteristics of water have an important role in supporting fish diversity freshwater ecosystems. The seasonal average values of physico-chemical parameters of the reservoir are depicted in the Table 3. In the present study, air temperature was recorded in the range of 28.00±2.16 during winter season to 33.50±0.58^oC at Site 1. Water temperature was varied between 25.25±0.50^oC at Site 1 and 2 during winter season and 30.25±1.71^oC at the site 3 during summer season. pH was in the range of 7.40±0.22 at site 3 during rainy season and 7.66±0.07 at site 3 during summer season. A minimum

value of electrical conductivity was recorded at site 120.68±35.93 mhos/cm during rainy season and a maximum of 169.50±9.75 mmhos/cm at site 3 during summer season. Free CO_2 was observed in the range of 1.41±0.52 at site 1 during summer season to 2.68±0.72 mg/L at site 3 during rainy season. DO level was varied between 6.19±1.16 mg/L at site 1 during summer season to 8.28±1.50 mg/L at site 3 during winter season. Biological oxygen demand for 3 days at 27°C was fluctuated between 0.83±0.04 mg/L at site 1 during rainy season and 2.48±1.41 mg/L at site 1 during summer season. Total hardness was registered in the range of 44.00±10.71 mg/L at site 1 during rainy season to 67.50±121.37 mg/L at site 3 during summer season. Nitrate concentration was observed in the range of 0.108±0.06 mg/L at site 2 during rainy season. Phosphate content was fluctuated between 0.012±0.01 mg/L at site 2 & 3 during summer season.

The diversity index indicates good correlation with overall species richness across the sites and could be utilized by the biodiversity conservation managers for prioritization of sites of conservation and habitat restoration. *Bergerot et al*. (2008) developed indices for fish biodiversity conservation concern, rarity index and fish magnitude values for prioritization of sites' for large scale European freshwater basin in France. *Lasne et al*., (2007) used fish zonation and identified indicator species for the evaluation of the ecological status of the Loire basin (France). They also carried out a discriminant analysis on environmental variables revealed that they could be mainly determined by the slope, temperature, and depth.

During study, it was found that abundance and diversity of fishes were found to be very high in respect to extent of water body. The maximum numbers of species were recorded from low land areas. According to *Shaikh et al*., (2011) in low and middle land areas fresh water fish diversity was found to be very high. It is due to deep water bodies that allow niche segregation in order to enable the fishes to live without facing more intra and inter specific competitions. During summer, when maximum level of water decreased due to hot air and high temperature most of the fishes migrated toward low land for survival. But during winter season, diversity of fish fauna is abundant due to clear water, preference of maximum amount of phyto and zooplankton as compared to rainy season. The presence of exotic species in the reservoir may be due to migration from nearby water bodies with flood water.

The mean air and water temperature values were similar and slightly high during summer season. pH is considered as an indicator of overall productivity that causes habitat diversity (*Raj and Jayasekhar*, 2007). The pH of most the natural water is varied between 6.0 and 8.5. In the present study, the seasonal mean pH was close to neutral. According to *Goldman and Horne (1983),* low pH < 5.0 can severely reduce aquatic species diversity. However, the range of water pH was reasonably good for fish population. The conductivity values were higher in the summer season due to decreases in the total volume of water, whereas; low in rainy season because of dilution factor (*Alam et al*., 2007). CO_2 has a great effect on photosynthesis which affect the fish growth. There is no much variation in the concentration of free CO_2 throughout the study period. *Fafioye et al.,* (2005) have observed similar trend in Omi waterbody Ago Iwoye (Nigeria). Further, free CO_2 more than 20 mg/L is toxic to fish (Battul et al., 2007). The observed values were below 20 mg/L of free CO_2 hence, water is more suitable for fish diversity. Dissolved Oxygen is a primary and comprehensive indicator of water quality in surface water. The decline of dissolved oxygen level has a serious implication for the health of aquatic system. The optimum value for good water quality is 4 to 6 mg/l of DO, which ensures healthy aquatic life in a water body (*Santosh and Shrihari*, 2008). In the present study, DO level was more than 4 mg/L at all sites. Further, DO level was high in winter followed by summer and rainy season. The reason could be low temperature, turbulence of surface water by high wind action etc. And its level drops in summer due to high metabolic rate of organisms (*Salve and Hiware*, 2006) and limited turbulence in the reservoir (*Mwaura*, 2006). BOD value can be used as a measure the degree of water pollution and is useful in evaluating self -purification capacity of a water body.

The seasonal BOD values were slightly high in summer, low during winter and rainy season. Higher values of BOD in summer season due to higher microbial activity and elevated temperature (*Patel,* 1999). As per the hardness values, water belonging to soft (0-60 mg/L) and moderate (60-120 mg/L) category. Seasonal variation of total hardness shows that its concentration was high in summer

and more or less similar in other seasons. The high hardness during summer season could be due to reduced water level and increased salts content in water (Kemdirim, 2005). Nitrate (NO_3) is an essential nutrient for aquatic plants in natural waters. An excess concentration of nitrate tends to stimulate algal growth and leads to eutrophic conditions. Seasonal nitrate concentration was high during rainy season at all stations and remained similar in winter and summer season. The higher concentration of nitrate during rainy season could be due to leaching of nitrate from agricultural fields. Phosphorus concentrations above 0.02 mg/L in reservoirs tend to produce algal blooms. However, in the present study, phosphate was below 0.02 mg/L. Its concentration was high in rainy season than summer and winter season. The overall water quality parameters indicated that water is suitable for fish life in the reservoir.

This ichthyofaunal study indicates that this waterbody is rich in diversified fish fauna consists of native species, economical, cultivable. Changes in fish community, directly or indirectly affect other components of the reservoir ecosystem including physical, chemical and biological characteristics. Habitat loss and environmental degradation has seriously affected the fish fauna. Conservation of fish diversity assumes topmost priority under changing circumstances of gradual habitat degradation. Therefore, a sustainable strategies needs to explore more fish species, utilization and save fish community of this reservoir. The study will provide future strategies for development and fish conservation.

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Effect of intragastric daidzein administration on blood testosterone level in male Wistar rat.

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Abstract : Daidzein is a phytoestrogen found in many plants consumed by humans and animals. The influence of isoflavone, daidzein on blood testosterone hormone level was studied in sexually mature male Wistar rats. Rats were divided into three groups: Control (received no daidzein), Sham control [received vehicle as ethanol: saline water (1:9)] and Experimental group [received daidzein 15mg/kg body weight (bw)]. Daidzein was administered intragastrically, i.e. by using a cannula inserted via oesophagus into the stomach once a day for 30 continuous days. In animals treated with daidzein blood testosterone hormone level was significantly reduced as compared to the controls after 15 and 30 days. This effect was probably due to the direct inhibitory influence of daidzein on testosterone hormone secretion from testis. Results obtained in the study indicate that daidzein affects testosterone hormone secretion and may therefore be responsible for infertility, cryptorchidism and reduced sperm concentration in Wistar rat.

Key words: daidzein, testosterone, Wistar rat, isoflavone

INTRODUCTION

Daidzein (4'7, dihydroxyflavone) is a phytoestrogen belonging to class of soy isoflavones, a sub-class of flavonoids that have received great attention for their potential on human health benefits. Daidzein aglycone is an isoflavone found at low concentrations in soybeans but at high concentrations in certain soy-derived food. Daidzein is much more abundant in the unprocessed soybean. Some phytoestrogens have been categorized as endocrine disruptors, i.e. exogenous substances that alter the functions of the endocrine system and consequently have adverse effects on the health of the organism, its progeny or its populations. They mimic or block hormones and disrupt normal endocrine functions by altering normal hormone levels, inhibiting or stimulating the production of hormones. They have been classified as natural environmental oestrogens[1] or natural endocrine- active agents[2]. Structurally, daidzein closely resembles 17ß-estradiol and it binds to estrogen receptors (ERs), the stronger affinity being for the ERß isoform[3]. Acting as a natural selective ER modulator, daidzein exerts its estrogen agonist or antagonist action in tissue in a dose dependent manner[4].

There is growing public concern regarding the adverse effects of environmental chemicals with an estrogenic influence on reproductive health. Daidzein and genistein are widely distributed in daily human diet[5]. In a typical Western diet, an average of 0.2 mg/kg isoflavones are consumed daily, whereas a typical Asian diet contains > 1.5 mg/kg isoflavones per day[6], which can raise individual human isoflavone serum levels to 500 Nm[7]. For infants fed soy- based formulas isoflavone intake can reach 9.3 mg/kg bw/day.[8] Phytoestrogens have been part of

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traditional diet in Asia for millennia. The negative effects of isoflavones, particularly on the male reproductive system have also been reported[9]. Previous studies demonstrated that isoflavones may produce male reproductive toxicity. Their main adverse effects on the male reproductive system include the disturbance of sex hormone release,[10,11] interference with the onset of puberty[12], altering penile corpus cavernosum structure weakening erectile function[13,14], suppressing the activity of some steroidogenesis associated enzymes[15] and decreasing the weight and epithelial height of accessory sex organs[11]. A high intake of soy-based food and soy isoflavones is associated with reduced sperm concentration, as demonstrated in animal experiments and human epidemiological studies[16,17].

Dietary isoflavones have caused infertility in male mice[18] and rats.[19] Genistein and ß- sitosterol had adverse effects on the reproductive organ weights in mice[20,21] and rats.[22,19] Reductions in the levels of plasma testosterone, estradiol and progesterone have been observed in rats after exposure to genistein[23,21,24]. Phytoesterol exposure caused an increase in plasma testosterone levels in the European polecat[25] and in the field vole[26].

Keeping the above in mind and considering the fact that there are limited reports on daidzein, the present study was conducted to observe the effects of short term intragastric administration of the isoflavone, daidzein on testosterone hormone in male Wistar rat.

Experimental Design

The experiment was performed according to the guidelines accepted by the Ethics Committee, Ranchi University, Ranchi for Investigations on Animals.

Male and female Wistar rats weighing about 150g were used in the experiment. The animals were kept under standard conditions, at a constant temperature (21±4°C) with a 12-h dark-light cycle. Rats were fed a soy- free diet *ad libitum*. Males were divided into three groups of ten animals each. Two female rats were introduced in each group. Animals in the control group received no daidzein, whereas, rats in the second group received the vehicle, i.e. saline water: ethanol mixture (1:9 v/v). The third group (experimental) received daidzein dissolved in the vehicle in the amount of 15 mg/kg BW. The vehicle and daidzein solutions (Sigma) were administered intragastrically (0.5ml/150g BW) once a day for 30 consecutive days. The males were anesthetized (di-ethyl ether) and their blood serum were collected after 15 and 30 days of the start of the experiment. It was stored (-80°C) until analysis.

ANALYSIS

Testosterone hormone was determined by Fully Automated Bidirectional Interfaced Chemi Luminescent Immuno Assay.

RESULTS

Results for the effects of daidzein on serum testosterone level of rat after 15 and 30 days administration are presented in Table 1. & Figure 1.

Serum testosterone levels in animals administered daidzein intragastrically for all durations tested, were significantly ($P<0.001$) lower than their respective controls and sham-controls. No significant difference was observed between serum testosterone concentration of sham-operated animals and control animals

Table 1: Effect of intragastric administration of daidzein on serum testosterone level (ng/dl) of male Wistar rat

Treatment	15 days	30 days
Control	22.43 ± 0.08	22.7 ± 0.02
Sham control	22.29 ± 0.04	21.54 ± 0.08
Treated	7 ± 0.10*	7.5 ± 0.09*

Daidzein was dissolved in saline water: ethanol mixture (9:1 v/v) and was administered intragastrically (0.5 ml; 15 mg/kg BW) for 15days and 30 days.

Values are means ±SEM

Significance of difference from control:*, p<0.001.

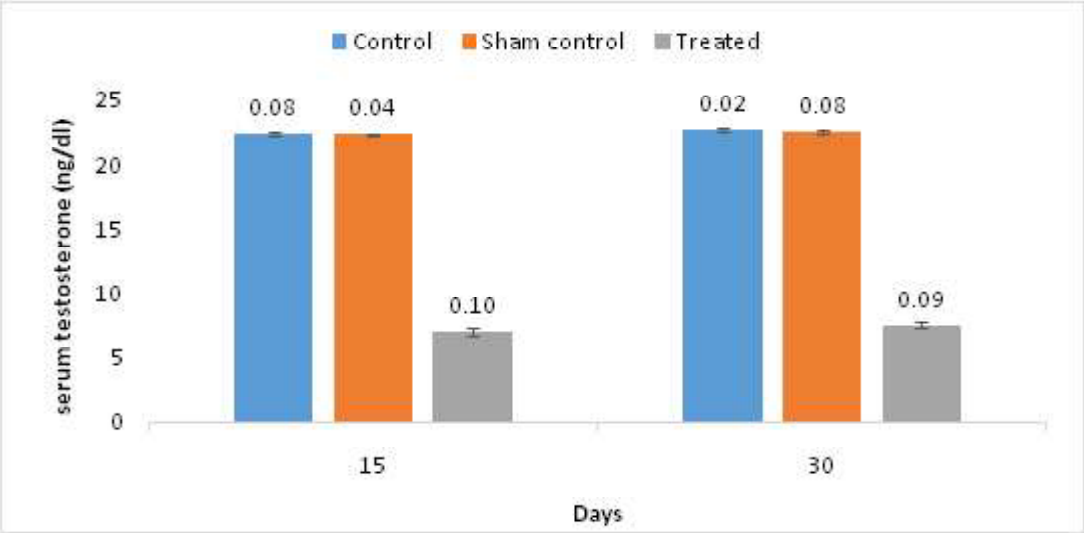


Figure 1: Effect of intragastric administration of daidzein (15mg/kg BW) for 15 days and 30 days on serum testosterone level (ng/dl) in male Wistar rat.

Significance of difference from control:*, p<0.001.

DISCUSSION

Studies for over the last 50 years reported an increased incidence of human male reproductive disorders like cryptorchidism, hypospadias, testicular cancer and low semen quality[27]. The rapid increase in disorder of the reproductive function suggests the involvement of environmental or life style factors. Exposure to endocrine disruptors appears to be an important cause.

Endocrine disrupting chemicals are exogenous agents that interfere with synthesis, secretion, transport, metabolism, binding action or elimination of natural blood -borne hormones that are present in the body and are responsible for homeostasis, reproduction and developmental processes. These endocrine disruptors are highly heterogeneous in structure and widespread in our environment. Less attention has been paid to the action of natural plant-derived endocrine disruptors (phyto estrogens) which are non-steroidal compounds having the ability to bind to estrogen receptors.

Daidzein, the major soy isoflavone glucoside is present at high concentration in soybeans and is a major source of xenoestrogen exposure in humans and animals. The results of the present study indicate an inhibitory effect of daidzein on the testosterone hormone level in the serum of Wistar rat when administered at a dose of 15 mg/kg body weight for a period of 15 days. Study on the long term administration of daidzein, at this dose, for over 30 days showed the same inhibitory action of daidzein on this hormone. The present result clearly shows that this isoflavone i.e. daidzein, under the present experimental conditions, is able to alter the testosterone level in the blood serum of this animal. This could have potentially detrimental effects on the fertility and reproductive functions of the animal.

Adverse effects of phytoestrogens have been observed in experimental animals but almost all of them focused on the effects of genistein and less attention has been paid to study the action of daidzein on reproductive functions. For example, the adverse effects of phytoestrogen, genistein in causing infertility in male mice were observed long back[18]. It has been shown that exposure to genistein adversely affects the reproductive organ weights in mice[20,21] and rats [19,22].

Reduction in the plasma testosterone, estradiol and progesterone levels, after genistein exposure have been shown in rats[21,23]. However, such reports are lacking for daidzein.

The present results are new and show that the isoflavone daidzein is capable of causing an effect on the testosterone level in the plasma, of this animal. These results are different from those observed for the European polecat, *Mustela putorius*,[26] in which it was demonstrated that

phytoesterol exposure increased the plasma testosterone levels.

Studies have reported that certain foods containing isoflavones, particularly infant formula, may have potential adverse effects on male reproductive function. Humans and animals are frequently exposed to products that contain low levels of isoflavones[28]. There are commercially available isoflavone containing products that have estrogen – like effects such as milk with added genistein and soy-based infant formulas[29,30].

Daidzein is found in many natural products that are marketed as promoting health and used to treat various diseases such as asthma, cancer, menopause, osteoporosis, atherosclerosis[31] as well as many commercial pet foods[32] and soy based infant formulas[33,29].

Isoflavones cause infertility in many animal species like sheep[34], the California quail[35], the mouse[36,18], the cheetah[37] and the rat[19]. Dietary phytoestrogens may have similar effects on the development and fertility of other species, including humans. But human data are very limited. There is paucity of studies available to evaluate the effects of soy and phytoestrogens on fertility or reproductive hormone in human males. Associations between exposure to phytoestrogens and hormonal disorders in children have been described[30].

The present results clearly demonstrated that dietary daidzein administration was able to decrease testosterone synthesis just after a period of 15 days in rats. These results are similar to those obtained by Zhu and Cowerkers[38] who reported a suppression of testosterone secretion by daidzein (30 µ mol /l for 24 hrs) in cultured testes and Leydig cells of neonatal mouse testes. These authors also suggested adverse effects of high doses of daidzein on Sertoli cells in neonatal mouse testes. A similar effect for genistein in impairing early testosterone production in fetal mouse testes *in vitro* has also been reported[39].

Genistein at a dose of 213 mg/kg bw/day from gestational day 7 to PND 13 also reduced plasma testosterone levels in rat offspring[40]. A reduction in plasma testosterone level was detected in primary Leydig cells by daidzein treatment in *Anser anser*[41,42]. The findings show a potential harmful effect of daidzein exposure on testis steroidogenesis function during the early neonatal period.

The results of the present study clearly define the association between daidzein exposure and reduced testosterone production in Wistar rat, under the tested experimental conditions. The mechanism of this action is still unclear. It has been suggested that testosterone production can be inhibited by exogenous compounds by suppression of steroidogenic enzymes, P450 scc, 3ß-HSD and P450c17ß[43-46]. It has been reported that genistein suppresses StAR, P450 scc, 3ß-HSD and P450C17ß expression and decreases testosterone production in fetal testis[39]. The observations in these studies may explain why testosterone levels in the testis or plasma are decreased. The expression levels of STAR and steroidogenic enzymes have been evaluated in daidzein- treated Leydig cells, in order to verify the mechanistic activities of daidzein on factors associated with steroid synthesis. The results showed decline in m-RNA and protein expression levels of STAR, P450scc, 3ß- HSD and P450C17. These results have been reported to be consistent with those of organ culture[47]. Testosterone is synthesized from cholesterol.

The conversion of cholesterol to testosterone involves series of key steroidogenic proteins, such as STAR protein, cytochrome p450 scc and 3ß-HSD[48,49,50]. The rate limiting step in the synthesis of testosterone is the transfer of cholesterol to pregnenolone via P450scc; StAR mediates this rate-limiting step in steroidogenesis[51]. The conversion of pregnenolone to progesterone is mediated by 3ß-HSD. The enzymatic action is essential for the production of all active steroid hormones[52].

Phytoestrogens including daidzein can modulate the endocrine system by altering the gene expression or activity of the enzymes involved in steroidogenesis[53]. The key enzymes involved in steroidogenesis have been reported to be important targets for phytoestrogens[54]. Isoflavones including daidzein can exert inhibitory feedback on the hypothalamus-pituitary gonadal axis[55].

The present results confirm the inhibitory effect of dietary daidzein on testosterone levels of blood in Wistar rat. It is suggested that this effect of daidzein may be exerted via the alteration of the activity of the enzymes involved in steroidogenesis. Further research in this direction will elucidate the complete mechanism involved in producing such an effect.

CONCLUSION

The results obtained in this study prove that 15 and 30 days intragastric administration of daidzein affects

testosterone hormone level in male Wistar rats. This compound essentially decreased testosterone concentration in blood serum. Mechanisms responsible for the detected activity of daidzein require further investigations.

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Comparative anatomy of ABO of estuarine gobies (*Monopterus cuchia)* of India

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Abstract - A wide variety of air breathing organs have been evolved in fishes along with different lineages. Most of these fishes remain in water and uses air as the source of oxygen at variable degrees. And with the evolution of air breathing organs the circulatory system of the fishes have been modified at different levels to accommodate blood to and from the newly developed air breathing surface. However in most of the air-breathing fishes, except snakeheads and lungfishes, seem to lack the ability of separating O_2-rich effluent blood of the air breathing organ during the passage of O_2 through the central cardiovascular system, although this has been inferred usually only from anatomical studies. The purpose of this paper is to review the current knowledge about the air breathing organs in selected species of air-breathing fishes (eel gobies, mudskippers, swamp eels, snakeheads and lungfishes).

Keywords: - ***Monopterus cuchia,*** **estuarine, hypopharynx**, **buccopharynx**

INTRODUCTION

Most of the fishes obtain oxygen from the water by using their efficient organs like gills but sometimes the low availability limits their oxygen intake. To cope with the hypoxic oxygen these fishes move to the water surface for the intake of oxygen.

From many years the air breathing fishes have attracted the different biologist for illustrations and understanding the evolution of different air breathing organs and their transition from water to land.[1] Many of the air breathing fishes occur in freshwater, estuarine and marine habitats.[2-4] Estuarine and marine air-breathing fishes are group of highly amphibious species such as mudskippers and rockskippers).

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The mud eel species *Monopterus cuchia* belongs to the family Synbranchidae and Synbranchiformes.[5] It inhabitats in holes, crevices muddy banks of swamps, lakes etc and slow running rivers[6] and it is an obligatory air breathing fish. The present article is basically focused on the structure and morphology of air breathing organs in the species of mud eel.[7]

In the respiratory regions there are quite extensive non-respiratory "Lanes" which would, therefore, reduce the overall values for respiratory surface. In addition, however, the respiratory islets themselves are often folded and this, together with the marked curvature of individual papillae would increase the surface area relative to that obtained by simple projection. A detailed study of the structure of vascular papillae of the respiratory islets and morphometry of the air-breathing organs of *M. cuchia* was undertaken.

MATERIALS AND METHODS

Fifteen species of *Monopterus cuchia*, having the body weight 100-300g and having the length of 550-700mm, were kept in an aquarium at 25^0 in the Department of Zoology, Madhepura. Several specimens were anaesthetized in MS 222 (1g/500ml) and were fixed by using vascular perfusion. Compressed air was fed into the air sacs at a pressure of 20cm H2O and a cannula was inserted into the bulbus arteriosus and perfusion was initiated with oxygenated physiological saline (0.9% NaCl) containing 2500IU heparin/ litre.

The process of perfusion was gradually followed by addition of 2.5 % of glutaraldehyde solution in 0.03 molar potassium phosphate buffer. Now the samples of tissues were taken from the several air breathing organs such as air sac, gills, hypopharynx, buccopharynx. Tissue blocks were post-fixed in 1% osmium tetraoxide solution (pH7.4,350 millosm.), stained in a 0.05 molar uranyl acetate solution, dehydrated in a graded series of ethanols and finally embedded in epon for transmission electron microscopy (TEM). Respiratory organs of a few *M.cuchia* were fixed, following anaesthesia by installation of 2.5% potassium phosphate buffered glutaraldehyde solution (pH7.4,350 milliosm.) through the mouth. Afterwards for 3 hours the fish were kept submerged in the same glutaraldehyde solution. Afterwards tissue samples were taken from the same positions and processed for TEM and SEM as described above. For preservation of mucus lining of respiratory tracts the fish that were anesthetized were decapitated and the cranial portions of the head were sectioned in a sagittal plane. Tissue samples were then taken from the same positions and post fixed in OSO4 (see above) to which 0.7% RHT was added. Tissues samples were also taken and post fixed in 0.7% RHT. Gills and air sacs of these fishes were also fixed in 2.5% glutaraldehyde and post fixed in 1% OSO4 in phosphate buffer kept at pH 7.4.

RESULT

While conducting the study on air breathing organs of *Monopterus cuchia,* the following important features were obtained:-

Morphology

The suprapharyngeal pouches are specialized air sacs that developed into dorso lateral extensions extensions of the pharyngeal cavity on each side. Sagittal section of the head shows the extensions of the buccal cavity, buccopharynx, hypopharynx and the air sac. The respiratory surface separated from the oesophagus that extends upto the gullet in hypopharynx. The various distributions of the vascularized respiratory islets on the inner mucosal linings of the pharynx, hypopharynx, and air sacs are also found. The posterior part of the air sac does not contain any respiratory outlets. Its surface is covered by non-vascular epithelium with many chromatophores.

When the air breathing organs was observed under SEM (scanning electron microscopy) the following were observed:-

a) Buccopharynx:

The buccopharynx when observed under SEM it contained the respiratory islets that were separated from each other by deep furrows. Each of the respiratory islets was also separated by slit like clefts into various parts. Many small and large islets and membranes lined by the membrane were seen in the gill clefts. These gills are found between the consecutive branchial arches.

The respiratory islets when seen under SEM were found to be separated from each other by deep furrows. The respiratory islets were also divided by slit-like clefts into several parts. The gill clefts appeared as narrow apertures lined by membrane with many large and small islets.

b) Hypopharynx:

The hypopharynx which leads into the oesophagus is also respiratory in function. Where the delicate tissue barrier is peeled off, the inner structure of the vascular papillae is exposed. The cell of the vascular papillae of the respiratory islet is honeycomb in structure.

c) Gills:

The gills are fingerlike structure that usually found in fused form with the branchial arch or holes that represents the interlamellar spaces. The second branchial arch bears a few finger shaped filaments. Microridged epithelial cell covers the surface of the filaments. No differentiations of secondary lamellae which are being represented by protubearances are found. Slit like structures are found near the base of the gill filament

representing interlamellar spaces and the surface of the filament is covered by epithelial cells.

d) Air sac:

The dorso-posterior regions of the air sac are non-respiratory in nature and are covered by microridged epithelial cells. Chemoreceptor is also seen in air sac that is bounded by microridged cells. Micro air pockets have been detected in air sac that penetrates deep into the air sac wall.

DISCUSSION

The gills found in *M. cuchia* are very much atrophied bearing few figure like structure. Filaments and lamellae are fused together to form pores through which the passage of water occurs during the aquatic ventilation. Secondary lamellae represent a bud like structure that is internally lined by endothelial cells.[8] The air sac represents the posterodorsal extended pouches of pharynx. Between the old islets and buccopharynx new respiratory islets develops. The posterior one third of the sac that serves as reservoir for residual air is non vascular in nature. In fishes since the regular gill filaments and lamellae are lacking but due to islets present at inner surface of the buccopharynx, hypopharynx and gill arches helps in exchange of gases and also in aerial respiration.

In *M.cuchia* air breathing organs are quite remarkable that functions not only for exchange of gases in both air and water but also for passage of different food such as earthworm, insects, mollusks etc. behind the respiratory hypopharynx is found which is a small chamber and leads into the oesophagus. This fish has a highly vascularised respiratory islets and the respiratory membrane is entirely covered by mucous which anchored by epithelial surface.

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Studies on the coleopteran pests infesting certain vegetables in Madhepura district, Bihar

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Abstract- Various studies are going on to restrict the extent of pest manifestations. Many varieties like cabbage, cauliflower, kohlrabi, broccoli, kale, etc. are derived from wild species of cabbage, via hybridization techniques. *Phyllotreta crucifera* is a major destructor, creating spot holes, on leaves of cabbage & cauliflower. By performing few prevention measures, large scale destructions can be avoided. Monitoring, especially during the seedling phase can help create healthy variants. In the present study a survey has been done in 7 various parts of Madhepura district, during the months of February, March and April to sample coleopteran pests infesting certain vegetables.

Keywords- ***Phyllotreta crucifera,*** **manifestations, spot holes, vegetables, flea beetles.**

INTRODUCTION

Madhepura is blessed with geographical conditions that suit a wide variety of vegetables, fruits & medicinal and aromatic crops. Though crops like rice, wheat, maize, jute & oilseeds occupy major area under cultivation, vegetables like potato, tomato, cauliflower, brinjal, cabbage and sponge gourd add extra income to the farmers. Packed with nutrients, vitamins and minerals these form a cheap staple, not only for the people of Madhepura but also for other parts across India.

The investment of finances and time required to gain maximum profit by increasing the yield as well as quality. Plant hybrid technologies and irrigation method has acted as a boon in increasing profit. But as known, in every boon there is a hidden curse. Same goes with growing these vegetables. These vegetables become a feast for never ending insect infestations. If not taken care of, these infestations can cause a great deal of loss, hampering even the entire batch. Cabbage and cauliflower are generally called as cruciferous vegetables belong to the family Brassicaceae. In its uncultivated form, it is called wild cabbage and is native to coastal southern and western Europe.[1]

Cabbage is a leafy green red or white biennial plant multi layered vegetable. Due to its high level of nutrient requirements is prone to nutrient deficiencies, and many diseases like internal tip burn, necrotic spot, wire stem, black leg and black rot. In India diamond back moth has caused several losses.[2] Cauliflower's head is composed of a white inflorescence meristem. Among major pests affecting cauliflower are aphids, root maggots, cut worms, moths and flea beetles.[3]

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Cabbage is damaged by flea beetles that belong to order coleoptera. Flea beetles are very small brown to black beetles that may have some yellow markings on their wing covers. The eggs are generally laid at the base of the plants. The white brown headed larvae have three pairs of legs. Flea beetles can cause damage to seedlings and small plants. Larvae are found in the soil and attack roots. Adult leaf beetles feed on cabbage plants making small round pits in the cotyledons and leaves. [4-5]

Phyllotreta crucifera is a species of flea beetle in the family Chrysomelidae[6]. The young larvae feed on the epidermal leaf tissues between the upper and lower leaf surfaces. The older larvae feed on all plant parts whereas they prefer to feed around the buds and leaves. They create irregular shot holes and disfigure the bud so that the cabbage head will not develop properly.

Flea beetles live through the winters as adults in leaf litter and become active in early spring. Female either lays cluster of eggs in small holes or single egg in roots, soil or leaves. Larvae transform into pupa in the ground.[7-8] The adults feed on leaves, scraping away leaf tissues and making characteristic perforation. In high populations, cause severe infestations even kill young plants.

MATERIALS AND METHODS

Protection against flea beetles can be achieved by proper monitoring, especially during springs, when seedlings have emerged. To determine the extent of effect & its control a survey was done in Madhepura via random sampling 7 agricultural plots were taken, during the months of February, March & also April. The survey was restricted to early morning hours (between 6 am to 8 am).

The foliage is rendered unfit for consumption, by larva. Larva skeletonizes it. It firstly feeds on leaves & then on the flower (in case it is a cauliflower). When larvae are small, they make irregular shot holes. In case of numerous infestations, they eat up the entire leaf leaving only veins.

Also immature forms (eggs, larvae) & adults (beetles) were collected from the vegetables crop fields & taken to the lab for further study. *Phyllotera crucifera* or as commonly called the crucifer flea beetle were found to be the main cause of damage, done by the coleopteran family.

OBSERVATIONS

Phyllotreta crucifera though very harmful, still isn't the cause of major destruction. Diamond back moths (*Platella xylostella*) of the family lepidopteron still hanks first in cabbage & Cauliflower destruction.

CONCLUSION

Several simple methods may be adapted to prevent serious damage. At first by checking the presence of flea beetles by putting yellow sticky traps in the field. If on an average more than 5 beetles are found in each plant, proper treatment must be given to the seedlings. Secondly, keep the flea beetles out of the vegetables crop, by using row covers or others screenings. Using pesticides contain active ingredients such as pyrethrins, carbaryl, malathion, spinosad, etc.

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Role of air breathing organs in Indian gobies of estuarine habitat

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Abstract- *Brachygobius xanthozonus* is an eel goby that inhabits in estuarine water. They usually reside in brackish water where plenty of plants and rocks are present which provides them with ample places for hiding, unlike the sympatric amphibious mudskippers it does not emerge but hides in rocks. Although they do well in a community setting, they will thrive the best in the company of only their own species. Even though they live in the brackish water, the addition of freshwater will induce spawning. The female will lay the eggs in the substrate beneath a root or rock where the male will fertilize them. After fertilization, the male will guard the eggs for approximately four days until the fry become waterborne. The Bumblebee Goby requires a diet rich in small, live foods such as tubifex worms, bloodworms, brine shrimp and daphnia.

Keywords: ***Brachygobius xanthozonus*****, goby,tubifex,teleosts**

INTRODUCTION

Gobies are fishes of the Gobiidae family, inhabiting saltwater, brackish water, and freshwater of tropical and subtropical ocean. The family of gobies comprises 1875 species in 212 genera in which 36 gobies are amphibious and are referred to as mudskippers.[1] Usually gobies lives in water and like other teleosts they also rely on gills for gas exchange, some of them are capable of air breathing (air breathers) and are generally found coexisting with the amphibious mudskippers. In tidal ponds the occurrence of air-breathers are usually limited. The gill of teleost fishes is well suited for water breathing, but upon air exposure the lamellae coalesce, greatly reducing the functional exchange surface.[2,3]

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Some fishes have the ability to breathe air effectively. Air breathing among fishes is associated with modification of the gill and the development of accessory breathing organs (ABO).[4] Air-breathing fish can exchange respiratory gases using an air-breathing organ (ABO), such as the oral cavity of the integument, during environmental hypoxia. The gill and integument surface area, diffusion capacity, and their diffusion barrier thickness were measured. Our results show that although gill surface area is smaller than observed in other facultative air-breathing fish, but it has all features necessary to perform aquatic gas exchange. Additionally the integument of the palate has a short diffusion barrier thickness and a large calculated O_2 diffusion capacity suggesting that it functions as the ABO. Air-breathing fishes continue to attract the attention of biologists because of their illustrative value as models for understanding the evolution of vertebrate air breathing and the transition from water to land.[5] Estuarine and marine

air-breathing fishes, in general, are dominated by highly derived groups of amphibious species (e.g. mudskippers and rockskippers), which routinely emerge during low tide[6]. The main purpose of the present study was to determine if the eel goby, *Brachygobius xanthozonus*, breathes air under natural conditions in the mudflat burrows. We also investigated the air-breathing behavior in order to obtain insights into the aerial respiratory performance of the fish.

MATERIALS AND METHODS

Two specimens of *Brachygobius xanthozonus* (Bleeker, 1849) were collected from the local water bodies of Madhepura. The capture location was in Madhepura (25.9240° N, 86.7946° E). Samples were collected by a drift gill net operated from traditional fishing craft. The specimens were preserved in 5% formalin and brought to the laboratory for a detailed examination.

OBSERVATION

Air-breathing behavior was observed in a fish chamber. The length (40 cm) and diameter (5.2 cm) of the horizontal bottom section provided enough space for the fish to move freely inside the chamber. The vertical section (length, 25 cm) had a tapered opening (diameter, 3.6 cm) at the top where fish could access air. Two cameras were mounted on the top and side of the chamber for video observation and recording.

Air breathing in the burrow

Seven out of the 15 burrows contained *Brachygobius xanthozonus* (Bleeker, 1849) but in only five burrows did fish perform air breathing. In the other two burrows with fish, water level intermittently moved up and down, suggesting activity inside. It is possible that the fish were breathing in other openings of the burrow. Air breathing was characterized by rapid inspiration followed by variable periods of breath holding either in the water or at the surface before expiration

Air-breathing capability

The present study demonstrated the capability of *Brachygobius xanthozonus* (Bleeker, 1849) to directly utilize atmospheric air for respiration. After inspiration, some individuals immediately retreat into the water whereas others retreat into the water and return or remain at the surface until expiration. During breath holding, the fish stays motionless with its head floating at the surface. Three types of air-breathing behavior were similarly observed in the field burrows.

Like most brackish water and marine air-breathing fishes *Brachygobius xanthozonus* (Bleeker, 1849) does not possess any special organ for aerial gas exchange. The gills therefore can be inferred as the potential sites of O_2 absorption, as similarly suggested in more than 30 species of gobies fishes.

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Histological study on the olfactory bulb of adult male albino rat exposed to different concentration of manganese chloride ($MnCl_2$)

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Abstract:-Manganese is named for pyrolusite and other black minerals from the region of Magnesia in Greece, which also gave its name to magnesium and the iron ore magnetite. Manganese (II) ions function as cofactors for a large variety of enzymes with many functions. Manganese enzymes are particularly essential in detoxification of superoxide free radicals in organisms that must deal with elemental oxygen. Manganese is regarded as one of the important elements for both plants and animals but sometimes when taken in heavy amount it becomes toxic in nature. Especially through inhalation, it can cause manganism, a condition in mammals leading to neurological damage that is sometimes irreversible. Manganese (Mn) is one of the trace elements that is required for human metabolisms and activity of several enzymes, but it also could induce toxic effect when it's ingested in high amount.

Keywords: olfactory bulb, histology, male albino rat, manganese chloride

INTRODUCTION

Manganese is a naturally occurring element found in rock, soil, water, and food. In humans and animals, manganese is an essential nutrient that plays a role in bone mineralization, protein and energy metabolism, metabolic regulation, cellular protection from damaging free radical species, and formation of glycosaminoglycans.[1] Manganese acts as both a constituent of metalloenzymes and an enzyme activator. Enzymes that contain manganese include arginase, pyruvate carboxylase, and manganese-superoxide dismutase (MnSOD)[2-4]. Manganese, in its activating capacity, can bind either to a substrate (such as adenosine triphosphate, ATP), or to a protein directly, thereby causing conformational changes.[5] Manganese has been shown to activate numerous enzymes involved with either a catalytic or regulatory function (e.g., transferases, decarboxylases, hydrolases). Although manganese is an essential nutrient, exposure to high levels via inhalation or ingestion may cause some adverse health effects.

Entrance of the manganese into the body usually occurs through the pulmonary route and as the constituents of the Mn enters the brain it remains there for longer period due to it its inability to get rid of it. High level of Mn intake can also lead to irreversible brain disease and the symptoms may be similar to those of Parkinson's disease.

The olfactory bulb is one of the structures located in the limbic region of the brain. It is an oval mass that lies below the anterior end of the olfactory sulcus. The olfactory bulb is supported and protected by the cribriform

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plate of the ethmoid bone, which in mammals separates it from the olfactory epithelium, and which is perforated by olfactory nerve axons. The olfactory bulb consists of a collection of nerve cells that receives impulses from the olfactory nerves of the nasal mucosa and continues as the olfactory tract [15].

MATERIAL AND METHODS

For the study of effect of manganese on the different olfactory organs of albino rat 15 adult male albino rats weighing about 180-200 g each were taken from the local market. All of these mice were kept in ventilated cages maintained on a 12:12-h dark–light cycle and provided with balanced food and water. Each of these mice was divided into 3 groups having 5 mice in each of it.

Group I:-known as control group:- the mice present in this group were fed with 1 ml of sterile saline which was given orally by a gastric tube once daily for 4 weeks.

Group II:-the mice present in this group was given $MnCl_2$ (10 mg/kg body weight) which was dissolved in 1 ml of sterile saline given orally by a gastric tube once daily for 4 weeks.

Group III:-the animals present in this group were simultaneously given $MnCl_2$ at the same doses and duration as given in group II

After completing the division of animals into groups and conduction of the experiment the experimented animals were anesthetized using intraperitoneal injection of sodium pento-barbital. After this through the left of the heart perfusion was done with fixative containing 2.5% glutaraldehyde and 4% paraformaldehyde in 0.1M phosphate buffer at Ph.7.4. The brain was then immediately extracted and the olfactory bulb was dissected for the further study.

For light microscopy the specimens of mice were

- fixed in 10% neutral-buffered formalin,
- dehydrated in ethanol
- cleared, infiltrated and embedded in paraffin wax.
- the sections were then cut in thickness of 5 ìm and finally
- stained with hematoxylin and eosin

For electron microscopy the specimens of mice were

- immersed in 2.5% phosphate buffered glutaraldehyde solution for 2 h at 4°C.
- washed with phosphate
- fixed in 1% osmium tetroxide solution for 1 h.
- specimens were then dehydrated, cleared in acetone and embedded in epoxy resin.

RESULT AND DISCUSSION

When the specimen of mice were observed under light microscope in control the species present in control group showed normal laminae olfactory nerve layer, mitral cell layer, internal plexiform layer and granular cell layer (fig1).when the ultrasections of the olfactory bulb was done it showed normal mital nerve cells with quite elomgated perikaryon with euchromatic nucleus and abundant cytoplasm containing free ribosomes amd mitochondria. In group 2 $MnCl_2$ treated animals the olfactory bulb was found to be abnormally distributed with loss of normal laminar organization. The cytoplasm was darkly stained, shrunken, small and appeared distorted (fig 2). The granule nerve cells were found disorganized and the neuropil showed vascular spaces. The nerve cells showed prominent ultra structural alterations such as accumulation of lysosomal dense bodies.

At last the animals fed with simultaneous $MnCl_2$ mild structural changes were observed in nerve cells.

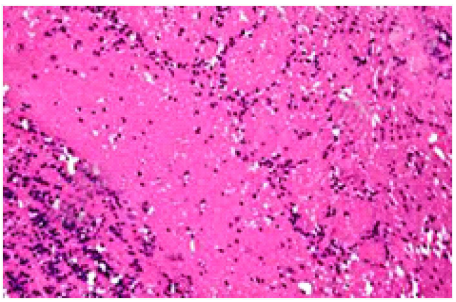


Fig : 1

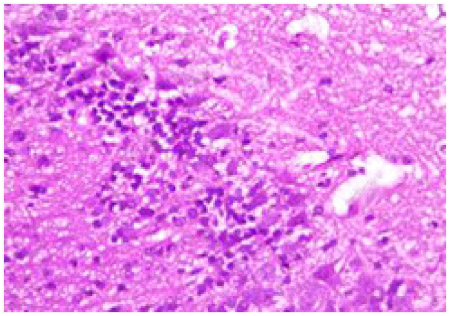


Fig: 2

The present work showed that administration of $MnCl_2$ induced structural changes in the olfactory bulb of adult albino rat in the form of loss of normal laminar organization, distortion of mitral and granule nerve cells, shrinkage of mitral cells with darkly stained cytoplasm and dark pyknotic peripheral nuclei as well as pericellular spaces and vacuolar spaces in the surrounding neuropil.

Several studies have shown that the olfactory bulb is one of the brain regions with the highest Mn concentration.[6-7]

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A survey of some coleopterans found in the cultivated vegetable lands of Madhepura, Bihar

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Abstract- Vegetables are an important element of the human diet and provides vital source of carbohydrates, vitamins, and minerals. Though it is grown throughout the world it faces extreme damage due to several pests. Some vegetables were grown in a specific area, whereas others have gained wide acceptance and have been transported to several continents where they are grown extensively. In the present study a survey have been conducted and different coleopterans such as blister beetles, flea beetles & Colorado potato beetles and their effects particularly on potato was observed. Two methods of trapping the insects were employed, namely hand capture for wingless insects as well as hand net for flying insects. Morphological identification of insects were also done using taxonomic keys, hand lens as well as light microscope for checking fine structures. One of the major drawbacks of such pests is the reduction in yield to such extent that it can incur huge losses. The entire potato plant can be affected by such beetles, which generally consume the leaves, roots & bore the tuber.

Keywords- Colorado potato beetle, tuber, blister beetles, flea beetles.

INTRODUCTION

Apart from staples like rice & wheat, potato a tuber, finds its place in almost every cuisine. In Madhepura potato cultivation covers almost 69000 hectares of area, when such large area is under cultivation any destruction caused by pests can adversely affect the economy of this place. The agricultural significance of pests on crop plant is the damage they cause which reduces the quality or quantity (or both) of yield. The colorado potato beetle is a major coleopteran that affects a variety of vegetables such as potato, tomato, brinjal etc. It is approximately 10 mm long with a bright yellow body along with five bold brown lines on its back. It has a strong association with the Solanaceae family. Both its larvae & adult relish on the potato plant causing 100% damage, if they infect before tuber formation. The most disappointing fact about this pest is that it is resistant to DDT & several other insecticides.

Potato (*Solanum tuberosum*) has great value in the Indian diet. It is a rich source of carbohydrates, (17%) cheap & easily available.[1] Insects can damage the crops by two ways either they damage the crops by biting and chewing their parts or by sucking the plant sap from different parts of plants.[2] Usually Orthoptera, Lepidoptera and mainly Coleoptera damages the crops by biting and chewing them & Hemiptera damages it by sucking the plant sap.[3] In this study we are discussing about the coleopteran infecting potato plants.

Though rare but the African black beetle (*Heteronychus arator*) also destroys the young plants of

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potatoes by chewing at or below ground level holes in tubers.[4] These beetles prefer clayey & loamy soil. It is generally 12-15 mm long and has shiny black colour and is oval in shape. *Epicauta funebris* Horn. is a margined blister beetle is also one of the coleopteran that affects potato a lot as it is native to North America but found in abundance here.[5]

MATERIALS AND METHODS

To determine the frequency of coleopteran species infesting the potato crop, random samples of infested plants were taken from different survey sites within Madhepura. The coleopteran were identified and segregated according to the species they belonged to. The coleopterans collected were kept under laboratory conditions providing them with proper food. Their life cycle was studies. Few farmers were interviewed and the impact of these pests on their financial conditions was studied. The sampling sites were usually between 10 km areas on the outskirt of North Madhepura. All the sampling areas were thoroughly studied. The parts that were affected were listed according to the preference of larvae and adults beetle.

Study area

The study was conducted on three irrigation sites within ten kilometers along the course of the Koshi River. The sizes of the farms studied ranged between 0.5 and 1.0 ha.

Sample collection

Each selected farm was divided into four quadrants and from each part four random plants were collected. After visualizing the insects on the plants they were captured and taken to laboratory for identification. Insects were collected with the help of swoop net for flying ones and forceps as well as hand picking for the larvae. Transparent containers and jars were carried for transference of insects to the laboratory. The plants were sampled at various stages of their growth such as seedling stage, vegetative stage, flowering stage, fruiting stage and finally harvesting stage.

CONCLUSION

The Colorado potato beetle was found to be the main reason of multiple crop failure in the area. Potato is a perfect crop that can be grown along with a variety of other crops (mixed farming) owing to its lower nutrient requirement. Though cheaper in rate compared to other vegetables its demand in market has never declined. Such coleopterans like Colorado beetle and the African black beetle has severe effects not only on potato cultivation but also on the other vegetables is grown along with potatoes.

The resistance against pesticides or insecticides which is the characteristic of such beetles further elevates the problem. The survey clarified the status of the potato crops damage and the financial lose of farmers which are usually caused by the outbreak of such coleopterans. They were advised to monitor their fields regularly specially during the initial stages of plant growth, to check any kind of pests attack. The result so obtained could be helpful for further research on control of insects pests in the affected areas and might be of great importance to both the farmers as well as the entire nation economically and for the purpose of food security.

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Assessment of water quality of Baya River, Samastipur, Bihar, India

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Abstract- Baya River is one of the most important river in Samastipur District of Bihar. This river enters through Mohiuddin Nagar in Samastipur district and finally meets into the Ganga at Teghra in Begusarai district of Bihar. During investigation physico-chemical characteristics of water of Baya River at three ghats - Barhauna Ghat, Dhobia Ghat and Teghra Ghat have been studied. Study revealed that quality of water changed due to seasonal variation. Physico-chemical characteristics of water are not static through whole year. The quality of water depends on Temperature, PH, DO, BOD, COD, TDS etc. All these parameters are variable according to seasonal variation. Variable affects the biological components.

Keywords- Baya River, Samastipur, Bihar, Seasonal Variation, Biotic Components.

INTRODUCTION

Water is one of the most important necessities of living organisms. Availability of safe potable water is still a problem of majority of people, due to steady increase in population, urbanization, industrialization, deforestation etc. Water resources have been adversely affected both qualitatively and quantitatively leading to as one of the major problems in developing country like India.

Baya River one of the rivers of Bihar of Samastipur district running between Mohiuddin Nagar to Teghraf. It is prominent for water supply, drinking, irrigation batting cattle, washing clothes & utensils etc. for the people of that area.

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MATERIALS & METHODS

Morphometry of Baya River from origin to mixing zone of Ganga River and formation of Bagulia wet land/ chor study area is divided into three sampling stations i.e. Barhauna ghat, Dhobia ghat and Teghra ghat. These sampling sites are westward to eastward respectively. The sampling period is also divided into three seasons- summer (March to June), Rainy (July to October) and winter (November to February).

The selected physico-chemical parameters applied for assessment of water quality i.e. temperature, turbidity, electrical conductivity, transparency, Total Dissolved Solids (TDS), pH, Dissolved Oxygen (DO), Biological Oxygen Demand (BOD), Chemical Oxygen demand (COD), free CO_2, total hardness, total alkalinity and chloride (Cl^-). Selected physico-chemical parameters are analyzed with the help of standard analytical methods.[1]

Temperature is measured by thermometer, pH by pH meter, total hardness by EDTA titration, TDS by TDS meter, transparency by Secchi disk and some other parameters are analyzed by different chemical procedures.

RESULTS AND DISCUSSION

Monthly observations are converted into seasonal by averaged i.e. observation in March, April, May and June are averaged and converted into 'Summer'.

Table -1 Physico-chemical characteristics of Water of Baya River Samastipur Bihar (March 2016- February 2017).

Sl. No.	Physico- Chemical Parameters	Summer			Rainy			Winter		
		S_1	S_2	S_3	S_1	S_2	S_3	S_1	S_2	S_3
1	Water Temp. (^oC)	32.1	32.3	32.9	29.1	29.3	29.7	19.6	19.4	19.9
2	Turbidity (NTU)	8.84	9.10	9.5	9.65	9.60	10.50	8.10	8.50	8.00
3	Electrical Conductivity (μ/s)	411.00	403.00	462.00	503.00	497.00	518.00	507.00	519.00	523.00
4	Transparency (cm)	18.5	13.6	17.8	19.00	15.2	13.5	26.00	19.00	18.50
5	Total Dissolved Solids (TDS)/ mg/L	275.7	265.8	281.9	213.4	217.6	210.5	305.2	301.6	351.5
6	pH	6.87	6.60	6.50	7.50	8.10	7.70	7.60	8.0	8.2
7	DO (mg/L)	5.50	5.68	5.20	6.35	6.65	6.10	7.40	7.70	7.2
8	BOD (mg/L)	8.50	7.80	9.0	8.20	8.50	8.00	7.80	6.80	8.2
9	COD (mg/L)	18.40	20.30	18.1	15.7	19.9	16.2	13.5	15.3	14.8
10	Free CO_2 (mg/L)	8.70	8.35	8.15	7.50	6.90	7.20	6.50	7.10	6.75
11	Total Hardneu (mg/L)	172.00	163.00	161.00	180.00	176.00	145.00	190.00	175.00	165.00
12	Total alkalinity (mg/L)	124.00	121.00	126.00	132.00	13.00	134.00	142.00	131.00	136.00
13	Chloride (Cl^-)	19.50	18.20	17.80	13.40	12.60	13.50	7.2	6.85	6.10

CONCLUSION

The study of Physico-chemical parameters of water of Baya River revealed that important parameters are favourable range of aqua-culture particularly for fish as well as also for phytoplanktons and other aquatic plants. Baya River may give high productivity of biomass. It may change life of peripheral people, if they form better association with this natural water body.

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Starch as precursor of bioethanol production through rdr baker's yeast, *Saccharomyces cerevisiae*

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Abstract- In bioethanol production saccharification of biomass to fermentable sugar is an important constraint because the enzyme production costs more and the complications associated with the removal of hearse acid, alkali and salts formed after neutralization. This led to the search for low cost enzyme and its combination with dilute acid to enhance biomass hydrolysis. In this study, the biomass of microalgae was hydrolysed using amylase and cellulase enzymes produced by solid state and submerged fermentation processes. Dilute tetraoxosulphate (VI) acid, crude enzyme complex and a combination of both were used to study saccharification of algal biomass. The highest yield of reducing sugar of 0.63 mg/ml was obtained with the co-combination hydrolysis of acid and enzyme, followed by acid hydrolysis (0.41 mg/ml) while the least was found with enzyme hydrolysis (0.36 mg/ml).

Keywords- Yeast, optimization, ethanol, starch, fermentation, nuclear petite

INTRODUCTION

Sources of alternative energy have attracted researcher's interests to focus on biomass as a cost efficient and effective feedstock for bioethanol production and can serve as a suitable transportation fuel replacing the limited crude oil[1]. The production of alcoholic beverages from fermentable carbon sources by yeast is the oldest and most economically important of all biotechnologies. Yeast plays a vital role in the production of all alcoholic beverages and the selection of suitable yeast strains is essential not only to maximise alcohol yield, but also to maintain beverage sensory quality. *Saccharomyces cerevisiae* strain of yeast dominates in the production of alcoholic beverages worldwide, and the particular strains of this species employed in fermentation exert a profound influence on the aroma and flavour characteristics of different beverages.[2,3]

Pure cultures of selected strains of *S. cerevisiae* are usually used for large-scale beverage fermentations, as in brewing, winemaking and distilled spirit production. Ethanol is very old chemical and has been made since old times due to the sugar fermentation.[4] Ethanol is produced from any fodder crop which contains simple sugar in abundance or their polymers.[5] The polymers like starch and cellulose are broken down into simple sugars through chemical hydrolysis or enzymatic hydrolysis (saccharification), and then converted by fermentation process to ethanol and carbon dioxide[6], in saccharification, process starch is

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converted into simple sugar (monosaccharide) using microorganism or enzymes such as glucoamylase and α-amylase.[7]

A two-step process includes the bioconversion of starch into ethanol. The first step is saccharification, where starch is converted into sugar using an amylolytic microorganism or enzymes such as glucoamylase and amylase. The second step is fermentation, where sugar is converted into ethanol using *Saccharomyces cerevisiae* .[8]

There are over 150 amylolytic yeast species, their industrial use is limited because of their low ethanol tolerance.[9] So, most of the research is focused on the development of genetically engineered amylolytic strains of *S. cerevisiae*, and in these strains, heterologous genes encoding amylase and glucoamylase from various organisms have been expressed and their products excreted.[10]

Several studies have pointed out the potential of utilizing respiration-deficient nuclear petites for the commercial production of ethanol.[11]

Despite the vast number of strategies adopted for the construction of amylolytic strains of *S. cerevisiae*, there have been no reports about the application of respiration-deficient nuclear petites for the production of ethanol from starch.

In time the development of a respiration-deficient nuclear petite *S. cerevisiae* strain excreting a bifunctional fusion protein that contains both *Bacillus subtilis* amylase and *Aspergillus awamori* glucoamylase activities can be seen. Since the nuclear petite strain generates its energy requirements by fermentation only, lower biomass concentrations and biomass yields from starch were observed with the NPB-G strain as expected.

This study constitutes a first step that provides the basis for utilizing nuclear petite mutants for the single-step bioconversion of starch into ethanol.

MATERIALS AND METHODS

Yeast strains

The 100% respiratory-deficient nuclear petite FY 23pet191 of the parental haploid S. cerevisiae FY23 strain (MATa, ura3-52, trp63, leu21) was generated using polymerase-chain-reaction (PCR)-mediated disruption of the PET191 gene, which encodes a protein required for the assembly of the polypeptide subunits that constitute the active cytochrome coxidase holoenzyme.[12]

Isolation *S. cerevisiae*

The *S. cerevisiae* NPB-G strain was generated by transforming (17) the FY23 pet191 strain with the pPB-G plasmid (5), which contains the *B. subtilis* amylase and the *A*. The *S. cerevisiae* WTPB-G strain was generated by transforming the parental haploid FY23 strain with the pPB-G plasmid and selecting the transformants on yeast minimal medium-agar plates without G418 sulfate.

RESULT AND DISCUSSION

By the formation of starch hydrolysis zones on plates stained with iodine Amylolytic activity was detected. Ethanol fermentation was carried out in an orbital shaker (Innova model no. 4340) at 30°C under aerobic conditions with agitation at 180.rpm. by using shake flask culture NPB-G mutant and WTPB-G parent cells were grown containing YEP-S medium (0.5% yeast extract, 1% peptone, 0.01% uracil, 0.01% tryptophan, 5% starch, 0.4% glucose). Finally the concentration ethanol, glucose and residual plasmid were determined. The highest yield of reducing sugar of 0.63 mg/ml was obtained with the co-combination hydrolysis of acid and enzyme, followed by acid hydrolysis (0.41 mg/ml) while the least was found with enzyme hydrolysis (0.36 mg/ml).

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Inter-relationship of fish culture, abiotic and biotic factor of river Sone

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Abstract- Animal and plants associated with water have from times immemorial signified life, fertility and abundance. The fishes, one of the first forms of evolutionarily higher life to appear in water is among the earliest vertebrates. It was therefore regarded as one of the "nighis" of water. The fish community of the Sone is very similar to that of the Alaknanda, although a few lowland species begin to appear, such as *Mastacembelus* and *Channa* so that the final species total is a little higher at 54. Again, however, numerically the snow trout and mahseers would tend to dominate the fishery. Fishes are considered as one of the important source of animal protein in all over the world and very popular in all economic classes. The equally in some water will provided embankment population have given most important nutritional food to the people of that were as well as to other parts of river that of the Ganga". Biological monitoring is a main aspect of today's water quality assessment of river and lakes. Both abiotic and biotic factors can have profound effects on population size in Sone. Zooplankton biomass and biodiversity were examined in a floodplain system of high hydrological dynamic. Temporal patterns were used to assess the significance of water retention time for the regulation of zooplankton population dynamics. The sampling period covered a board range of hydrological conditions ranging from lotic to lentic situations. Retention time could be identified as the driving variable governing total zooplankton biomass following a sigmoidal relationship. It also proofed to have different effects on rotifers and crustaceans. Rotifer biomass followed a polynomial 2nd order relationship showing a maximum at medium retention time while crustacean biomass was positively linear relation. The present work has presents inter-relationship of fish culture, abiotic and biotic factors of river Sone.

Keywords-

INTRODUCTION

Animal associated with water have from times immemorial signified life, fertility and abundance. The fishes, one of the first forms of evolutionarily higher life to appear in water is among the earliest vertebrates. Fishes are considered as one of the important source of animal protein in all over the world and very popular in all economic classes. The equally in some water will provided embankment population have given most important nutritional food to the people of that were as well as to other parts of river that of the Ganga". Biological monitoring is a main aspect of today's water quality assessment of river and lakes.

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The Sone River at 784 kilometres (487 miles) long is one of the largest rivers of India. Its chief tributaries are the Rihand and the Koel. The Sone has a steep gradient (35-55 cm per km) with quick run-off and Ephemeral

regimes, becoming a roaring river with the rain waters in the catchments area but turning quickly into a fordable stream. The Sone being wide & shallow leaves disconnected pools of water in the remaining part of the year. The channel of the Sone is very wide (about 5 km at Dehri) but the flood plain in narrow, only 3 to 5 wide

The division stretches between 24-0' to 24-32' North latitude and 83-32' to 83-52' East longitude. This forest division is bounded on the North by Sone River, on the East by North Koel, on the South by Garhwa South Division and at the west by Mirzapur district of U.P. There are 163 villages in Nagar register of Garhwa North Division. The fish community of the Sone is very similar to that of the Alaknanda, although a few lowland species begin to appear, such as *Mastacembelus* and *Channa* so that the final species total is a little higher at 54. Again, however, numerically the snow trout and mahseers would tend to dominate the fishery.

Eight different methods of fishing were documented and 23 species of recorded. The most common were the snow trouts, Schizothorax (3 species) contributing between 61 and 74 percent of the catch over a year. The sites ranged from 1855 masl to tehri at 770 masl with seasonal temperatures kept between 10.2^0 and 19.4^0 C by snowmelt near the source.

Notable contribution are those of Ahsan S N (1966), Baggerman, B. (1957), Ball, J. N. (1960), Cope, O.B. (1965), Herbert, D.W.M. and Shurben, D.S. (1964), Hoar, W.S. (1969), Idler, D.R. and Clemens, W.A. (1959), Khanna, S. S. and Sanwal, R. (1971), Loft, B. (1975), Robertson (1958), Sinha. V.R.P. and Srivastava, N.C. (1991) and Tan. C.E. Chong. H.K.S. and Moulton.T. (1972).

MATERIALS AND METHODS

The primary productivity in the River Sone was estimated at two sampling sites, during the present investigation from July 2005 to June 2007 on monthly basis. The sampling sites were Rihand (S1) and Koel (S2) representing the middle and lower section of the river, respectively. Initially , the classic method of Gaarder and Gran (1927) was adopted for in situation measurement of the rate of primary production for phytoplankton in the river (Table 4.2.1) the experiment could not materialize and was, therefore, terminated .

The primary production by Sone, was estimated by three standard methods, as follow:

1. Chamber Method: Of the several methods methods available for primary productivity measurements in streams (Bott *et al.*1978), polystyrene Chamber method of Rodgers *et al.* (1978) was selection as the most appropriate. The substrate bearing periphytons were incubated in a 1.93 liters molded polystyrene chamber for 4- hour incubation period (0800-1200 hr). A submersible pump powered by an attached battery resistor allowed variable flow I the chamber. Black plastic tape was used to cover dark (opaque) chamber.

Direct measurement of undisturbed periphyton in chamber in which the water is kept moving offers an accurate determination of primary productivity (Hynes, 1970).

The dissolved oxygen technique was applied in conjunction with light and dark incubation chamber for estimation of primary productivity.

Calculation for the oxygen produced over a given time period were made after running Winkler's test on each of the samples. Calculations for respiration, gross primary productivity were made as follow;

Respiration (R):

OC= OS- OE(black tamped chamber).

Gross Primary Productivity pg:

TOP=OE(chamber not covered)-OE(blacks taped chamber).

Net Primary Productivity (pn)= pg - R

OC= Oxygen Consumed; OC= Oxygen at Start

OE= Oxygen at End; TOP= Total Oxygen Produced

pg= Gross Primary Productivity;

pn= Net Primary Productivity

The values obtained in mg 1^{-1} oxygen were converted to milligrammes of carbon per square meter (mg C m^{-2}) multiplying the values by 375.36 (Strickland and Parssons, 1960). These were further expressed as grammes of carbon per square meter (g Cm^{-2}). The values in mg C m^{-2} can be converted to grams of dry weight (g dry wt. biomass m^{-2} surface) by multiplying the milligrammes of carbon by 2 and dividing by 1000. The value of dry wt. was converted to calories of energy multiply by 5.5.

Daily rated were obtained by multiplying the average hourly rate with the sunshine values. The date for the daily average sunshine hours were obtained from the field station

observatory of Central Soil & Water Conservation Research & Training Institute (ICAR), Dehra Dun (approximately. 60.0 km from the study area). Table 5.1 depicts the average daily average daily sunshine over the area for the study period. Monthly and annual rates were also estimated accordingly

2. Chlorophyll Estimtion: The chlorophyll content of attached communities is a useful index of the biomass of the phyto-periphyton. The pigments are extracted with aqueous and the optical of the extract is determined with a spectrophotometer.

The algal samples were collected from known area of 0.1x 0.1 m^2 and stored frozen and kept I dark (Grzenda and Brehmer, 1960) and brought to the laboratory.

3. Efficiency of light Energy Conversion : The overall efficiency under variety of ecological condition has been studied and it is found that the solar energy conversion seldom exceeds 10 or 12% of the available light energy (Oswald and Gotaas, 1957). Since the probable values for the amounts of visible light energy can be obtained from Kimaball (1935) or Smithsonian Meteorological Table (1951) or from a nearby observatory as in the present case for corresponding north latitudes of the Indian towns in the sampling sites are sites are sites are situated (Table 5.2), and photosynthetic efficiencies may be assumed, the quality of oxygen that will be produced for a given efficiency may be calculated from the equation:

RESULTS AND DISCUSSION

Both abiotic and biotic factors can have profound effects on population size in sone. Zooplankton biomass and biodiversity were examined in a floodplain system of high hydrological dynamic. Temporal patterns were used to assess the significance of water retention time for the regulation of zooplankton population dynamics. The sampling period covered a board range of hydrological conditions ranging from lotic to lentic situations.

Retention time could be identified as the driving variable governing total zooplankton biomass following a sigmoidal relationship.

It also proofed to have different effects on rotifers and crustaceans. Rotifer biomass followed a polynomial 2nd order relationship showing a maximum at medium retention time while crustacean biomass was positively linear relation.

Biodiversity patterns of the two taxonomic groups differed significantly. Rotifer species number declined linearly along the hydrological gradient while crustacean species number increased exponentially, showing a maximum at periods of long lotic conditions.

We exaplained this "hydrological succession" by alternation of flood caused mortality (e.g. washing out effects) and taxa specific potential of reproduction along the hydrological gradient. Perioda of long retention, favoured crustacean development, most likely due to competitive advantages, where rotifer biomass and species number declined.

General multivariate models have appeal when analyzing ecological system because they have the potential to shed light on the relative importance of multifaceted factors. We hypothesized that plant community composition is a function of the interactions among three general constructs : abiotic and biotic factors and disturbance history. We sampled vegetation and soils on 75 randomly located 0.05 ha plots across a board soil gradient within an 110,000 ha ponderosa pine forest landscape in northern Arizona to evaluate this hypothesis.

The variation in abundances of 271 under story plant species was reduced to two orthogonal axes using non-metric multidimensional scaling. We evaluated the relationship between ordination results and environmental conditions using two types of structural equation models. 'Specific' models provided a details perspective of hypothesized relationship among observed variables. A 'general' model was evaluated to address the general hypothesis stated above. The general model employed composite variables, which specify the combined effects of multiple factors on a response, in order to evaluate the relative importance of the three theoretical constructs.

The results of the final specific model suggested that organic matter and nitrogen explained unique variation in composition, but the 'total effects' of organic matter were not significant due to offsetting direct and indirect pathways. Soil texture and Ph also constrained where many plant species occurred. Elevation was a strong determinant of species distributions, and its effects could be propagated through a network of processes involving soil mineral and organics properties.

The results of the general model provided new insight into the underlying structure of independent plat community

gradients in ponderosa pine forests. The model suggested that abiotic factors generated one gradient in community composition, whereas biotic factors, and to a lesser degree, disturbance history, generated a second independent gradient in community composition. Future evaluations of this composition model across a diversity of ecosystems could determine the generality and utility of this multivariate hypothesis. Water quality change is discussed in details, with emphasis on acidity formed from exposed pyrite material and on increase in dissolved soils. Preventive and treatment measures are recommended.

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Histopathological study on the effect of cyclophosphamide on the spleen of adult albino rat

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Abstract- Cyclophosphamide is a widely used chemotherapeutic drug which has a range of adverse effects on normal cells also. The present study was aimed to investigate the effect of cyclophosphamide on the spleen of adult male albino rat. Rats were divided into two groups of 6 individual each. Group I (Vehicle Control) rats were treated with distilled water, while group II rats were administered with a single dose of cyclophosphamide (200 mg/kg b.wt) intraperitoneally. After 6 days, rats were evaluated for spleen weight and histology of spleen. Cyclophosphamide administration significantly reduced the absolute and relative weight of spleen as compared to vehicle control group rats ($p<0.05$). Histopathological examination of spleen showed that cyclophosphamide induced marked alterations manifested by loss of clear distinction between red and white pulp, highly expanded red pulp, reduction in number and cellularity of white pulp, enlargement of central arteriole and presence of megakaryocytes in red pulp region. These findings suggested that cyclophosphamide administration may have toxic effects on the spleen resulting in immunosuppression.

Keywords- Cyclophosphamide, Spleen, Relative weight, White Pulp, Megakaryocytes, Immunosuppression.

INTRODUCTION

Cyclophosphamide is the most widely used alkylating agent in chemotherapy with a high therapeutic index and broad spectrum of activity against a variety of cancers since the late 1950's.[1] It is widely used in the treatment of several human cancers including solid tumours, lymphomas and leukaemia, as well as many non-neoplastic diseases, such as systemic lupus erythematosus and rheumatoid arthritis.[2] The anti-tumour effect of cyclophosphamide is in proportion to the dose administered, which often results in immunosuppression and cytotoxic effects.[3] As with all alkylating agents, rapidly proliferating cells are most sensitive to cyclophosphamide.[4] Being the largest secondary lymphoid organ, spleen is also greatly involved in host response against blood borne antigens.[5] An individual will be prone to infection when the spleen is damaged or removed. This fact is important to reflect that spleen acts as a suitable parameter for monitoring immune system function.[6] The current study was aimed to investigate the toxic effects of cyclophosphamide administration on weight and histopathology of spleen.

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MATERIALS AND METHODS

Male albino rats (150-160 g) were obtained from local animal supplier in Ranchi. They were housed in plastic cages and maintained under standardized environmental conditions (22-24° temperature and 12 hr light/dark cycle) with *ad libitum* access to standard pellet diet and tap water. Rats were acclimatized for 1 week before using in the experiment. Cyclophosphamide (Endoxan-N; 200 mg) was obtained from Azad Pharma, Ranchi. Rats were weighed before dividing into two groups (N=6). Group I served as vehicle control and were given distilled water. In group II (CYP-treated group), rats were administrated with a single dose of cyclophosphamide (200 mg/kg body weight) intraperitoneally. For final administration, CYP was dissolved in distilled water (1.6 ml). After administration, rats were maintained for six days with provision of food and water *ad libitum*. On 7th day, rats from both groups were sacrificed and spleens were collected. Body weight of rats was measured before sacrifice. Spleens of rats from both the groups were weighed and absolute and relative weights (% of body weight) were calculated. Small portions of spleen were cut and fixed in Bouin's fixative. Sections of about 3 µm were stained with Haematoxylin and Eosin (H & E) for histological evaluation. Photomicrography was done by using image analyser (Olympus CH20i). Obtained Data were statistically analysed by using Student's t-test and $p<0.05$ were considered significant. The values were expressed as mean ± SD (Standard Deviation).

RESULTS

Table 1: Effect of single dose administration of cyclophosphamide (200 mg/kg b.wt) on absolute and relative weight of spleen.

Sl. No.	Parameters	Group I (Vehicle Control) (Distilled Water)	Group II (CYP) (200 mg/kg b.wt)	% change Increase (+) / Decrease (-)
1.	Body Weight (g)	153.33 ± 11.54	126.67 ± 5.77*	(-) 17.39
2.	Absolute Weight of Spleen (g)	0.373 ± 0.095	0.194 ± 0.006*	(-) 47.99
3.	Relative Weight of Spleen (% of body weight)	0.253 ± 0.055	0.153 ± 0.004*	(-) 39.53

*p<0.05 Group II rats were statistically compared with Group I rats.

Table 1 showed the effect of cyclophosphamide administration on absolute and relative weight of spleen in male albino rats. In vehicle control group rats (Group I), the absolute and relative weight of spleen were 0.373 ± 0.095 g and 0.253 ± 0.055 % of body weight respectively. In group II, rats were intraperitoneally administered with single dose of cyclophosphamide (200 mg/kg body weight). After 6 days of administration, the absolute weight and relative weight of spleen was found to be 0.194 ± 0.006 g and 0.153 ± 0.004 % of body weight respectively. Cyclophosphamide administration significantly reduced the absolute weight of spleen by 47.99 % and relative weight of spleen by 39.53 % ($p<0.05$).

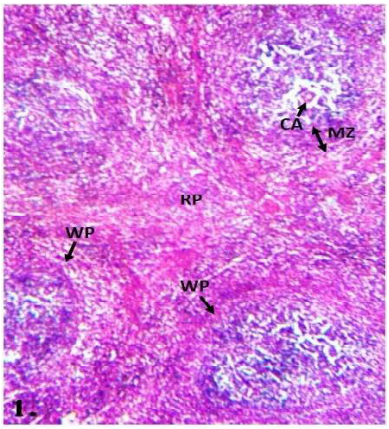


Fig 1. Photomicrograph of T.S. of spleen of vehicle control (Group I) albino rat showing distinct white pulp (**WP**) and red pulp (**RP**), well-defined marginal zone (**MZ**) and central arteriole (**CA**) (H & E; X100).

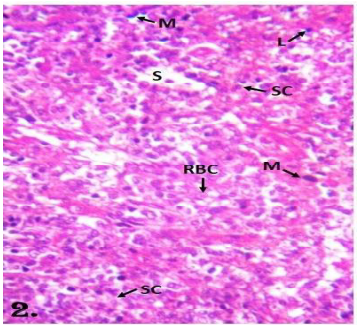


Fig 2. Photomicrograph of T.S. of spleen of vehicle control (Group I) albino rat showing normal architecture of red pulp with splenic cords (**SC**), blood sinusoids (**S**), red blood cells (**RBC**), macrophage (**M**) and lymphocytes (**L**) (H & E, X400).

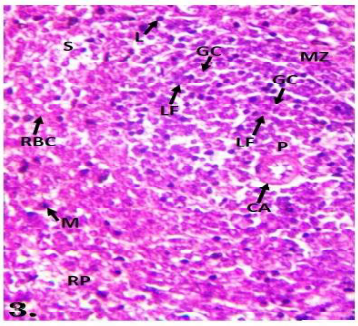


Fig 3. Photomicrograph of T.S. of spleen of vehicle control (Group I) albino rat showing normal architecture of white pulp with central arteriole (**CA**), PALS region (**P**), Lymphoid Follicles (**LF**), Germinal Centre (**GC**), lymphocytes (**L**) and well-defined marginal zone (**MZ**) (H & E; X400).

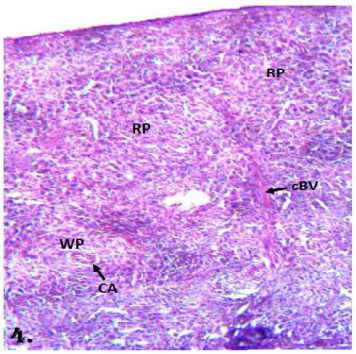


Fig 4. Photomicrograph of T.S. of spleen of cyclophosphamide-treated (Group II) rat showing indistinct differentiation of red (**RP**) and white pulp (**WP**), reduced number and size of white pulp and congestion in blood vessel (**cBV**) (H & E; X100).

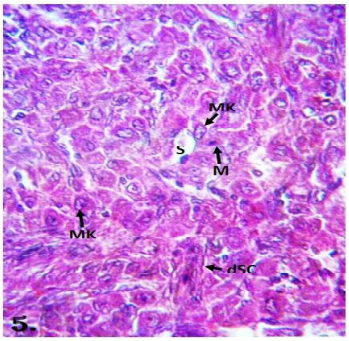


Fig 5. Photomicrograph of T.S. of spleen of cyclophosphamide-treated (Group II) rat showing large number of megakaryocytes (**MK**) but degeneration of splenic cord (**dSC**) and red pulp (H & E; X400).

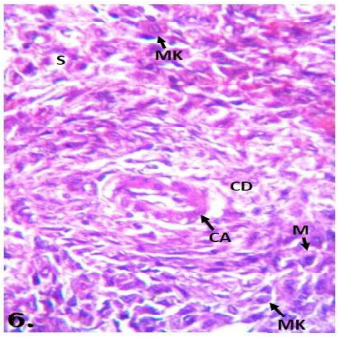


Fig 6. Photomicrograph of T.S. of spleen of cyclophosphamide-treated (Group II) rat showing enlargement and thickening of central arteriole (**CA**), areas of cellular degeneration (**CD**), loss of marginal zone and reduced cellularity in white pulp (H & E; X400).

Fig 1. showed the photomicrograph of T.S. of spleen of vehicle control albino rat. It showed normal architecture of spleen having distinct red pulp and white pulp. A well-defined, broad marginal zone was present between red pulp and white pulp; separating them into two distinct functional zones. In the spleen sections of vehicle control group rats, red pulp was composed of branching and anastomosing splenic cords and blood sinusoids in between them. The cells forming the splenic cords included lymphocytes, macrophages and other blood elements such as, red blood cells (Fig 2). H & E stained sections of vehicle control group spleen displayed multiple white pulp regions, located around an eccentric central arteriole. White pulp region was composed of periarteriolar lymphatic sheath (PALS), lymphoid follicles and marginal zone. The

lymphoid follicles consisted of a large number of lymphocytes with darkly stained nuclei. Germinal centres were also seen within some lymphoid follicles. The marginal zone was made up of few small and medium-sized lymphocytes and macrophages (Fig 3). Many small and medium-sized lymphocytes with darkly stained nuclei, macrophages and other blood elements were found scattered in the various regions of splenic pulps in the vehicle control group rat (Fig 2 & Fig 3). Rats in group II were administrated with single dose of cyclophosphamide (200 mg/kg body weight; i.p).

After six days of treatment, its effects on histopathology of spleen were studied. The spleen of cyclophosphamide-treated rat showed marked alterations within their white and red pulp. There was indistinct differentiation of red and white pulp due to loss of well-defined marginal zone. There was a visible reduction in the number and size of white pulp. Highly expanded red pulp with congested blood vessel was also observed (Fig 4). Large numbers of megakaryocytes were observed in the red pulp region of spleen of cyclophosphamide treated rat (Fig 5).

After treatment with cyclophosphamide, marked degeneration of splenic cords and loss of blood elements; such as lymphocytes, macrophage and red blood cells, were also seen in the red pulp region (Fig 5). In cyclophosphamide-treated groups, enlargement and thickening of central arteriole was observed. There was a hyaline degeneration of the wall of arteriole. There was a loss of cellularity in the PALS region. No lymphoid follicles and germinal centres were observed in the white pulp region after treatment with cyclophosphamide (Fig 6).

DISCUSSION

The present study showed that single dose administration of cyclophosphamide (200 mg/kg body weight; i.p) induced several pathological changes in the spleen of albino rat. Cyclophosphamide administration significantly reduced the absolute and relative weight of spleen as compared to vehicle control group rats ($p<0.05$).

In accordance with the present study, Kim[7] also reported a significant decrease in absolute and relative weight of spleen after intraperitoneal administration of cyclophosphamide at the dose of 150 mg/kg and 110 mg/kg. In the same context, Hou[8] also observed a significant decrease in relative weight of spleen after daily oral administration of 10 mg/kg cyclophosphamide for 30 days in wistar rats. These pathological changes in spleen were also confirmed by studying histology of spleen sections in the control and experimental group rats. In group II (Cyp-treated) rats, there was no distinct differentiation of red and white pulp due to loss of well-defined marginal zone (Fig 4).

Corroborating the present findings, Banerjee[9] reported that after administration of cisplatin, margin between white pulp and red pulp was disorganised indicating the toxic effect of the drug on the tissue architecture. After six days of intraperitoneal administration of cyclophosphamide, there was a decrease in number of white pulps in the spleen (Fig 4).

Similar effects were reported by Yoon[10] when 150 and 110 mg/kg of cyclophosphamide was used to induce immunosuppression in mice. They explained that the decrease in spleen weight observed in cyclophosphamide treated mice can be considered to be the result of the atrophic changes detected on histopathological examination.

The depletion of lymphocyte population in the red and white pulp and loss of germinal centre in the white pulp region was consistent with the previous observations that revealed decreased cellularity in the spleen of adult wistar rats given cyclophosphamide at a dose of 10 mg/kg/day for 4 weeks[11].

They also reported the congestion in blood vessels and thickening of central arteriole in agreement with the present study (Fig 4 & Fig 6). Corroborating present investigation regarding changes in red and white pulp region after treatment (Fig 5), Nabila[12] reported a highly expanded red pulps accompanied by reduced white pulps with multi small degenerated areas; as well as presence of megakaryocytes in the red pulp after treatment with colchicine.

The abundance of megakaryocytes in the red pulp region can be correlated with the red pulp increase and white pulp decrease as a result of the breakdown of normal splenic structure.[13] In a study made by Kabat-Koperska[14], a visible reduction in the white pulp and an increase in red pulp area were observed in spleen of rats while treating with combinations of immunosuppressants during pregnancy.

CONCLUSION

In conclusion, single dose administration of cyclophosphamide resulted in immunosuppression as evidenced by significant decrease in absolute and relative weight of spleen. Severe atrophic changes were also observed in the spleen sections of cyclophosphamide treated rats; such as, loss of marginal zone, reduced cellularity, congestion and presence of megakaryocytes. Therefore, a strict regulation must be taken into account before prescribing cyclophosphamide treatment.

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Studies on the energy flow and physico-chemical parameters in some ponds of Madhepura, Bihar

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Abstract- Energy flow is vital for the establishment of any living form on this planet. The energy flow is always unidirectional, usually following the 10% low. The 10 % low states that only 10% of the energy is that is available in any trophic level is transferred to the next trophic level. The energy flow is usually continuous and requires the involvement of almost all biotic & abiotic components. Earth is massive so the energy flow is complex as well as unique for every ecosystem. The ponds form a small part in this interacting ecosystem, representing more or less ideal example. In the present study the ponds studied were observed for all aspects, such as the flora, fauna, water acidity & alkalinity and the presence of inorganic and organic matter usually their percentage. The affects of these components on the energy flow was studied. The most focused aspect of this study was the change in the biological oxygen demand & chemical oxygen demand and their effect on energy flow.

Keywords-physico-chemical studies, biotic components energy flow

INTRODUCTION

The biotic components of the pond ecosystem include phytoplanktons, zooplanktlons, fishes, aquatic plants algae, fungi, bacteria & other decomposers that help in nutrient cycle. The producers are the aquatic green plants which may be divided into two groups; microphytes (microscopic autotrophs) which fix solar energy such as spirozoa, zygnema, volvox etc. & macrophytes which manufactures complex food such as pistea, hydrilla etc. Ecosystem O_2 uptake, ecosystem respiration, and ecosystem metabolism became synonymous with ecosystem energy flow, now regarded as an ecosystem function.[1] Biotic factors are materials produced by living organisms that affect the ecosystem. That is why the energy flow from a producer to a consumer is essential for balance within the ecosystem. Organisms can be either producers or consumers in terms of the flow of energy through an ecosystem. Energy flow in aquatic ecosystems is a consequence of metabolic activity of all organisms producing heat in accordance with the second law of thermodynamics.[2] Community primary production and the respiration of many species as well as of entire biotic communities are still being measured at present in terms of oxygen production and uptake.[3] In this paper energy flow in an aquatic ecosystem of all forms of matter in plankton and intertidal sediment samples, ecosystem

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respiration and energy flow are greatly underestimated by measurements of oxygen uptake.

The energy is required for the performance of all kind of life activities. The existence of living world depends upon the flow of energy and circulation of materials through the ecosystem. The ultimate source of this energy is sun. Nearly 8 per cent of solar energy which enters the earth surface strikes the plants, out of which 80-85 per cent is absorbed, and only fifty per-cent of it is utilized in photosynthesis.[4] The energy is then stored in plants and represent first trophic level in the ecosystem. The energy is passed to the next trophic level through the consumers. A food chain in any ecosystem usually consists of maximum four steps, the producers, primary consumers, secondary consumers and tertiary consumers. The energy flows from the producers to consumers and at every level 80-90 % of the stored energy is dissolute as heat produced during the process of respiration.[5]

Ecologists have long debated what regulates the trophic structure and dynamics of ecosystems. [6] This is important because trophic structure and dynamics regulate many of the goods and services that ecosystems provide to wildlife and humankind, such as the production of harvestable food and energy, carbon sequestration and modulation of climate change, and nutrient uptake and control of global biogeochemical cycles.[7,8]

MATERIALS & METHODS

To study the biotic norms of prey-predator relationship in a pond ecosystem several ponds have been visited. Ponds are situated in Murliganj and Parwa Bishanpur area. An area of approximately 3 acres is occupied by these ponds. The various physical, chemical and biological parameters were studies. Physical factors studied were temperature which on average ranged between 22-31 ^{o}C. RH% average ranged about 93%, average rainfall about 169.95mm .The transparency was measured and was found out to be good. The ponds were less turbid. Rainfall is highest during the months of July – August about 290mm-396mm. Chemical factors was also measured such as pH, alkalinity, acidity, hardness, carbonates, phosphates, nitrates, sulphates etc. In biological parameters zooplanktons, phytoplanktons nektons were found to in plenty, benthic organism usually found in the bed of the ponds. The ponds were rich in benthic organism such as aquatic insects, snails, aufwuchs (mixture of algae, fungi & bacteria)

Table 1- Physico-chemical parameters of Murliganj and Parwa Bishanpur area during the investigation

Sl.No.	Year	Month	Temperature(^{o}C)		RH(%)	Rainfall (mm)
			Min.	Max.		
1.	2017	July	24.5	32.1	95	358.3
2.		August	23.4	30.1	92	266.3
3.		September	22.9	29.8	96	233.2
4.		October	23.1	29.9	95	256.2
5.		November	22.3	31.5	93	244.2
6.		December	20.2	29.3	95	123.2
7.	2018	January	20.3	30.2	94	5.4
8.		February	21.2	31.0	92	2.3
9.		March	23.5	34.1	91	40.2
10.		April	24.2	34.2	92	102.3
11.		May	23.3	33.2	93	152.3
12.		June	23.5	31.8	95	255.5

CONCLUSION

On conducting the above study it is found that all the ponds were rich in biotic components. The food webs were highly complex and involvement of varieties of organism was there. It was seen that even by removing one aspect of the pond, the dynamic stability of the ecosystem was disrupted. These ponds formed an ideal example of how our whole ecosystem works. Even a

microscopic zooplankton holds as much importance as a large fish.

To keep these ponds pollution free, a lot of effort has to be taken. Though self cleaning, human involvement and inappropriate garbage disposal effects the entire organism and simultaneously the food chain and webs. In all, various factors contribute equally in maintaining this dynamic equilibrium between prey and predator. Increase in either of the two can cause a misbalance, adversely affecting all the biotic components.

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Studies on the diversity of *Solenopsis* species from Mathahi block of Madhepura District, Bihar

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Abstract- The study has been carried out in the Mathahi block, of Madhepura District as no adequate information on *Solenopsis* species diversity of this region has been available yet. The present study was carried out during November 2017 - May 2018. During the study diverse subfamily of ants were observed. Out of which Myrmicinae was the dominating. The ants were sampled by employing intensive all out search method (AOS method). The sampled specimens were collected carefully. Through this study availability and abundance of ants in this Mathahi block region were revealed. This study also shows that ants can also survive in odds and adverse condition.

Keywords-*Solenopsis,* Mathahi block, diversity, AOS

INTRODUCTION

Solenopsis species are widely distributed in almost every possible terrestrial area of India. These are you eusocial organisms working continuously together to sustain their lives as well as maintaining the survival of their species. They are usually beneficial for the ecosystem showing active participation in measure ecosystem stabilizing activities. These activities vary from minor roles in decomposition to major roles in pollination. They show mutualistic behaviour with most plants and animals.[1]

They show high level of interaction among themselves. This coordination in maintaining their family of billions forms the milestone. Their communication method generally consists of scents pheromones, touch body language and sound antennae forms of key component in communicating.[2] *Solenopsis germinata* generally called red ant owing to its light brown reddish colour it is also called fire and due to the burning sensation it causes, when it stings.[3]

Other than *Solenopsis germinata* ants such as *Solenopsis indica*, pharaoh ants, black garden ants, weaver ants, harvester ants, grassland ants, carpenter ants and flying ants are found in India.[4]

S. germinata undergoes metamorphosis the first stage is egg which is followed by larva, pupa and finally adult stage. *S. germinata* flourishes in a hot and arid climate. It now seems to flourishes in climate controlled environment like buildings, human dwellings etc. This method restricts their spread/ dispersal to far areas but forms a continuous establishment. Their abundance is seen to decrease in areas where they remain under shade for more time.[5]

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They derived nutrition from various sources such as grains seeds decaying vegetables. *Solenopsis germinata* infestations are major in agricultural areas. They are usually feared for their stinging habits. The sting is venomous enough for an invertebrate to be preyed upon. They respond rapidly and are very aggressive under even a slight disturbance to their habitat or food.[6] It gives multiple stings even after its venom depleted. The sting is followed by inflammation and reddening of nearby tissue.

MATERIALS AND METHODS

This field work was conducted in various regions of Matahi block of Madhepura district. Both well ventilated and shaddy areas were examined thoroughly.

Ant sampling method

Ant field sampling was done from November 2017 to May 2018. Intensive All Out Search (AOS) method was employed for raw data collection. The ants were manually collected preferably during the daytime with the help of forceps and brush. The data comprising of date time, habitat and locality of collection was also recorded and kept for future references.

The samples were transferred in a bottle containing ethyl alcohol. They were then taken into the department's lab for identification labeling and preservation.

Cleaning

- Any kind of foreign material such as mud or pollen which adhered to the ant's body was removed. To do this process a small brush dipped in water was employed.
- The ants were sorted very carefully according to their location.
- Samples were separated from dirt and they were washed before preserving them.
- Species were identified and grouped accordingly.
- Different levels were put on the jars for easy identification.

Identification

The genus was identified using basic taxonomic keys prepared by Balton in 1994.[7]

Experts were also referred to avoid any confusion regarding the species.

Labeling

The ants were permanently preserved using the preservation techniques and provided with identification and data labels. The data labels contained the date and time of collection along with the habitat and locality they were found. Identification labels consist of species name, common name, and scientific name with author, genus, family and order.

The ants were preserved both by wety preservation and dry preservation.

Wet preservation

The ants were directly put in 70% ethyl alcohol separate vials were used for different species.

Dry preservation

The ants are glued down to the apex of a small triangular strip of thick paper, between the fore and middle coxae. Then a pin is pushed through it. This then stored in an insect box.

CONCLUSION

The diversity of *Solenopsis* species was analyzed in this comprehensive survey. Approximately 1,000 specimens were captured. *Solenopsis germinata* & *Solenopsis invicta* dominated the area under study. The distribution of these two species depended on several factors such as food resources, nesting habitats etc. Mathahi is rich in *Solenopsis* species owing to the ant favouring climate throughout the year and availability of food resources. Their ecological niches had overlapping. Ants have generally shown resistance to pollution and harsh climatic condition. Their body covering consisting of chitin helps them to do so. The diversity of ants was seen through extensive data collection for 8 months. This data clearly shows increase in ant activity from winters to summers showing its peak during the hottest month of the year April- May. Though tiny, the number of ants balances the hectic lifestyle they maintain for survival. They play a vital role in maintaining the dynamic equilibrium of stability. This study provides a model for future surveys and establishes a foundation for continuing research.

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Certain endangered and threatened ethnobotanically important plants of Ranchi District, Jharkhand

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Abstract : Jharkhand came into existence as a separate state of Indian Republic on 15th November 2000. It includes the geographical territory of Chotanagpur & Santhal Pargana division of the Old Bihar State. Jharkhand literally means "Land of Forest" and it has presently 29.61% forest area (the total area of Jharkhand is 79,712 Sq.Kilometer). Ranchi is the capital and the largest district of Jharkhand. It is situated in the centre of Chotanagpur plateau of Jharkhand at an elevation of 652 meters above the sea levels approximately between 20.0°-23.60° North Latitude and 84.90°-85.90° East Longitude. Chotanagpur plateau the new Jharkhand where Ranchi, situated in full of forest, rivers and plants. So the people of this remote area are very much attached with forest and mountains. They love to live within nature. They behave with nature as their mother because all the necessary sources of their food, shelter and home are found in forest. So they worship some plants as God. They totally depend in forest for food cloth shelter and medicines also. Tribals dominant forest areas are playing an important role in ethnobotanical aspect in their day-to-day life.

Key words: endangered, threatened, ethnobotany.

INTRODUCTION

The plateau of Ranchi district is full of Mountains Rivers and Forest. The plants are finding place as pharmaceuticals cosmetics and food supplements. Even as traditional source of medicines and they continue to play pivotal rule. The World Health Organization (WHO) estimated the 80% of the population of developing countries still survive on traditional medicines, mostly plant drugs, for their primary health care needs. Also modern pharmacopocia contains at least 25% drugs derived from plants. Many other are synthetic analogues built on prototype compounds isolated from plants. Demand for Medicinal plants is increasing in both developing and developed countries due to growing recognition of natural products, being non-toxic having no side-effects easily available at affordable prices. Medicinal plant sector has traditionally occupied an important position in the socio-culture, spiritual and medicinal arena of rural and tribal lives of Ranchi District.

Millions of rural households use medicinal plants in self help mode. Number of ethnic people viz.-Oraon, Munda, Kharia, Ho, Santhali, etc. Tribal people have used plant species in their day-to-day life, such as food fodder medicines, vegetables, fibers and domestic purposes.

Literature reveals that certain places of Central India have been explored for their ethnobotanical utility.[1-4] Due to anthropogenic activity numbers of certain species are

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also going to be disappear and included in the list of endangered. Present investigation have recorded certain ethnobotanically important plants and enumerated their uses in day-to-day life.

MATERIALS AND METHODS

Ethnobotanical explorations were undertaken during 2000-2008 in different tribal dominant forest areas of Jharkhand. The uses of plants in different aspects of life especially medicinal purpose and causes of endangered were gathered from knowledgeable and elderly persons, because they are the only source to collected of information about local plants name and their uses.

RESULTS AND DISCUSSION

Plant species were enumerated alphabetically with family followed by their local name, habit, uses and treats to biodiversity.

Andrographis paniculata (Burm.f) Wall ex. Nees. Kalmegh (Acanthaceae) Herbs, wild. The plant is used for controing sugar of diabetic patient, urinary trouble etc. Endangered.

Achyrantes aspera - Linn. Chirchira (Acanthaceae) Annual herbs.

Decoction of herb is useful in pneumonia, cough, kidney stone, leaf extract is used for Leprosy.

Amorphophallus campanulatus. Blume (Araceae) Herbs

It cures piles, abdominal tumours, intestinal worms and dysentery. Vegetable of tuber is useful during intestinal disorders.

Asparagus racemosus Linn. (Asparagaceae) Shatawaree Climber. Satavari is powerful drug capable of improving memory power intelligence, physical strength and youthfulness. It is sweet bitter in taste. Roots are used to cure diarrhoea as well as in case chronic colic and dysentery. Tubers are refrigerant demulcent aphorodisiae, antidysentery and are useful in fever.

Ageratum conyzodies Linn. (Asteraceae) Osaaddee. Aromatic, erect annual herb is used internally as a stimulant and tonic. Juice is a good remedy in prolapsusani. Oil applied in rheumatism, Leprosy, Skin diseases. Juice of leaves is antilithic.

Abelmoschus moschatus - Bhindi (Malvaceae) Annual herbs. Commonly known as Lady's finger. Although it's different parts are used as traditional medicine and the traditional healers. It is heavy demand at national and international markets. The farmers are not growing it for the traditional healers. The healers are fulfilling their demand from wild population as well as from very small - scale cultivation.

Barleria prionitis Linn. (Acanthaceae) A much branched prickly shrub. Katsareya has antiseptic properties; its decoction is used in dropsy to wash the body. A decoction roots is used as mouth wash in toothache and paste is applied to disperse boils and glandular swelling leaves are chewed to relieve toothache.

Cissus quadrangularis Linn. (Vitaceae) Boneseter, Harkankani. It is useful in indigestion, piles, worms and asthma. It has ability to rejoin broken bones. Juice of plant is beneficial in scurvy; Fresh shoots are applied for bums and wounds.

Centella asiatica L. Urban. (Apiaceae) Beng Sag. *Centella* is used as leafy vegetables. It is most often prepared as vegetable and dal. *Centella* appears to act on the various phases of connective tissues development, which are part of the healing process and increases Keratinization which allows it to stimulate healing of ulcers, skin injuries, connective tissue and to decrease capillary fragility.

Ficus benghalensis Linn. Moraceae Banyan tree. An infusion of bark cures dysentery nervous disorders, diarrohea and reduces blood sugar in diabetes, Milky juice is beneficial as local application in toothache, sores and ulcers. Infusion of young buds is used in diarhoea and dysentery.

Ficus religiosa Linn. Moraceae Peepal tree. Leaves and young shoots are purative. Bark is astringent and found efficacious in gonorrhea, Infusion of bark given internally in scabies, ulcers and skin diseases.

Psidium guajava, Linn. Myrataceae Amrud. Leaves are used as astringent for bowels and wounds and ulcers, Young leaves are used as a tonic in the diseases of digestive functions. Flowers are said to cool the body and are used in bronchitis. Perukam fruit is tonic, cooling and laxative. It is good in colic and for bloody gums. Fruits and its conserve are astringent and used in diarrhoea and dysentery.

Phyllanthus niruri Linn. Euphorbiaceae. *Bhoomi anwalla*. It is used as a diuretic in dropsical affection and

other trouble of genito urinary tract. Herb is bitter, astringent, deobsruent, diuretic, febrifuge and antiseptic. Fresh root is a remedy for jaundice. Milky juice used as applications to sore. Leaves are a popular remedy against fever. Fusion of young shots given in dysentery.

Ziziphus mauritiana Lamk. Rhamnaceae tree. Ber. Ber roots are bitter and cooing, biliousness and headache. Decoction of roots is used in fever. Fruits are mucilaginous, pectoral, styptic considered to purify blood and aid digestion.

Ethnic people of Jharkhand are well known for its heritage regarding medicinal plants growing near forest areas. The ethno medicinal values differ from species to species in composition and properties. Some plants have characteristic of quality timber, edible fruits and some are poisonous and also medicinal.

Many species are used in medicinal purposes by different tribal people and have other uses as food, fodder, fiber, timber and other purposes.

Due to different anthropogenic causes many of those specimens enumerated in the list becoming threatened as *Centella asiatica*, *Asparagus recemosus* Vitex, Lantana and very rare are clerodendren species.

Different measures have been adopted for conservation of these categories species. So that it may give maximum benefit to the present generation while maintaining its potential to meet the needs and appertains of the future generation.

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Ethnomedicinal plants of Jharkhand

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Abstract : The tribes of Jharkhand are men of forest. They exist within the vicinity of nature. The land itself is bounty of forest and mineral wealth. The people of this area have learnt to live in harmony with nature. They have acquired deep knowledge about the existing flora and fauna of Jharkhand. This knowledge has been passed down from several generations through Pahans, Ojha and other medicine men who provide treatments to various ailments, prevailing in interiors of Jharkhand. We are listing 15 such ethnomedicinal plants found in Jharkhand state with their uses. The present article puts on record some important ethnomedicinal plants sampled from different places of states of Jharkhand.

Key words: ethnobotany, nature, uses, flora and fauna

INTRODUCTION

The present paper is documentation of some important ethnomedicinal plants of terrestrial habitats of state of Jharkhand adjoining perennial rivers like Damodar, Deonar, Barakar, Koel, Karo, Sone & Swarnrekha.

Ethnomedicines

The traditional knowledge folk medicines and herbal medicines have been providing panacea and relief of millions of people across the country. The renewed interest in the medicinal plants might be attributed to the fact that modern medicines are not only expensive but in many instances have proved to be a complete failure due to drug resistance and unpredictable complications to the patients. It is no denying fact that the herbs and herbal cures is taking over its popularity in the last few decades wider populace providing relief to people and at the time they are safe economical, and powerful alternative to number of ailments. Under the topic of ethnomedicines 'The Ethnomedicinal plants of which are used by the tribes of Jharkhand must be discussed.

Treasure of Ethanomedicines-Jharkhand

The state of Jharkhand is the 28th state of India located between 21°59' North latitude to 25°8' North and 83°20' East to 87°57' East longitude and an area of 79. 714 sq. kms with population 26,907,428 representing 2.62% population of India.

The state consists of 32 ethnic and tribe groups consisting of 11.85 % of the population. Its 8 primitive tribal groups (PGT) include Asur, Birhor, Birjia, Korwa, Malpaharia, Paharia, Saurya Pahariya and Savar. Jharkhand is articulated withperennial rivers like Damodar, Deonar, Barakar, Koel, Karo, Sone and Swarnrekha.

The state is endowed with richest mineral and forest wealth with pristine natural landscape and bountiful of plant herbs of 7,50,000 plants of which few have been discussed below.

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Ethnomedicinal plants of Jharkhand

1. Name- *Euphorbia hirta*
   Family- Euphorbiaceae
   Common name- Dudhi

Uses -used in treatmemt of Gastrointestinal disorders, bronchial and respiratory diseases
-stem sap is used in treatment of eyelid
-leaf poltice is used in swelling and boils
-Plant extract has anticancerous properties
-Decocation of dry herbs is used in skin diseases
-Root decoction is useful for nursing mother deficient in milk

2. Name- *Androgrphis paniculata*
   Family- Acanthaceae
   Common Name- Kalmegh

Uses-whole plant is blood purifier, cures jaundice and -dermatological diseases
-acts to dispel heat and removes toxin
-plant is antibacterial, reduces diarrhoea
-plant is best remedy for malaria
-it also cures flu, cough and bronchitis

3. Name- *Boerhaavia diffusa*
   Family- Nyctaginaceae
   Common name- Punarnava, khapda sag

Uses -leaves act as carminative
-increases appetite, reduces abdominal pain, improves digestion and relieves constipation
-reduces inflammation and pain in joints in arthritis
-plant is an excellent natural remedy for guinea worm
-plant is used as vegetable to cure kidney stones
-its decoction rejuvenates and detoxifies liver

4. Name *Phyllanthus niruri*
   Family Euphorbiacae
   Common name Bhoomi amla

Uses-Plant extract is liver tonic, and medicine for hepatitis
-leaves extract cures jaundice, viral hepatitis and alcoholic cirrhosis of liver
-whole plant dried powder is given as antibacterial, antihepatotoxic and antiinflamatory, laxative, stomach ache,vermifuge
-when given along with honey, plant is the best medicine for jaundice

5. Name *Centella osiatica* L
   Family Apiaceae
   Common name- Brahmi, Beng Sag

Uses-plant has traditional use in leprosy
-whole plant revitalizes the brain increases the attention span and concentration. It is a memory tonic
-leaves juice has antioxidant properties. It reduces hypertension and it is also a coolant.

6. Name- *Datura metal* Linn
   Family Solanaceae
   Common name- Dathura

Uses-the young leaves and seeds contain the drug Hyoscine, Hyoscymine and antropine
-the dried leaves and twigs are smoked as an antispasmodic in asthama whooping cough and bronchitis etc.
-the juice of the fruit is a usefuldressing for the scalp and falling of hair.
-seeds are astringent, antispasmodic, narcotic,anodynes
-intoxicating, aphrodizac, bitter, karminative, stomach ache
-paste of seeds is used for preventing decay of teeth, piles, tumours Fistula and skin diseases.

7. Name *Ravolfia serpentine* Benth
   Family Apocynaceae
   Common name Sarp gandha

Uses-tribes of Jharkhand use it as strong insect repellent.
-the plant contain indole/ajmaline group of alkaloids are reserpine and serpentive which are hypertensive, neuropsychiatric, gynaecological and geriatrics
-the drugs is useful in certain gynaecological conditions like menstrual, moliminea, frigidity and women complaining of menopausal syndrome.
-the plant has been considered as antidote against snake bite in Charak Samhita.

8. Name *Terminalia arjuna* Roxb
   Family combretaceae
   Common name arjun

Uses-bark cures heart diseases, improves functioning of cardiac muscle
-Leaves juice cures dysentery and ear ache
-maintains cholesterol level

-bark powder possesses diuretic properties

9. Name *Terminalia belleria* Roxb
   Family Combretaceae
   Common name baehra

Uses -ripe dry fruit is astringent tonic, laxative, cures piles and leprosy
-fruit cures headache, dropsy, premature hairfall
-fruits cures cough and sore throat
-seed oil is applied in skin diseases and woolen part
-half ripened fruit is purgative and fully ripened astringent
-dried fruit powder taken with luke warm water for stomach clearance
-the plant is among the three ingredients of Trifala choorna

10. Name *Termenalia chebula* Retz
    Family Combretaceae
    Common name- Harara

Uses-fruits is laxative, astringent, stomachic tonic
-fruits pulp cures chronic diarrhoea, dysentery, asthama, urinary disorders, enlarged spleen and liver
-fine powder of fruits cures teeth bleeding, ulceration of gums chief ingredient of Trifala choorna.

11. Name *Achyranthes aspera* Linn
    Family Amaranthacae
    Common name chirtchita

Uses-plants extract is used as laxative, antihelmenthic, antiviral, Antiplasmodic and antihypertensive
-treatment of cough, renal dropsy and skin rash
-the juice of the plant is used in boils diarrohea, dysentery, haemorrohoids, rheumatic pain and itches

12. Name *Bacoba moneiri*
    Family Planta ginacaea
    Common name- Brahmi

Uses-plant is used for improving memory
-it contains powerful antioxidants
-it is used in reducing anxiety, removing stress and in treating epilepsy

13. Name *Aloe vera* L.
    Family Asphodelaceae
    Common name- ghritkumari, gowarpatha

Uses-the gel has antioxidant and antibacterial properties
-the gel speeds up the healing of burns
-It reduces dental plaque
-the exudate of the leaf improves skin and prevents wrinkles
-the leaf gel reduces constipation
-the gel lowers the blood sugar level

14. Name *Vitex negunda* L
    Family Verbenaceae
    Common name nirgundi, sindwair

Uses-bathing with boiled water alongwith the leaves of this plant relieves muscle aches and joint pains
-the extract of leaves, seeds and roots are used in treatment of excessive vaginal discharge
-the leaf extract, seeds and root are used in treatment of edema skin diseases, pruritus, helminthiasis

15. Name *Justicia adhatoda*
    Family Acanthaceae
    Common name- bakas, vasak

Uses-leaves are used for multiple disorders as cold cough, bronchitis and asthama
-Leukoderma
-Jaundice
-Gonorrhea
-Heart trouble,
-Leprosy
-Mouth trouble
-Fever
-Sore eye

Jharkhand could be a hot medical tourist spot with abundance of medicinal plant at every corner as an unexplored resource. Only the fringe of the subject has been touched. Even then hundreds lie unidentified and unexplored which could be used for benefit of Indians and people of Jharkhand.

Plants remedies are harmless provided that they are selected carefully and taken under medical guidance as a Chinese saying "A potion in hands of inexperienced and quack may behave like a poison, while a poison in the hands of an experienced and gifted Physician becomes a potion."

The tribes of Jharkhand have traditional knowledge about use of medicinal plants. The curative effects of the

medicinal plants were known to the people long before scientific explanations came into being. This wealth of acquired knowledge was passed on verbally amongst the tribals. The medicinal properties are known to general people and also to medicine man among the tribes such as Pahans, Bhagats, Ojha. They pass on these traditional knowledge's to the new generation, but gradually these practices are distorting.

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Physico-chemical analysis of effluents of Sri Ram Paper Mill, Forbisganj, Araria

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Abstract : Present study reveals that Sri Ram Paper mill, Araria causes high water pollution. The effluent contains of straw, cellulose fiber and dissolved lignin in addition to a complex mixture of chemicals are the cause of pollution. TDS, BOD and COD of the effluent were recorded as 3000mg/l, 1075mg/l and 7825 mg/l respectively.

Key words: Effluent, TDS, DO, BOD, COD.

INTRODUCTION

Sri Ram Paper mill was setup in 1992. It covers an area of about 2.5 acres. This industry has both pulp and paper production unit and produces strong brown wrapping paper, bag making papers, paper board used for cartons, corrugated boards, containers and strong white paper form bleached Kraft.

This Paper mill discharges large amount of waste water of paper manufactured having very strong colour, high BOD, high COD, high suspended solids and high pH. Crude paper and pulp waste causes very serious pollution consequences. This affects the survival, mortality and growth rate of fish present in tank of the effluents.

MATERIAL & METHODS

Waste water samples from effluent was collected in 1 liter plastic bottles and preserved for laboratory analysis as per standard procedures recommended in APHA (1975). The preserved samples were brought to the laboratory and were kept at 4°C in refrigerator before the start of chemical analysis. The time interval between collection, storage and analysis was intentionally shortened to avoid any major change in chemical characteristics of the sample.

Estimation of four physical parameters and five chemical parameters were performed. Among physical parameter colour, odour, temperature and pH were estimated and among chemical parameters TDS, DO, BOD, COD and Chloride were estimated as per the standard procedures recommended in APHA (1975).

RESULT

Colour

Almost all industrial effluents were coloured, highly disagreeable odour, turbidity high BOD and COD value.

Odour

The odour of investigated industrial effluents was of disagreeable odour, which in an important physical parameter.

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COD

It is much useful in finding out the pollution strength of industrial effluent. COD value of this paper mill was recorded as 4825mg/l.

Total Dissolved Solids

Determination of dissolved solids is useful in deciding the mineral content of effluent. The present investigation revealed the presence of 2800 mg/l total dissolved solid.

Dissolved oxygen

The measurement of dissolved oxygen indicates the purity of effluents and is important for maintaining aerobic condition in the system and in the treatment of industrial waste waters. The value of dissolve oxygen was nil in the effluent.

BOD

It is test of great value in the analysis of industrial effluent. BOD is the quantity of oxygen required by bacteria and other microorganisms, during the biochemical degradation and transformation of organic matter under aerobic condition. The value BOD was recorded 1075mg/l (5days at 20^0C)

pH

pH was measured using digital pH meter in the laboratory.

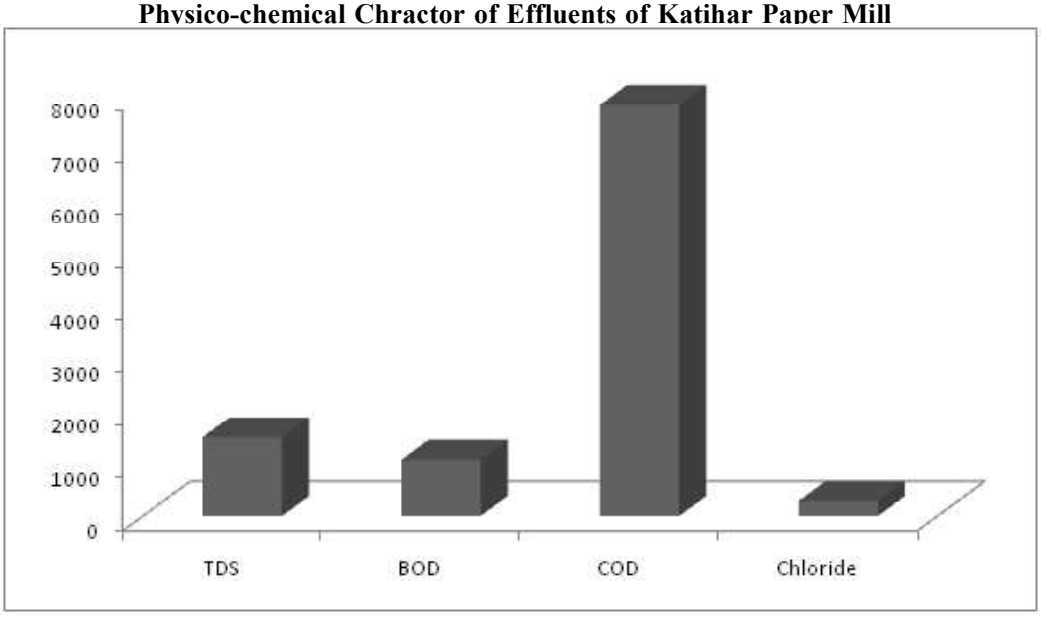
Chloride

Chloride is the common anion found in the industrial effluent. The chloride value of effluent of this paper mill was analysed to 282 mg/l. the physico chemical analysis is mentioned in table- 1.

Table 1:- Physico chemical characteristics of effluents of Sri Ram Paper Mill, Araria

Parameters	Value	Tolerance Limit ISI
Colours	Dark Brown	Should be Absent
Odour	Highly disagreeable	Should be Absent
Temperature	41^0C	31^0C
pH	7.8	5.5- 9.0
Total Dissolved Solids (mg/l)	3000	1690
Dissolved oxygen	Nil	5.8
BOD (mg/l)	1075	29
COD(mg/l)	4825	225
Chloride (mg/l)	282	30

Physico-chemical Chractor of Effluents of Katihar Paper Mill



CONCLUSION

Present study reveals that Sri Ram Paper mill causes a great amount of water pollution. The effluent contains of straw, cellulose fiber and dissolved lignin in addition to a complex mixture of chemicals. About 50% of the raw material used for the process is rejected as waste material (Manivasakam, 1984). This fact is evident by studying the physico chemical analysis of effluent of the paper mill. The effluent had total dissolved solid, BOD, COD, and high amount of Chloride.

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Ethnomedicinal study of some wild herbs of Nawhata block of Saharsa district

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Received :12th May, 2019 ; Revised : 25th June, 2019

Abstract : Altogether 18 wild herbs were identified for different diseases. Out of 18 plants both *Tinospora* and *Boerhaavia* were informed to be suitable for Jaundice, for skin diseases *Argemone* and *Achyranthus* are being used by different informants. Similarly for urinary disorders both *Achyranthus* and *Sida* are being used.

Key words: Ethnobotany, informants, jaundice

INTRODUCTION

Ethnobotany is the direct inter relationship between plants and human. Man has been using flora and fauna since ancient time. He gained the knowledge of useful and harmful plants. This knowledge became an integral part of his culture.

In Rig Veda and Athar Veda there is mention of use of plants. Charak samhita is an old literature in which medicinal use of several plants is mentioned.

Some wild plant resources are severely threatened by habitat loss and species selective overexploitation. In addition indigenous knowledge about the use of plant resources is rapidly disappearing from traditional communities. In the context of conservation and sustainable use of plants resources can contribute to the scientific base for management decisions.

Today ethnobotany plays a crucial role in the study of traditional medicine as it has an interfacial function linking nature with culture and traditional knowledge with modern technology. The main aim of ethnobotany is to document the knowledge about plants that have come through generations and use the knowledge for the benefit of society. The ethnic communities have significant customary knowledge on utilization of plant and there is a long tradition of transferring this indigenous knowledge from generation to generation.

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MATERIAL & METHODS

A survey was conducted in four villages, Bakaunia, Ashnahi patti, Chandrain and Baksaur from 2013 to 2014 at regular intervals. People of this area were consulted. People of different group were selected as informants. Included informants were both males and females. Questioners were asked to all informants about-

(a) Local name of the wild herbs to which the use for medicinal purpose and for other economic values.
(b) How they use the herb or part of the herb.

(c) If the herb is of medicinal value what part which aliment.
(d) Flowering and fruiting season of the herb.
(e) Season of the herb to which they use.
(f) How they knew the property of the particular herb?

Their knowledge regarding medicinal use of herb for application in different ailments was asked. All together 50 informants were interviewed.

RESULT

Altogether 18 herbs were recognized for their medicinal value. Herbs were collected, identified and herbariums were prepared. The herbs were described and there medicinal value noted.

1.*Tinospora cordifalia*

Therapeutics

It is a tonic, ant periodic, diuretic, cures fever, Jaundice, diabetes, piles and respiratory disorders.

2. *Argemone mexicana L.*

Therapeutics

Ant parasitic drug is obtained from *Aremone mexicana*. Its root is used in skin diseases. It is also suitable in the treatment of syphilis and Piles. Seeds are used diarrhoea and dysentery.

3. *Gynandropsis pentaphylla*

Therapeutics

Tilaparni or Suryavarta is used in scorpion sting and snake bite. Ether extract of plants is anticancer and spasmolytic.

Decoction of roots is used in fever. Juice of leaves and seeds oil used to cure skin diseases. Leaves are vesicant in rheumatism; remedy for muscular pain, headache and intestinal wounds.

Seeds are antispasmodic, sudorific, anthelmintic and carminative. They are applied as poultice to sores having maggots and given as infusion for cough. Powder of seeds is a good remedy for piles.

4. *Sida cordifolia L.*

Therapeutics

Juice of root is used for healing wounds Infusion of root is given in nervous and urinary diseases and disorders of blood and bile. Seeds are given in gonorrhoea, useful in neurological disorders.

5. *Acalypha indica (Willd.) L.*

Therapeutics

The plants is bitter, acrid and possesses diuretic, cathartic, expectorant, emetic, anthelmintic, anodyne and hypnotic properties. But it causes gastro-intestinal irritation. It is used as a substitute for ipecac and Senegal. A decoction of the herb is used as a cure for tooth and ear ache and is safe and speedy laxative. It is useful in bronchitis, pneumonia and asthma. Root is cathartic.

Leaves are laxative, used in scabies and in snake bite. Fresh leaf juice is useful in rheumatoid arthritis and skin affections. Juice with salt applied on eczema.

Paste of leaves applied on burns; with juice of lime, useful in early cases of ringworms. Powder of leaf is used in bed-sore and maggot infested wounds. Extract found useful in cardinal symptoms of whooping cough; the raised eosinophil count also came down to normal.

6. *Euphorbia hirta L.*

Therapeutics

It is demulcent, antispasmodic, anti-asthmatic pectoral, anthelmintic. Extract of plant is effective in cardiovascular disorders. Plant juice is recommended for dysentery. *Euphorbia hirta* have an alkaloid which is effective in respiratory disorders. Milky latex is present in plant which is a cure for diseases of urino-genital tract.

7. *Phyllanthus niruri L.*

Therapeutics

Bhudhatri or Bhumyamalaki is effective in gonorrhoea. Herb is bitter, astringent, diuretic, febrifuge and antiseptic. Fresh root is a remedy for jaundice. Milky juice of plant is applied in sore. Powdered leaves along with rice water are a remedy for ulcer. Lives are applied in fever and young shoots in dysentery.

8. *Boerhaavia diffusa*

Therapeutics

Whole plant is used for the medicinal purpose. Fresh juice of root is applied in eyes for conjunctivitis. Paste of root is applied on wounds. It boosts up the filtration and renal functions.

9. *Achyranthes aspera*

Therapeutics

It cures ascites, enlargement of cervical glands & piles. It is used in dropsy and urinary diseases, skin eruptions, Colic and snake bite. Decoction of herb is useful in Pneumonia, cough and Kidney stone. Root is given in stomach pain.

10. *Amaranthus spinosus*

Therapeutics

Used for strengthening uterus and ameliorating its pain. It is also used against burning sensation, dyspepsia and fever. Decoction of root is used in gonorrhoea.

11. *Centella asiatica*

Therapeutics

It is nervine and Cardio-tonic. It maintains youthful vigour and strength useful in dermatosis, anaemia, diabetes and cough. It is useful in skin diseases.

12. *Ageratum conyzoides*

Therapeutics

Juice of leaves boiled with oil is externally used in rheumatism. Decoction of leaves recommended in diarrhoea and dysentery.

13. *Vernonia cinerea*

Therapeutics

It purifies blood, bite and semen. It is used for leucorrhoea and excessive bleeding. Used in chronic skin diseases, bladder stones piles, worms and Haematological disorders.

14. *Leucas aspera*

Therapeutics

It is a reputed remedy for worms, fever and intestinal catarrh in children. It is antipyretic, antiseptic, carminative. Used in Jaundices, respiratory and skin diseases Leaves are considered useful in chronic rheumatism.

15. *Ocimum americanum*

Therapeutics

It possesses aromatic, carminative, diaphoretic and stimulant properties. Decoction is used for cough, that of leaf for dysentery. It is also used as mouth wash for toothache.

16. *Chenopodium album*

Therapeutics

It is laxative, blood purifier, Anthelmintic. Leaves used for intestinal ulcers and considered useful in piles, throat and eye troubles. Flowers and buds are used in stomach troubles. Seeds are prescribed for hepatic disorders.

17. *Hygrophila spinosa*

Therapeutics

It is a reputed remedy for Arthritis. It promotes strength and appetite. It cures oedema, ascites, bladder stone eye diseases and dysentery.

18. *Rungia repens*

Therapeutics

It is used as diuretic vermifuge which is given in snake bite. Fresh leaves bruised and mixed with Caster oil are applied to scalp in cases of *Tinea capitis*.

CONCLUSION

A survey was conducted in four villages, Bakaunia, Ashnahi patti, Chandrain and Baksaur from 2013 to 2014 at regular intervals. People of this area were consulted. People of different group were selected as informants. Included informants were both males and females.

Altogether 18 wild herbs were identified for different diseases. Out of 18 plants both *Tinospora* and *Boerhaavia* were informed to be suitable for Jaundice, for skin diseases *Argemone* and *Achyranthus* are being used by different informants. Similarly for urinary disorders both *Achyranthus* and *Sida* are being used.

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A survey on identification and classification of wheat plant disease, in Madhepura, Bihar

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Abstract : Plants are infested with several diseases and it is of great importance to identify the disease and its causative agents. Identifying disease from the sample of plant is one of the interesting research areas in agriculture research field. This paper deals with the process of identification and classification of wheat plant disease based on the survey. This paper not only deals with the survey of wheat fields but also concisely discusses important concepts of plants disease detection and classification.

Key words: ***Alternaria triticina,*** **wheat plants, identification, leaf blight pathogen.**

INTRODUCTION

Agriculture is one of the main sources of income in India. Farmers can grow variety of crops but disease hamper the growth of crops. Agriculture is the major sector of the Indian economy. About 70-75% of countries population is agriculture dependent. Wheat is the staple good for about one third of the global population. Wheat is commonly known as "Genhu". Its cultivation covers an area of about 12 million vectors (Britannica online 2015)[1]. One of the major factors responsible for the crop destruction is plant disease. The main part of plant to examine the disease is leaf.[2]

The present survey has been carried out in different wheat field around Madhepura. In Madhepura agriculture is the main source of income for the natives. Leaf blight is the most serious biotic constraint of the wheat. It occurs as a complex spot or black spot caused usually by *Alternaria triticina.*[3] Farmers were experiencing heavy economic loss due to their less crop harvest. The crops were getting damaged by several factors. One of such factors is fungal disease. The farmer fails to identify the proper disease and suffer with loss. Yield loss was directly related to area under the disease progress curve.[4] Leaf blight occurs to some extent wherever wheat is grown. Losses in grain yield are primarily attributed to reduced floret set and grain shrivelling.[5] In this paper authors provides the way to identify such diseases especially leaf blight. The leaf disease detection and classification of leaf disease is the key to prevent the agricultural loss. there are different methods to detect plant leaf disease. Due to possibility of various alternatives at different states of plant disease identification shown in fig. 1 & 2, researchers have attempted various methods to do so. Furthermore utility of the presented survey and findings is shown in our proposed work in the same direction.

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MATERIALS AND METHOD

Survey and collection of wheat seeds from different localities of Madhepura were done during 2017. The study has been carried out in the following manner.

Study Area

The present study was carried out in different localities of Madhepura usually in the outskirts of the city. Madhepura is situated 25°31'N latitude & 86°36' E langitude. The wheat fields are visible from far off distance as abundant of farming takes place. The total areas of



Fig. 1- Surveying the wheat fields having tender plants



Fig. 2- Collecting the infected plants



Fig. 3- Infected Leaf



Fig. 4- Wheat field



Fig. 5- Infected wheat plant



Fig. 6- Infected wheat grains

land used for cultivation is about 1,36,646 Hectare. Farming is in ample amount just because no any big industries has been setup in this district so far.

Sample Collection

Samples were collected from different fields where the infected wheat plants were found. Samples were first of all identified and then collected very carefully. The climatic conditions of the area were regularly monitered. The infected plants were collected and were carried to the laboratory for further studies. Figure given below shows the collected plants and survey.

Testing

The test set for leaf prediction as healthy/unhealthy with its disease name will be used in this stage to assess the classifier's output.

CONCLUSION

Diseases of wheat plants can cause tremendous amount of loss in the field of agriculture if proper attention is not given. Using proper technologies these disease can be identified and cured. The authors have provided several images to identify the leaf blight pathogen which makes it possible to identify and differentiate with other diseases. It is found that extraction of disease region from the leaf image is the first step. The authors utilized the survey and study presented in this paper, to propose their work in the same direction. At the proposed work, for the benefit of society directly as well as indirectly.

ACKNOWLEDGEMENT

The authors are thankful to the University Department of Botany, B.N.M.University, Madhepura, Bihar for the necessary equipments provided in completing the research.

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Nutritional study of some local strain of blue green algae from Saharsa district

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Abstract : Three local strains of Blue-green algae were selected for the nutritional study. Growth was observed in different media. Effect of nitrogen source and phosphorus was also observed. *Nostoc linckia* showed best growth in CHU-10 medium and *Anabaena macrospora and Oscillatoria limnosa* in BG-11 medium. Best growth of the culture of *Oscillatoria limnosa, Nostoc linckia and Anabaena macrospora* was seen in the medium supplemented with $NaNo_3$ followed by NH_4NO_3 and NH_4Cl. Growth was concentration dependent. Linear increase in growth yield with increasing concentration of nitrogen source was observed. Maximum growth of *O. limnosa* was seen in the medium supplemented with 2.5 mg/l phosphate followed by 0.5 mg/l. Minimum growth was reported at 6.5 mg/l of phosphate. Maximum growth of *A. macrospora and N. linckia* was recorded at 2.5 mg/l phosphate.

Key words: Blue-green algae, *A.macrospora, O.limnosa, N.linckia.*

INTRODUCTION

Most of the blue-green algae grow well in mineral solutions devoid of organic substance, a slight alkaline reaction proving most suitable. Nitrogen can be supplied in the form of nitrates, ammonium salt or nitrites, the first in general affording the best and last the least satisfactory growth; calcium is essential and there is considerable uptake of both calcium and magnesium.

Diverse earlier workers established that blue-green algae often grow well in nitrogen free solutions, with or without organic matter, and this has usually been ascribed to the cooperation of nitrogen fixing Bacteria, which almost certainly occurs within the enveloping mucilage.

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Monochromatic light of the visible spectrum are involved in many functions such as photosynthesis, growth, morphogenesis, chromatic adaptation and heterocyst differentiation of the Blue-green Algae.

Much information is available regarding the role of phosphorus in bloom formation and species diversity (Kumar *et al.*, 2000; Thakur, 1996). N:P ratio pays vital role in determining the phytoplankton growth in any water body.

MATERIAL & METHODS

Blue-green algae were collected from local area of Saharsa district and identified. 3 local strains were selected for nutritional study. Selected strains were *Anabaena microspora, Nostoc linckia* and *Oscillatoria limosa*. Axenic

culture of each strain was prepared and cultured in five different media (BG 11 medium, Allen Arnon medium, Chu-10 medium and JM). Growth response in each medium was recorded. For the study of effect of nitrogen two nitrogen source Ammonium chloride and Ammonium nitrate were added in culture media while for phosphorus different concentration of dipotassium hydrogen phosphate was used.

RESULT

For optimizing the growth in vitro, growth response was tested in BG-11 medium, Allen and Arnon's medium, Modified Jaworsk's medium, and Chu-10 medium. Initially the growth behaviours were tested 25±2^{0}C and 2200±200 Lux light intensity. *Nostoc linckia* showed best growth in CHU-10 medium and *Anabaena macrospora and Oscillatoria limnosa* in BG-11 medium (Fig. no. 1-3).

Best growth of the culture of *Oscillatoria limnosa, Nostoc linckia and Anabaena macrospora* was seen in the medium supplemented with $NaNo_3$ followed by NH_4NO_3 and NH_4Cl. Growth was concentration dependent. (Fig. no.-4-6) shows the impact of various concentrations of nitrogen sources on *O. limnosa, N. linckia* and *A. macrospora*. Linear increase in growth yield with increasing concentration of nitrogen source was observed.

Maximum growth of *O. limnosa* was seen in the medium supplemented with 2.5 mg/l phosphate followed by 0.5 mg/l. Minimum growth was reported at 6.5 mg/l of phosphate. Maximum growth of *A.macrospora* and *N. linckia* was recorded at 2.5 mg/l phosphate (Fig no.-7-9).

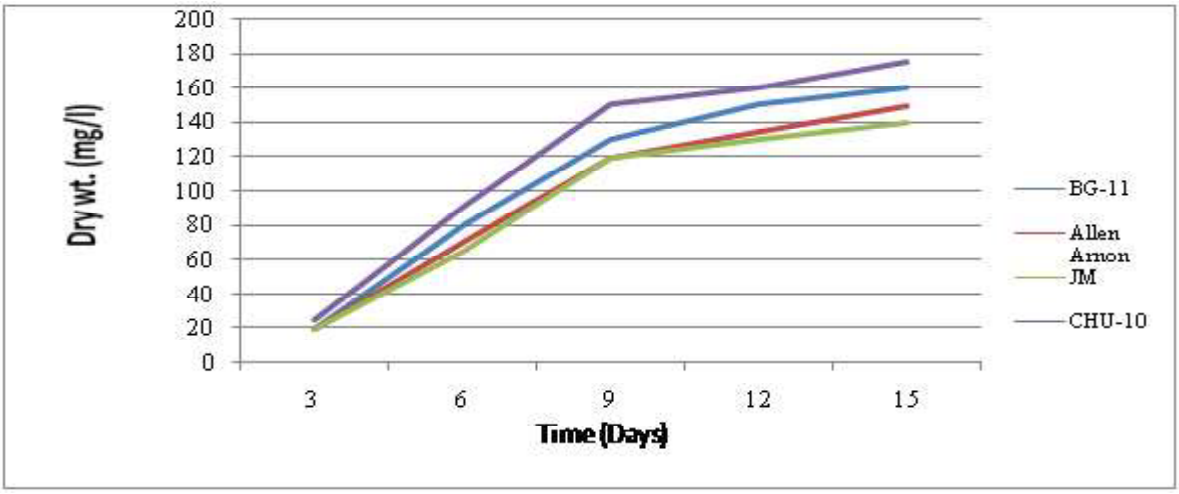


Fig. no. 1: Growth behaviour of *N. Linckia* in different media

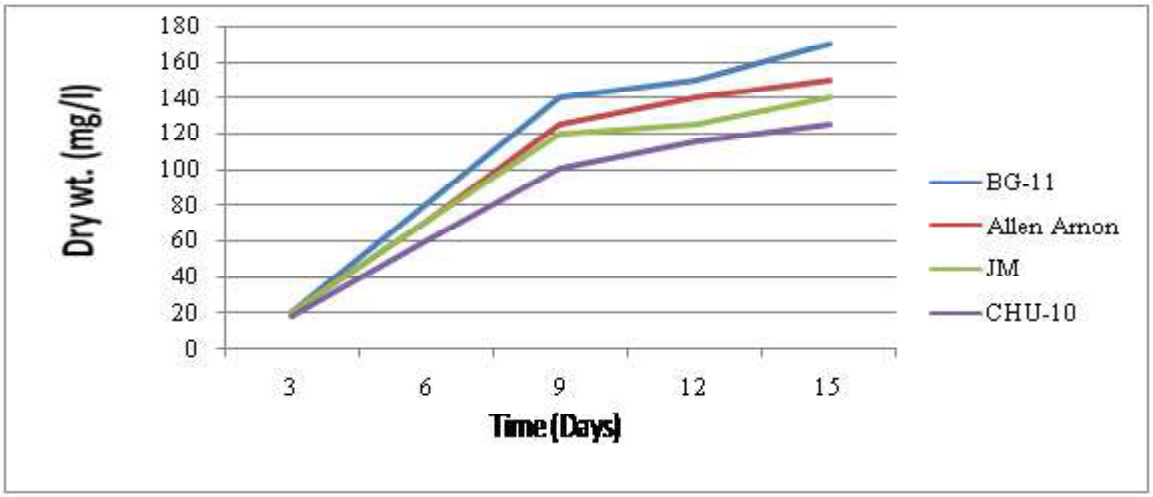


Fig. no. 2: Growth behaviour of *A. microspora* in different media

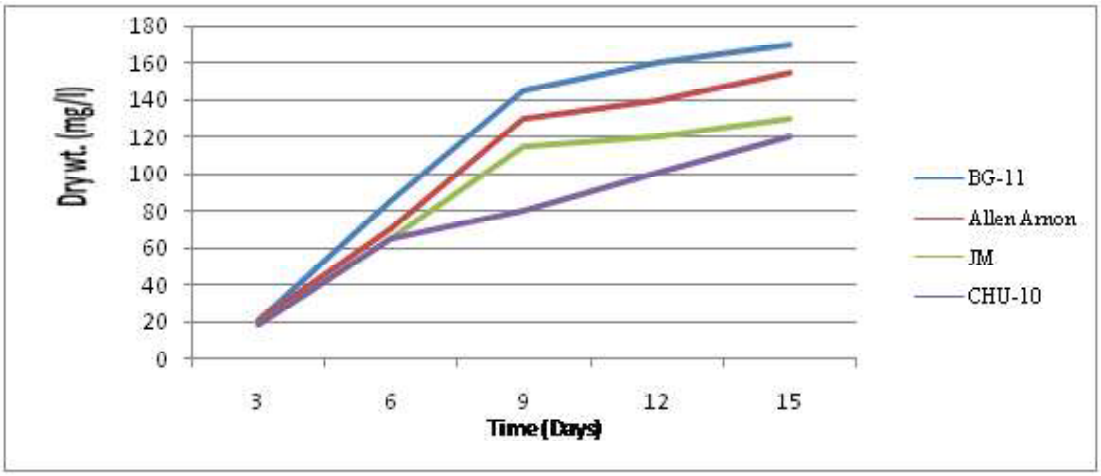


Fig. no. 3: Growth behaviour of *O. limnosa* in different media

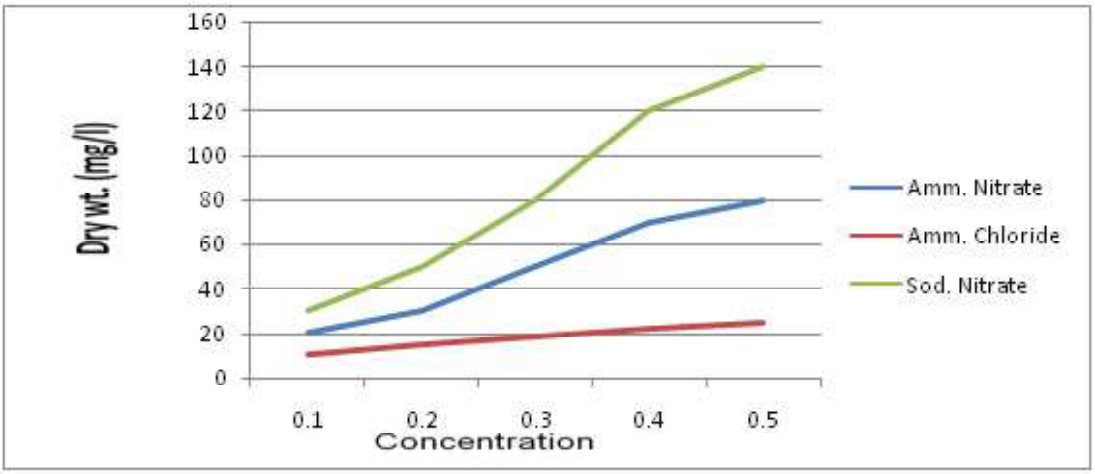


Fig. no. 4: Growth behaviour of *O. limnosa* in different Nitrogen concentration

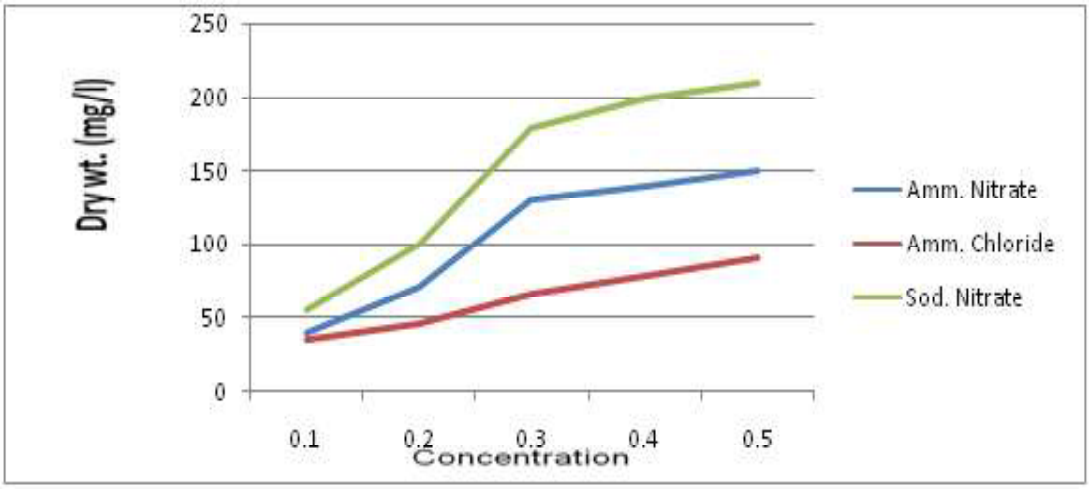


Fig. no. 5: Growth behaviour of *A. microspora* in different Nitrogen concentration

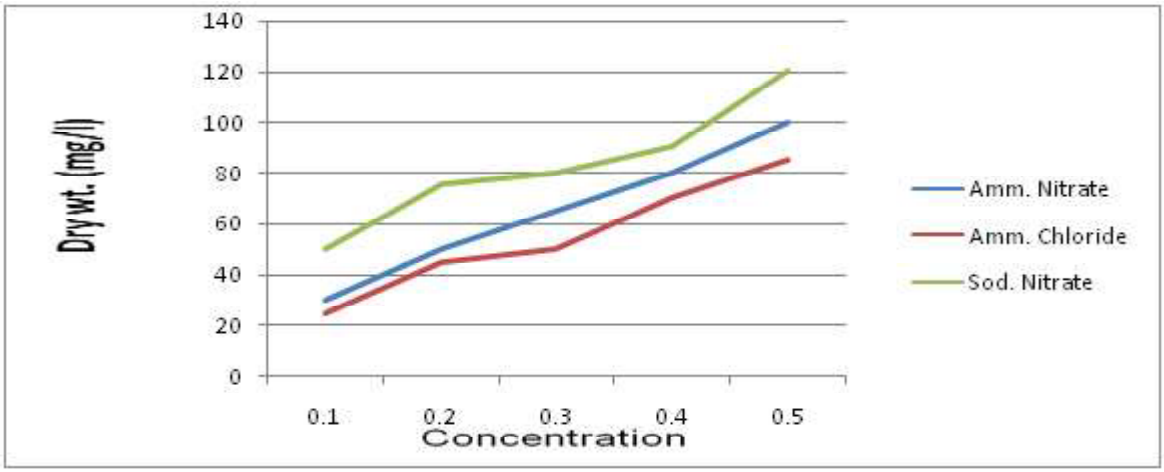


Fig. no. 6: Growth behaviour of *N. linckia* in different Nitrogen concentration

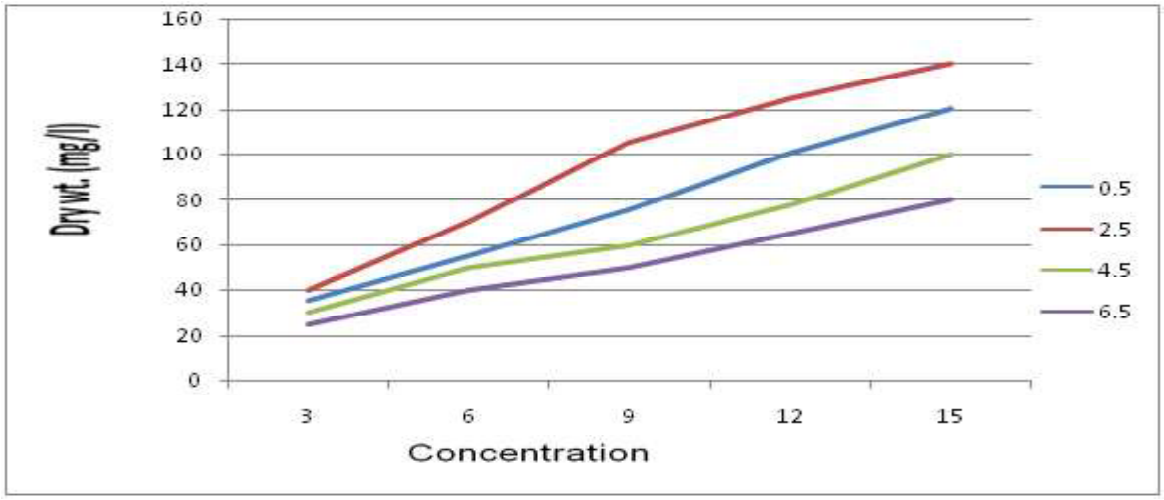


Fig. no. 7: Growth behaviour of *A. microspora* in different Concentration of K_2HPO_4

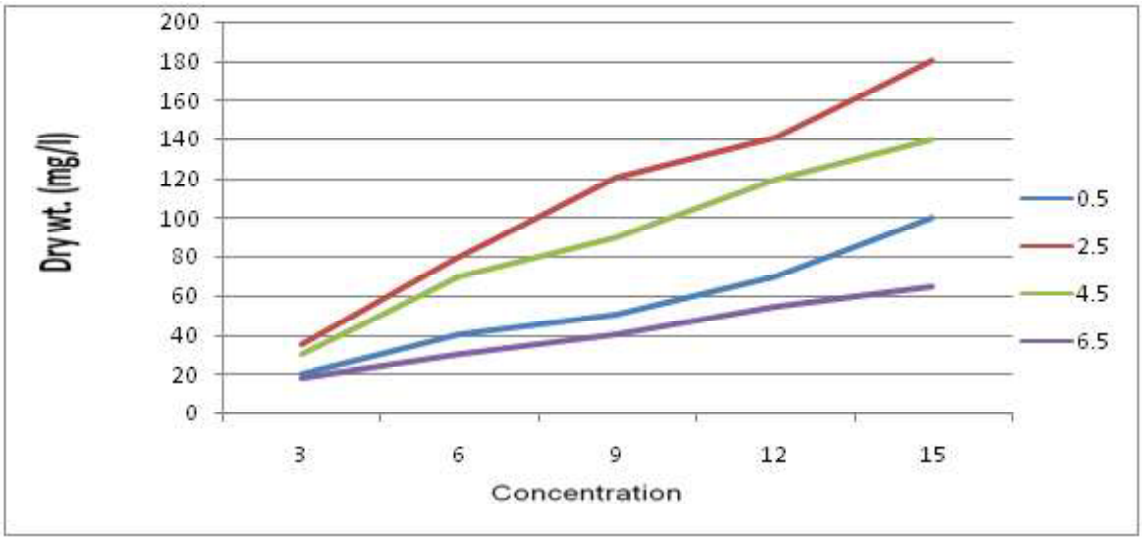


Fig. no. 8: Growth behaviour of *N. lickia* in different Concentration of K_2HPO_4

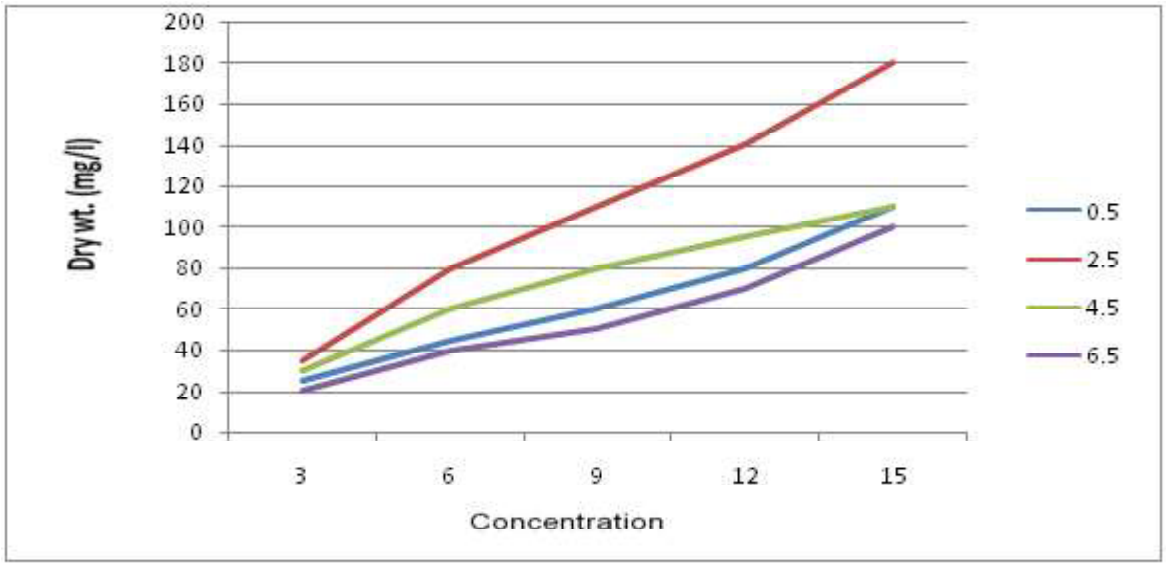


Fig. no.- 9 Growth behaviour of *O. limnosa* in different Concentration of K_2HPO_4

CONCLUSION

3 local strains were selected for nutritional study. Selected strains were *Anabaena microspora, Nostoc linckia* and *Oscillatoria limosa*. Axenic culture of each strain was prepared and cultured in five different media (BG 11 medium, Allen Arnon medium, Chu-10 medium and JM). Growth response in each medium was recorded. For the study of effect of nitrogen two nitrogen source Ammonium chloride and Ammonium nitrate were added in culture media while for phosphorus different concentration of dipotassium hydrogen phosphate was used.

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Ethnomedicinal values of some indigenous plants for treatment of skin diseases in Mandar block of Ranchi District, Jharkhand

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Abstract : Medical ethnobotany deals with the nature and application of plants used traditional medical systems. There has been increasing interest in recent years in traditional herbal medicine particularly "tribal medicine". There are number of herbs which have come to light in the recent years with miraculous therapeutic effects. Medical ethnobotany deals with the nature and application of plants used traditional medical system and it involves the identification of plants used in traditional remedies. In this paper a survey list of some traditional plants used against skin diseases is being along with their curative values.

Key words: Traditional medicine, skin disease.

INTRODUCTION

The skin is very sensitive part of the body and protects from sun, dust, rain etc. The skin is accessible to indicate solar radiation and has a structural heterogeneity that permits the absorption of some wavelength and transmission of other. Human skin is a sandwich of two distinctive components, the epidermis and dermis, separated by the basement membrane. The epidermis is major component of the skin is dermis. In contrast to the epidermis rich vascularisation of the dermis allow it to play an important role in temperature regulation. Skin disease generally affected by environment like dust and sun.

MATERIALS AND METHOD

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The method used in collecting the ethnobotanical data the one involving field study, meeting and interviewing practitioners of ethnomedicine and identifications of plants, practicing local herbalists and knowledgeable elderly people belonging to each of the traditional communities who were convinced regarding the objectives of this study formed the source of information.

RESULT AND DISCUSSION

Many people belonging to various indigenous communities of this Mandar area have nature rich knowledge about medicinal uses of plants in their surroundings. All plants are found abundantly in Ranchi district of Jharkhand. A list of some important medicinal plants and their parts used for skin disease is given below. All plants are identified by Herbarium of Dr. T.K Ghosh of Ranchi University.

Table 1. Ethnomedicinal plants found in the Mandar block of Ranchi, Jharkhand

Sl.No.	Botanical Name	Common Name	Family	Parts used
1.	*Cassia alata*	Dadmari	Fabaceae	Leaves
2.	*Cassia tora*	Chakunda	Fabaceae	Leaves
3.	*Contella asiatica*	Beng Sag	Apiaceae	Leaves
4.	*Acorus calamus* L.	Ghorbach	Acoraceae	Roots
5.	*Amona squamosa* L.	Sarifa	Annonaceae	Bark
6.	*Curcuma longa* L.	Haldi	Zingiberaceae	Stem
7.	*Azadirachta indica* Juss.	Neem	Meliaceae	Flower, Leaves
8.	*Pongamia pinnata* L.	Karanj	Fabaceae	Seed
9.	*Citrus aurantifolia* Christm & Panz	Nimbu	Rutaceae	Fruits
10.	*Allium cepa* L.	Lahsun	Alliaceae	Stem
11.	*Calotropis gigantea* R.Br.	Madar	Apocynaceae	Leaves, seed
12.	*Bacopa monnieri*	Brahmi	Serophulariaceae	Whole plants
13.	*Holarrhena antidysentrica* Wall.	Kurchi	Apocynaceae	Bark
14.	*Datura metel*	Dhatura	Solanaceae	Leave , Seed
15.	*Ocimum sunctum*	Tulsi	Lamiaceae	Dry Leaves

CONCLUSION

The skin disease is very much popular in Mandar block of Ranchi District. It has many causes like pollution, dirty water and pollen allergy.

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